# Desperate Capital Breeds Productivity Loss: Evidence from Public Pension Investments in Private Equity

**by**

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**Abstract**

I study investments of U.S. public pensions in private equity (PE), and trace them to ultimate micro assets—target firms which PE funds invest in, using micro-data on private investments combined with confidential U.S. Census data. I show that more severely underfunded public pensions receive lower average PE returns, and match with smaller GPs on average, than less underfunded pensions. Consistent with matching and returns, firms financed by most underfunded public pensions and smallest PE funds face labor productivity decreases. I introduce a novel instrument—public unionization—in support of underfunding positions driving selection into funds. Lastly, I evaluate alternate mechanisms.

**Keyword:** public pensions, underfunding, private equity, productivity

**JEL Classification:** G23, G24, H55

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# 1 Introduction

U.S. public pensions are one of the largest institutional investors in private assets having $5.3 tn. assets under management as of 2022. The reported underfunding positions of U.S. public pensions, that is, the amount by which assets are unable to cover their present value of liabilities, have significantly worsened over time, especially post the financial crisis. This has become a huge concern for state and local public finances, given that public pensions provide benefits to 14.9 mn. active workers and 12 mn. retirees, making it of key economic importance to understand their asset allocation decisions.[1] Given their need for high expected returns, public pensions increased their allocation to alternative assets by three-fold from 2001 to 2021, with private equity (PE) occupying a substantial share.[2]

In this paper, I study the outcomes of public pensions' investments in PE based on the degree of their underfunding positions. First, I find that more severely underfunded public pensions exhibit lower average PE returns, and allocate capital to smaller GPs (general partners, parent entities of PE funds), on average, than less underfunded pensions. Second, I study how public pensions' allocation decisions ultimately impact real outcomes at target firms financed via PE funds. I show that the smaller GPs, matching with the most underfunded pensions on average, invest in target firms which do worse in terms of labor productivity than firms which receive capital from the least underfunded pensions. With larger funds getting the best deals and limited good projects, the inflow of pension capital ends up financing marginal deals on average via small funds. Next, I provide support for the underfunding channel using public union rates as a novel instrument for pension underfunding. Further, I evaluate alternate channels, such as, ruling out union board representation, local investment bias, political motives, and the influence of a few large states. Lastly, I discuss policy implications on both the pension side, where liability discounting should use risk-appropriate rates, and the firm side, where the influx of public pension capital may finance marginal deals, thereby pulling down average target-firm outcomes.

The finding of a positive relation between average PE returns and average GP size is specific to public pension LPs, and not universal to all PE funds. It is consistent with prior evidence of a positive relation between fund size and returns for this investor type (Andonov et al. (2018)).[3] This also does not

[1] Statistics from Boston College Retirement Center Public Plans Data (PPD).

[2] Alternative assets include private equity (PE), venture capital, private debt, and other privately traded assets. PE is two-thirds of alternative assets. The growing importance of PE for the economy is visible as the number of firms supported by PE increased by 106% from 2006 to 2020 providing 11.7 mn. U.S. jobs (Ernst & Young report), while the publicly listed firms have decreased over time.

[3] Refer to Table IX, Andonov et al. (2018). The paper also notes that larger PE funds tend to exhibit higher performance given their ability to raise large sums of money, and is positively correlated with manager quality.

invalidate the literature showing a negative relationship between a fund's size and its returns, as the returns exhibited by public pension LPs are across a portfolio of PE investments. For instance, multiple investments by well performing small PE funds might not be enough to overturn an under performing investment by a large fund.

To study these questions, I construct a comprehensive micro-dataset on PE buyouts, combining detailed investments by public pension limited partners or "LPs" (e.g., CalPERs) to general partners or "GPs" (e.g., Blackstone Group), their corresponding PE funds (e.g., Blackstone Capital Partners VI), and the target firms (e.g., Hilton) financed by the individual PE funds, from Preqin, along with real outcomes at target firms from the U.S. Census Bureau micro-data. I supplement characteristics of public pensions from the Public Plans Data (PPD) and the Comprehensive Annual Financial Reports (CAFRs). Preqin obtains most of its data from FOIA requests, less likely to suffer from selective disclosure bias for non-public investors, and comprehensive for public pension investors.[4] Recent academic literature (Andonov et al. (2018), Begenau and Siriwardane (2020), Korteweg et al. (2023)) has used the Preqin data to study public pensions' investments in alternative assets. Sparse data availability on private firms, and potential name and ownership changes following a PE deal, makes the study challenging. To overcome these hurdles, I merge the PE transactions along with the LPs, GPs, and PE funds, with the U.S. Census Longitudinal Business Database (LBD), providing comprehensive information to track 6,700 PE targets over time, covering close to 8% of total U.S. non-farm business employment (10.9 mn. jobs) and 11% of revenue ($3.1 tn.) in 2018. This allows me to track the entire chain of capital from the end investor via the PE fund to the end recipient, along with real outcomes at target firms.

In the first part of the paper, I study the average PE returns realized by public pensions and their investment allocations within PE by the degree of their underfunding positions. After accounting for pension characteristics, such as, size as measured by assets, current and past asset allocations, and investment consultants, who are potentially key important players in asset allocation (Andonov et al. (2022)), I find that a 10 percentage point increase in underfunding ratios of public pensions is associated with a 1.36 percentage point decline in average annual PE performance, as measured by annual net IRR. This corresponds to an average decline of 11.4%. Further, I find that even after controlling for various PE fund and pension characteristics, the more underfunded public pensions match with on average smaller GPs. A one standard deviation increase in underfunding is associated with a 9.5% decrease in

[4] Begenau et al. (2020) compares Preqin data with self filed FOIA requests and confirms the comprehensive coverage for U.S. public pensions.

the average size of the GP. Size of the GP is the sum of the sizes of constituent PE funds existing at each point in time. This is a useful measure, especially since majority of the GPs are non-traded. Larger PE book assets represent bigger scale, access to better deals, and more broadly, better PE quality within the PE industry. Managerial skill is broadly reflected in the cross-sectional distribution of fund size and assets under management (Berk and van Binsbergen (2015)). On average, the small GPs may either not have the expertise or experience, relative to the large GPs. On the other hand, because I observe the average PE returns received by pensions across a portfolio of PE investments, it is possible that multiple small GPs perform well on the margin, but it is not enough to cover up the under performance by an individual large GP, and on average the more severely underfunded pension matches with small GPs and observes lower PE returns. The matching between the most underfunded public pensions and the smallest GPs can exist for multiple reasons. Large GPs have stable investor relationships, while the smaller GPs now get more capital from the most underfunded pensions, which are in greater need to close funding shortfalls.[5]

I discuss and account for various measurement challenges. U.S. public pensions use the average assumed return on their investments to discount their liabilities, which distorts the representation of their actual unfunded liabilities (Novy-Marx and Rauh (2011)). Following Lenze (2020), I calculate the Bureau of Economic Analysis (BEA)-Adjusted underfunding ratios using a common and lower discount rate across plans, reflecting the risk of cash flows.[6] I find that a 10 percentage point increase in the market value underfunding ratio is associated with a 1.50 percentage point $(12.6\%)$ to 1.57 percentage point $(13.2\%)$ decrease in average annual net IRR, larger than using the reported pension underfunding ratios. Similarly, coefficient estimates using the market value of underfunding ratios predict a larger decline in magnitude of average GP size, as compared to the accounting measures. The strengthening of the magnitude highlights that the accounting underfunding measures understate the incentives of public pensions. Further, I account for varying fiscal end dates and actuarial cost methods used by pension plans. Overall, the measurement robustness exercises confirm the main results that more underfunded public pensions realize, on average, lower returns from PE and match with smaller GPs.

In the second part of the paper, I investigate the returns to underfunded public pensions by examining real outcomes at the target firms underlying the PE deals. PE funds receive returns from deals based

[5]Another explanation is that small PE funds have to engage in marketing efforts to attract capital, and accept low quality capital by the most underfunded pensions. There is also a large overlap between the small and new PE funds. Lerner et al. (2022) document preferential access of capital between top investors and top funds.

[6]The discount rates used are stipulated by the BEA, and used by the Federal Reserve Board (FRB) Financial Accounts. I am grateful to Irina Stefanescu for helpful conversations.

on the performance of target firms they invest in, and public pensions receive returns from their PE investments, net of management and performance fees. Since majority of the PE target firms are privately held and small, I do not observe returns of these companies. Hence, I use the Census Bureau micro-data to observe real outcomes — employment, revenue, and wages — at the firm level from the Longitudinal Business Database (LBD), following the literature. Detailed cost measures are available only for manufacturing firms, which is a small subset. Hence, labor productivity is the standard productivity measure in the literature, to study firms across industries.[7] Because Preqin draws primarily on public reports and is well represented for public pensions (Begenau et al. (2020)), the majority of target firms in the sample are financed by PE funds with public pension investors, which is the focus of this study. Using all the PE-backed target firms matched to the LBD, I find no significant or economically meaningful increase in labor productivity following the buyout, after controlling for observable firm characteristics: industry, size, age, single/multi-unit status, pre-buyout growth, and year fixed effects. In the smaller, less representative subsample of manufacturing firms, I similarly find no substantial increase in total factor productivity (TFP) or labor productivity following a PE deal. These estimates are suggestive of a PE buyout's role rather than causal, given the potential for selection on unobservables into a PE deal.

In the next section, I correlate real outcomes at target firms with public pension underfunding positions and GP size, which supports the matching between underfunded pensions and GPs, documented above. On average, 20 LPs are involved in financing a deal via commingled PE funds. Firms need to receive capital from U.S. public pension LPs to appear in the sample, which helps ensure that the analysis reflects firms financed by U.S. public pension PE capital. I categorize target firms into terciles based on the degree of the underfunding positions of pensions (at the time of capital commitment) financing the firm via a PE fund, weighted by the capital committed by the pension. I compare very similar target firms but differing by the type of investor in the deal, by using a fully saturated specification of industry, size, age, type of firm, pre-buyout growth, and year controls. I show that the deals financed by the bottom most tercile of funded public pensions (i.e., the most underfunded pensions) experience a $-5.5\%$ significant change in labor productivity per year, whereas, firms financed by the top most tercile

[7] I study labor productivity at the firm level, which is a standard productivity outcome in the literature using the Census micro-data for firms *across* industries (Davis et al. (2019)). Total factor productivity (TFP) would have been ideal, however, only manufacturing firms have TFP data from the Census of Manufactures (CMF) and the Annual Survey of Manufactures (ASM). Over time, the share of PE backed firms has shifted away from manufacturing to other industries such as technology and healthcare. The sample of target firms financed by public pension invested PE funds have very few manufacturing firms, which makes the cross-sectional analysis with the Census TFP data not feasible. However, when studying broader implications for the average firm and pooling the firms together, I find no significant increases in both labor productivity and TFP for the very small subsample of manufacturing firms (refer to Appendix D.5).

of funded pensions (i.e., the least underfunded public pensions) face a $-1.3\%$ insignificant yearly productivity change. The cross-sectional differences are statistically different across pension underfunding categories. This suggests that the greater the share of PE capital sourced from underfunded pensions, the larger the subsequent productivity loss at the target firm.

Since capital flows from LPs to target firms via GPs, and the decision of investments in firms is made by the GPs and not the LPs, I study the differences in target firms' outcomes based on GP characteristics. I split target firms based on the size of the GP financing the deal. Bigger GPs also have larger teams, and management staff, and broadly represent better fund quality, following the mutual fund literature (Berk and van Binsbergen (2015)). I use size as the sum of book value of capital committed by investors to PE funds, additional market value of investments based on performance, and capital yet to be called, i.e., dry powder.[8] I also construct a book value measure of size. Rankings of GPs are consistent over time and across measures. When more than one GP is financing a deal, I weight the GP characteristics by the number of funds (per GP) involved in the deal. I find that targets financed by the smaller GPs experience the largest productivity declines. For instance, firms financed by GPs in the bottom 25th size percentile experience a -2.9% annual labor productivity change as compared to firms in the top 75th percentile which face a +1.4% productivity change. The negative productivity changes are larger, further down the GP size distribution. I only observe total wages at firms as one of the costs incurred by target firms, and find the patterns hold after taking them into account. Similar to the investor-level analysis, I saturate the regression with a full set of observable firm characteristics to compare similar firms differing only by GP, though I cannot account for unobservables. Decreasing labor productivity along the PE GP size distribution suggests a decreasing returns to scale at the industry level, different from a decreasing returns at the fund level. Capital from underfunded public pensions fuels the growth of more, on average, smaller PE funds, resulting in more projects being financed, pulling down the average over time. Cross-sectional differences in labor productivity post PE deals emerge in splits of firms based on both, pension and GP characteristics, supporting the assortative matching between the more underfunded pensions and on average, smaller GPs documented in the first part of the paper. The decrease in labor productivity at firms is consistent with the lower average PE returns exhibited by the most underfunded pensions, and matching with the broadly interpreted lower quality GPs.

In the third part of the paper, I provide further evidence for the channel of underfunding positions of public pensions. One potential confounder is that more underfunded pensions might be less skilled in

[8] This is directly supplied as one measure by the data provider.

selecting investments, leading them to match with on average, smaller, lower quality PE funds. To rule out alternative mechanisms, I conduct multiple tests. First, I use a novel instrumental variable (IV) for the underfunded positions of pensions: public unionization rates, which predict underfunding through higher public-sector bargaining power and wages (Booth and Chatterji (1995)), while plausibly affecting PE allocations only through underfunding ratios of pensions. I ensure the instrument's strength and validity is not driven via few large states by removing one large state at a time or few large states together. To ensure the exclusion restriction is not affected by union board members, I use data on public pension board composition from Andonov et al. (2018) to exclude plans with union representatives on the board. I also lag the instrument by one year to be conservative regarding reverse causality. Further, I control for investment consultants, who may make investment decisions for public pensions (Andonov et al. (2022)). Using this instrument, I find that more underfunded pensions allocate capital to on average, smaller GPs, and earn lower PE returns. Second, I perform additional checks to evaluate alternate hypothesis. These results are robust to excluding states with large public pension plan coverage or political heterogeneity, and pensions with concentrated political or in-state investment exposure.

Finally, I discuss broader implications of public pensions' investments in private equity. The paper has important policy implications for the fragility of state and local retirement systems, and lends support to the argument for discounting public pension liabilities using a common, risk-appropriate discount rate across plans (Novy-Marx and Rauh (2009)). This paper shows that the most underfunded pensions, which are in severe need to close their funding shortfalls, realize, on average, lower returns from PE than less underfunded pensions, which could further worsen their funding shortfalls. On the firm side, I evaluate all PE target firms matched to the U.S. Census micro data against a control group of non-PE target firms, based on existing literature (in the same industry, firm age and size category, multi/single unit, and buyout year), and find that labor productivity declines by an insignificant 0.4% at target firms post buyout relative to controls. This raises important questions for future research: whether U.S. public pension LPs are a contributing source of the excess capital supply in PE, such that, at the margin, lower quality deals are being financed?

**Related Literature.** The paper contributes to four main strands of literature. First, is the literature studying pensions' investment decisions and its incentives (Andonov et al. (2015), Andonov et al. (2017), Andonov et al. (2018), Andonov and Rauh (2022), Andonov et al. (2022), Andonov et al. (2025), Hoops et al. (2016), Shivdasani and Stefanescu (2010), Stefanescu et al. (2018), Rauh et al. (2020), Ivashina and Lerner (2018), Hochberg and Rauh (2013), Chemla (2004), Giesecke and Rauh (2022)). Andonov et al.

(2018) examines the relation between representation of state officials on pension boards and PE returns, broadly defined, and finds a negative relation for venture capital and real estate, but not for buyout funds. Korteweg et al. (2023) isolates the risk-compensation from PE returns for public pensions. Peng and Wang (2019) shows that pensions' investments in private assets might be a short term solution. This paper substantiates the existing field of work by focusing on public pensions' investment decisions driven by their underfunding positions, and further, connecting them to real outcomes at target firms, the end receivers of capital.

Second, this paper complements the existing literature on relationships between investors and funds (such as, Lerner et al. (2007), Sensoy et al. (2014), Da Rin and Phalippou (2017) document heterogeneity in returns by investor types, Andonov et al. (2021) examines infrastructure funds, Begenau and Siriwardane (2020), Phalippou (2020) study fees paid). This paper shows assortative matching between investors and funds, with a focus on cross-sectional distribution of public pensions based on their underfunding. Moreover, the existing literature either studies effects of private funds on firms or investments by LPs to funds. This paper connects both, LP investments with outcomes on PE target firms.

Third, the paper contributes to a vast and mixed body of literature on fund size, deal flow, and performance in the PE buyout industry. Kaplan and Schoar (2005), Robinson and Sensoy (2013), Harris et al. (2014), Rossi (2019), and Harris et al. (2023) document mostly an insignificant relation between fund size and performance for buyout funds. Robinson and Sensoy (2016), Chung et al. (2012), and Metrick and Yasuda (2010) show a positive relation and scalability, while some papers show a negative relation (Lopez-de Silanes et al. (2015), Braun et al. (2026), Bhardwaj et al. (2025)).[9] Kaplan and Schoar (2005) also discusses better GPs having proprietary deal flow and ability to invest in better investments, substantiating this paper's results. It is worth highlighting key differences between the existing literature and this paper. These papers are agnostic on the LP, while this paper focuses on public pension invested PE funds. The closest paper in this strand of literature is Andonov et al. (2018), which finds a positive relation between fund size and PE performance among funds invested in by public pension LPs — consistent with the results in this paper.[10] Further, size here is measured as the average across the GPs, encompassing PE funds, a pension invests in. Similarly, PE returns exhibited by pensions reflect a portfolio of investments. The positive relation between average GP size and pension-level PE returns does not invalidate the literature showing a negative relation between a fund size and its return, since several

[9]Refer to Brown et al. (2024) for a further overview on this topic.

[10]Refer to Table IX, Andonov et al. (2018). The paper also notes that larger PE funds tend to exhibit higher performance given their ability to raise large sums of money, and is positively correlated with manager quality.

well performing small funds may be insufficient to offset a large under performing fund. Another difference is that I study the cross-section of funds, and not performance within a fund over time. This paper builds on the existing literature by relating pension characteristics, specifically, underfunding positions, to average PE returns and size of their PE investments.

Lastly, I contribute to the literature studying financial and real outcomes of private funds. The PE industry is opaque, involving many layers of financing, data is limited, and finding identification which caters to all industries has been difficult. Existing research studies specific industries (Davis et al. (2014), Bernstein and Sheen (2016), Faccio and Hsu (2017), Liu (2021), Howell et al. (2022), Johnston-Ross et al. (2025)), or relies on survey data and case studies (Baker and Wruck (1989), Jensen (1999), Metrick and Yasuda (2010), McCourt (2017)), and other countries (Amess and Wright (2007), Boucly et al. (2011), Antoni et al. (2019)). Davis et al. (2019) studies PE target firms across industries and finds heterogeneous results based on the type of deal, with employment changes varying from $-12.6\%$ to $+12.8\%$. There is also a vast literature studying other financial outcomes (Ivashina and Kovner (2011), Kaplan et al. (2008), Bernstein et al. (2018)). I build on the existing work by studying target firms financed by public pension LPs, and importantly, examining real outcomes at target firms based on LP and GP heterogeneity. I study a large and longer sample period, tracking 6,700 U.S. target firms across 22 industries.

**Overview.** Rest of the paper is structured as follows. Section 2 gives an overview of the data and presents institutional details. Section 3 discusses public pensions' investments in PE. Section 4 documents real outcomes at target firms based on public pension LPs and GPs. Section 5 presents evidence in support for the channel of underfunding positions of pensions via an instrumental variable strategy, and tests for alternate mechanisms. Section 6 discusses broader policy implications. Section 7 concludes and discusses areas for future research.

## 2 Data and Institutional Background

### 2.1 Data

In this section, I provide an overview of the data used for the analysis. I construct a comprehensive dataset of investments in PE by U.S. public pensions, combining pensions' investments in PE funds, and deals between the PE funds and target firms. Additionally, I track the target firms over time.

### 2.1.1 Public Pensions and Private Equity Transactions

**Preqin.** On the investor side, I obtain investments by institutional investors ("LPs"), focusing on U.S. public pensions, in individual PE funds and GPs. Preqin obtains most of its public pension data through FOIA requests, which means that U.S. public pension coverage in Preqin is comprehensive. Begenau et al. (2020) compares Preqin's data with self-filed FOIA requests to U.S. public pensions and confirms this. Since this analysis relies solely on public pensions, it is less likely to suffer from the selective-disclosure bias, which is prevalent among non-public investors. Further, I am able to manually match most of the U.S. public pension LPs from Preqin to the Public Plans Data (PPD), which is a complete and public data source for the largest U.S. public pensions (discussed below). Moreover, recent academic literature (for example, Andonov et al. (2018), Begenau and Siriwardane (2020), Korteweg et al. (2023)) has used Preqin's data for studying U.S. public pensions' investments in alternative assets.

The main advantage of this data is the connection between investors and individual PE funds within the GP, which is an extremely sensitive part of the data, and allows me to study capital flow to firms accurately and at a granular level. On the deal side, I obtain transactional data on deals between PE funds and target firms. I observe the PE fund and the corresponding GP financing the deal, target firm, and the deal date. I also obtain a comprehensive list of PE fund attributes, including their location, vintage, GP, and industry focus. I consider private equity funds whose main strategy is "buyout".[11] Due to differences in structure, I do not consider venture funds that invest in startups. The data on investors, PE funds, and firms, is from 1979 to 2021, with better coverage post 2000. The main sample period is 2001-2021. In this paper, I focus on U.S. public pensions and U.S. target firms.

I merge the investment side and deal side data, to obtain the full chain of capital flow in private markets from end investor (U.S. public pensions) to PE funds, and subsequently to end recipients (target firms). There is no one dataset which covers PE transactions comprehensively. I supplement Preqin with news sources and manual web searches to verify deals, identify names of target firms before and after buyout, and ensure accurate addresses, which I use for the matching with the U.S. Census Bureau micro-data (discussed below).

**Public Plans Data (PPD), Pension Websites.** I obtain data on U.S. public pension characteristics from the Public Plans Data (PPD) and 75 Freedom of Information Act (FOIA) responses from individual state pensions shared with me by Korteweg et al. (2023).[12] This provides financials and investment

[11] The strategy is as defined by Preqin, and is fairly standard and consistent across datasets, such as, SEC form ADV filings.

[12] The Public Plans Data (PPD) (link: https://publicplansdata.org/) is maintained through a partnership between Center for Retirement Research at Boston College (CRR) and MissionSquare Research Institute. I am grateful to Korteweg et al. (2023) for

allocations by asset class of U.S. public pensions over time. I track 204 U.S. public pensions, covering 95% of state/local public pension assets. I connect data on public pension financials from PPD and FOIA requests with pensions' investment allocations to individual PE funds from Preqin via a manual match on public pension name. I obtain the hierarchy of state pensions from state websites, and merge exact entities if available in both datasets, and consider the parent entity, if not available.

**Comprehensive Annual Financial Reports (CAFRs).** I collect data on total pension liabilities and net pension liabilities for pension plan rate plus/minus one percentage point from the public financial statements of the U.S. public pensions. According to the Governmental Accounting Standards Boards (GASB), pensions are required to report the liability sensitivities from 2014 onwards. This enables me to construct the market value of pension liabilities following the BEA methodology (Lenze (2020)). I also use the CAFRs to collect the reported total and net pension liabilities at the plan rate, the plan assets, and the fiscal end dates for each plan-year, to verify them against the PPD. Further, I use the CAFRs to cross-check PE returns collated by the PPD for a subset of plan-years.

### 2.1.2 Matching with Census Micro-Data for Target Firm Data

To track the underlying firms of public pensions' investments in PE over time, I merge the individual public pension investments and the subsequent PE deals with the U.S. Census Bureau micro-data. As most of the target firms are private, studying these firms encounters several limitations. First, basic balance sheet, income statement, and market valuation statistics, which is available for public firms, is rarely available. Second, studies have relied on surveys, which might depict an incomplete representation. Third, changes in firm name or ownership following a PE deal are not uncommon. To deal with these central concerns, I turn to the Longitudinal Business Database Revenue Enhanced (LBDREV or the LBD) of the Census Bureau, which has the most comprehensive information on the universe of private firms in the U.S., along with time consistent firm identifiers enabling me to study a firm over time.[13]

First, I merge the target firms with the Standard Statistical Establishment Listing (SSEL) database. The SSEL, also known as the Business Register (BR), provides and updates names and addresses of all establishments in the U.S. each year, along with establishment and firm identifiers connecting entities over time. I use name and address fields in the SSEL and the Preqin's PE target firm data to merge these

sharing their FOIA requested data with me.

[13]Longitudinal Business Database Revenue Enhanced (LBDREV) is the revenue enhanced and revised version of the original Longitudinal Business Database (LBD). The major improvement of LBDREV over LBD is consistent longitudinal firm and establishment identifiers across time. I will refer to LBDREV as LBD going forward. I give a full description of the data and matching in detail in Appendix E and F.

two datasets. Since target firms might undergo name and entity changes following the buyout, I use names and addresses one year pre-buyout in the SSEL.[14] After merging the buyout deals with the SSEL, I use Census firm-establishment linkages to combine all establishments at the firm level across years for the matched target firms.

Second, I link the merged PE investments by public pensions, PE deal transactions, and the SSEL data to the LBD. The LBD covers all business establishments in the U.S. private non-farm sector with at least one paid employee (Jarmin and Miranda (2002)), covering approximately 7 million firms and 9 million establishments as of 2019. Connecting the target firms with the LBD allows me to observe granular changes in employment and revenue at target firms over time, along with their industry affiliations. The revised LBD with revenue and employment, which I have access to, is available for 1997 to 2018. The PE target firms in the sample matched to the LBD account for close to 8% of the total U.S. non-farm business employment and 11% of revenue in 2018. This corresponds to 10.9 mn. jobs and $3.1 tn. revenue (Appendix Figure F.1).[15]

Third, the Annual Survey of Manufactures (ASM) and Census of Manufactures (CMF) provide detailed cost data only for manufacturing firms. This allows me to study total factor productivity, an additional and commonly used productivity measure, for this small, though unrepresentative, subset of target firms. In the recent years, PE transactions have evolved away from the manufacturing sector, towards other industries, such as, healthcare and information technology.

#### 2.1.3 Other Data: Unionization Rates, Pension Board Composition

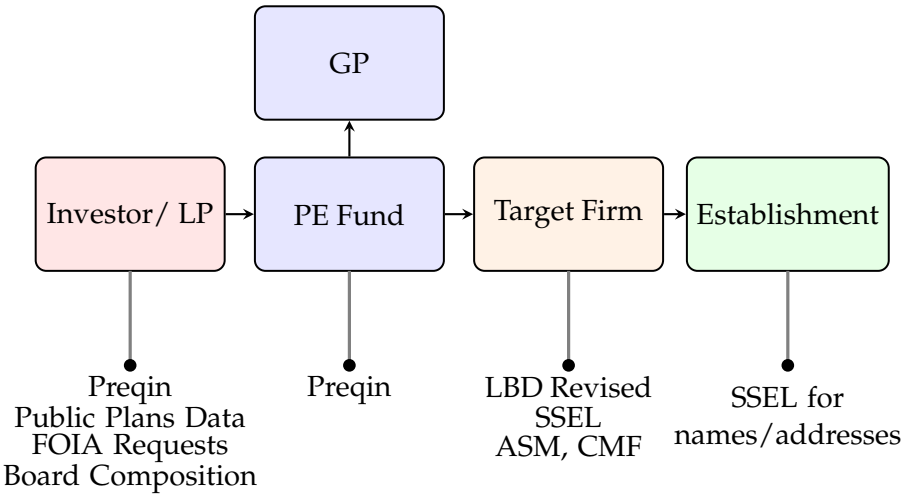
For the instrumental variable strategy, I obtain unionization rates at the state-year level from the Current Population Survey (CPS), made available through Union Stats and the U.S. Bureau of Labor Statistics (BLS).[16] I also obtain the board composition of public pensions from Andonov et al. (2018) and the National Association of State Retirement Administrators (NASRA). Figure 1 shows a visualization of the data structure and its sources. Appendix E describes the datasets in detail, and Appendix F describes the merge process across datasets.

[14] I implement a very strict matching criteria, described in Appendix F. Appendix Figure F.3 shows match rates of PE target firms over time. Appendix Figure F.4 shows match rates of PE target firms by industry and state. The merge is robust based on firm characteristics one to two years pre-buyout.

[15] Appendix Figure F.2 shows total employment and revenue for the Census matched target firms as a percentage of total U.S. non-farm payroll employment and revenue in the year of buyout.

[16] More information on Union Stats can be found here: http://www.unionstats.com. Union Stats and BLS provide consistent data from CPS.

Figure 1: Visualization of Data and its Sources




*Notes:* The figure draws connection between the data and its sources. FOIA – Freedom of Information Act. Board composition data is obtained from Andonov et al. (2018) and NASRA – National Association of State Retirement Administrators. LBD – Longitudinal Business Database. SSEL – Standard Statistical Establishment Listing. ASM – Annual Survey of Manufactures. CMF – Census of Manufactures.

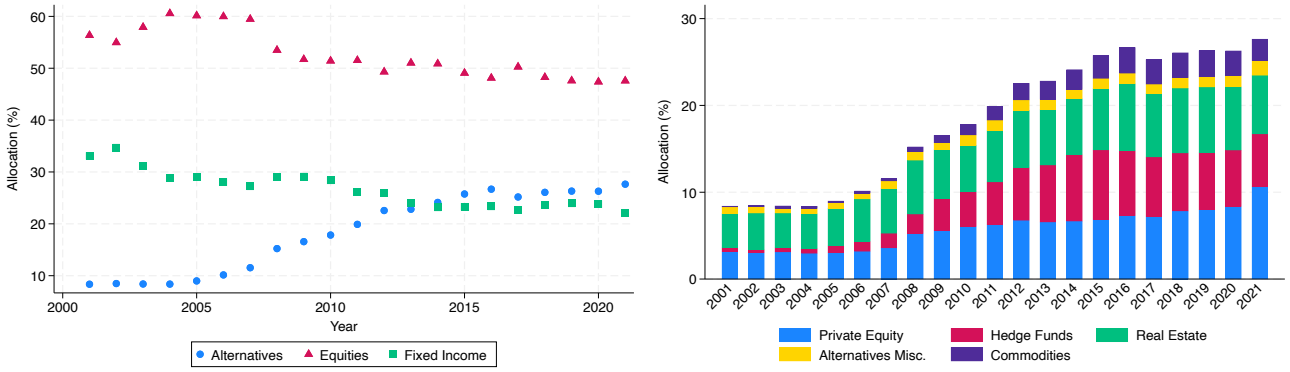
### 2.1.4 Summary Statistics

Figure 2 shows that, on average, U.S. public pensions increased their allocations to alternative assets from 8.4% (2001) to 27.6% (2021), while their allocation to public equities and fixed income decreased over time. Within alternatives, PE has occupied a substantial share, averaging 10.6% of total assets in 2021. Table 1 shows summary statistics for 2001 to 2021, for the U.S. public pensions in the sample. The sample covers 204 pension plans from the Public Plans Data (PPD), covering close to 95% of state/local pension assets and members in the U.S.[17] The average underfunding ratio reported by public pensions, calculated as one minus assets divided by reported liabilities, over the sample period is 22.06%, with a standard deviation of 19.22%. This implies that for every $1 of liabilities due, the pension plans will fall short by 22.06 cents. When I split pension plans into terciles each year based on their underfunding ratios, the cross-sectional dispersion is even more stark. The average underfunding ratio for the most underfunded tercile of pensions is 40.68%, and for the least underfunded category is 4.35%. Appendix Figure E.2 shows that the share of capital committed to PE funds by the bottom most tercile of funded public pensions, defined in the year prior to commitment, as a percentage of all capital committed by public pensions increased from 9.6% in 2001 to 29.0% in 2018. This corresponds to a commitment amount of $919 mn. in 2001 and $14 bn. in 2018. According to the Governmental Accounting Standards Board (GASB), pensions calculate the funding measures using an assumed rate of return on their assets, which

[17] Refer to: https://publicplansdata.org/.

averaged to 7.66% with a standard deviation of 0.52% during the period. These numbers obscure the true market value of unfunded liabilities. The true underfunding ratios, calculated using an appropriate common discount rate reflecting the risk of cash flows, are much larger than the reported underfunding ratios. I will discuss this further in Section 3.4. Pension assets are the actuarial assets measured under the GASB rule and average $14.3 bn. The PE performance measure is the net internal rate of return (IRR) averaging across the PE investments of the pension's PE portfolio.[18] The average net IRR is 11.81%. This is comparable to the average net IRR of 10.4% exhibited by U.S. public pensions in Andonov et al. (2018), which is obtained from Preqin.

Figure 2: Portfolio Allocation of U.S. Public Pensions Over Time




**(A)** Across asset classes **(B)** Within alternative assets

*Notes:* The y axis represents portfolio allocation of U.S. public pensions as a percentage of total assets from 2001 to 2021. Panel A shows asset allocation of U.S. public pensions *across* asset classes over time. Panel B focuses on *within* the alternative asset class. Data are sourced from Public Plans Data (link: https://publicplansdata.org/).

Table 2 provides a summary of the public pension financed PE target firms matched to the U.S. Census micro data. I match 9,300 target firms and 190,000 establishments, representing close to 8% of the total U.S. non-farm payroll employment in 2018 (Figure F.1). For 6,700 firms, I am able to construct labor productivity, defined as real revenue per employee. The sample period for studying the real outcomes is 1997 to 2018, the years for which the revised LBD has both, revenue and employment measures. Panel A shows PE target firms have on average 1,700 employees, $571 mn. revenue in 2020 U.S. dollars, and generate $400,400 revenue per employee.

On average, 20 investors finance a private equity deal through 1.4 funds. In Panel B, I split the public pension financed target firms into terciles based on underfunding positions of pensions, weighted by the capital committed by the pension. There is another category of firms not reported here, that is financed

[18]Majority of the pension plans report performance net of fees in their annual reports. For example, Teacher Retirement System of Texas reports returns "net of all fees" in the investment section of its 2015 CAFR. I manually cross check a subset of plan-years from the PPD with the CAFRs.

Table 1: Summary Statistics of U.S. Public Pensions, 2001-2021

| Variable | Tercile Split | Mean (1) | Median (2) | SD (3) | P25 (4) | P75 (5) |
|---|---|---|---|---|---|---|
| Underfunding ratio (%) | | 22.06 | 21.37 | 19.22 | 9.60 | 33.50 |
| | Most underfunded | 40.68 | 39.00 | 14.75 | 32.50 | 46.81 |
| | Medium underfunded | 20.83 | 22.40 | 8.35 | 15.10 | 27.20 |
| | Least underfunded | 4.35 | 6.20 | 12.65 | -0.68 | 13.20 |
| Pension assets (mn.) | | 14,351.57 | 4,712.94 | 29,393.76 | 1,382.50 | 14,463.70 |
| PE return (%) | | 11.81 | 12.70 | 15.87 | 5.10 | 19.19 |
| Pension discount rate (%) | | 7.66 | 7.75 | 0.52 | 7.50 | 8.00 |
| # Public pensions | 204 | | | | | |

*Notes:* This table presents summary statistics for U.S. public pensions from 2001 to 2021. The data are sourced from the Public Plans Data (https://publicplansdata.org). The sample covers 204 of the largest U.S. public pensions. The observation level is pension plan by year. The reported underfunding ratio is calculated as one minus the funding ratio, where the funding ratio is calculated as (plan assets/reported liabilities). Public pensions are split into terciles based on the underfunding ratios in *each* year, and categorized as most underfunded, medium underfunded, and least underfunded. Pension assets are the actuarial assets measured under the GASB, and represented in million dollars. The PE return measure is the internal rate of return, and the pension discount rate is the assumed rate used by pensions to discount their reported liabilities, both as reported on the PPD, and verified from the CAFRs.

by other investors, which is not the focus of the study. The most underfunded public pension financed target firms have an average revenue per employee of $381,200, while the least underfunded public pension financed target firms have an average labor productivity of $454,900.

In Panel C, I split firms by a measure of GP size. This measure is proxied by the market value of the GP, including the book and market value of private investments. This measure is directly provided by the data provider. The fund size measure, which also includes the market value of investments, is suggestive of PE fund quality, borrowing from the mutual fund literature which shows manager skill is visible in the cross-sectional distribution of fund size (Berk and van Binsbergen (2015)). These target firms are financed by 850 GPs and their 2,200 PE funds. Firms financed by the bottom 25th percentile of GP size have an average $391,000 in labor productivity, and those financed by the top 75th percentile generate $407,600 per employee on average.

## 2.2 Institutional Background

Private equity as a form of financial intermediation has gained prominence over the past 20 years. Capital is provided by institutional investors, also called the limited partners or “LPs" to firms or “targets" via PE funds. Institutional investors like public pensions, insurance companies, sovereign wealth funds, private pensions, endowments, and family offices are suppliers of capital. The PE fund (for example, Blackstone Capital Partners VI) receives capital from LPs, and finances a target firm (for example, Hilton). The PE fund is part of a larger entity, called the general partner or the “GP" (for example, Blackstone Group). Appendix Figure E.1 provides a schematic depiction of the institutional structure.

Table 2: Summary Statistics of Public Pension financed PE Target Firms, 1997-2018

| | Count (1) | Mean (2) | Median (3) | Std Dev (4) | 25th Pct (5) | 75th Pct (6) |
|---|---|---|---|---|---|---|
| **Panel A: All Targets** | | | | | | |
| Employment | 9,300 | 1,500 | 62 | 11,000 | 15 | 300 |
| *Targets with Productivity* | | | | | | |
| Employment | 6,700 | 1,700 | 76 | 11,000 | 22 | 350 |
| Revenue (000s) | 6,700 | 571,000 | 19,000 | 4,712,000 | 4,900 | 88,500 |
| Revenue/Employment (000s) | 6,700 | 400.4 | 235.7 | 1,300 | 139.5 | 417.3 |
| **Panel B: By Pension Category** | | | | | | |
| *Most Underfunded Pensions* | | | | | | |
| Employment | 1,200 | 1,200 | 69 | 6,100 | 20 | 322 |
| Revenue (000s) | 1,200 | 427,000 | 17,000 | 3,015,000 | 4,800 | 73,000 |
| Revenue/Employment (000s) | 1,200 | 381.2 | 238.4 | 644.8 | 144.8 | 402 |
| *Medium Underfunded Pensions* | | | | | | |
| Employment | 1,300 | 1,300 | 65 | 7,000 | 19 | 292 |
| Revenue (000s) | 1,300 | 499,000 | 17,500 | 6,069,000 | 4,100 | 77,000 |
| Revenue/Employment (000s) | 1,300 | 389.7 | 233.5 | 834.9 | 138.6 | 420.8 |
| *Least Underfunded Pensions* | | | | | | |
| Employment | 1,400 | 3,500 | 171 | 18,500 | 30 | 1,057 |
| Revenue (000s) | 1,400 | 1,108,000 | 45,500 | 6,071,000 | 7,100 | 267,000 |
| Revenue/Employment (000s) | 1,400 | 454.9 | 252.5 | 2,534 | 142.1 | 446.7 |
| **Panel C: By GP Category** | | | | | | |
| *Bottom 25th Percentile* | | | | | | |
| Employment | 700 | 1,000 | 74 | 4,800 | 26 | 287 |
| Revenue (000s) | 700 | 385,000 | 19,000 | 2,396,000 | 6,000 | 63,000 |
| Revenue/Employment (000s) | 700 | 391.0 | 236.3 | 616.2 | 132.9 | 438.6 |
| *Top 75th Percentile* | | | | | | |
| Employment | 4,800 | 1,900 | 80 | 11,500 | 21 | 424 |
| Revenue (000s) | 4,800 | 653,000 | 20,500 | 5,255,000 | 4,800 | 104,000 |
| Revenue/Employment (000s) | 4,800 | 407.6 | 240.2 | 1,443 | 141.8 | 421.9 |

*Notes:* PE deals from 1997 to 2018 are considered. Medians and percentiles are calculated according to U.S. Census Bureau disclosure rules. Observations are rounded to meet Census disclosure requirements. Panel B splits targets based on the underfunding ratios of public pensions financing the deals. Underfunding ratios are aggregated at the firm level using commitment amounts as weights. Panel C splits targets based on size of the GP financing the target, proxied by average market value of assets.

PE funds obtain majority of the capital for deals, approximately 95%, from the LPs, while the rest is financed by themselves. The Limited Partnership Agreement (LPA) specifies the contractual terms between LPs and GPs, including returns and fees. Fees includes a management fee and performance fee, and are negotiated between the LP and the GP. LPs give a commitment of capital to PE funds. This capital is generally committed at inception of the fund. Over time, PE funds call portions of the committed capital, and LPs make their contributions to funds. On receiving the capital, PE funds invest in target firms, earn cash flows from operations or from disposition of investments, and make distributions to their investors. These distributions are net of management and performance fees. The returns net of

fees follow a waterfall structure where the PE fund's portion of returns (or "carried interest") becomes larger as performance hurdles are reached. Investors are residual claimants on the net asset value of the fund.

# 3 Public Pensions' Investments in Private Equity

## 3.1 More Underfunded Pensions Exhibit Lower PE Returns

According to the Governmental Accounting Standards Board (GASB) rulings, the U.S. public pensions use the average assumed return on their investments to discount their unfunded liabilities. This means that investing in private assets decreases underfunding ratios both ways, by increasing the expected growth rate of assets and by decreasing the present value of liabilities. The incentive to invest in private assets is naturally greater for the most underfunded public pensions. Having shown that public pensions' allocations to alternative assets have increased drastically over the past two decades, in this section, I study the PE performance of public pensions based on the degree of their underfunding positions.

I begin the empirical analysis in Table 3 with a regression of PE performance on the underfunding ratios of public pensions. The time period studied is 2001-2021, when public pensions increased their allocations towards alternative assets (Figure 2). The observation level is public pension-year. The performance measure capturing PE returns is the net internal rate of return (net IRR) reported by public pensions in their Comprehensive Annual Financial Reports (CAFRs) and collated by the Public Plans Data (PPD). This captures the average PE performance, net of fees, obtained by public pensions across their portfolio of PE investments. The main independent variable is underfunding ratio, defined as one minus the funding ratio reported by public pensions. In Section 3.4, I show that the results strengthen when using the market value of underfunding positions of public pensions, which reflects the true accrued pension promises. I account for varying discount rates, fiscal years, and actuarial cost methods used by plans (Novy-Marx and Rauh (2011)), following the BEA methodology (Lenze (2020)).

In Table 3 models (1) to (4), I estimate the regression specification 1 measuring how PE returns respond with pensions' underfunding positions, captured by $\beta$.

$$y_{pst} = \gamma_t + \beta \cdot \text{Underfunding Ratio}_{pst} + \text{Controls} + \epsilon_{pst} \tag{1}$$

$y_{pst}$ is the total realized PE return, measured as net IRR, for public pension $p$ in state $s$ at time $t$. Underfunding Ratio$_{pst}$ is the reported underfunding ratio of the corresponding plan. The control variables increase gradually from models (1) to (4). All models include public pension and year fixed effects. In model (1), I control for the natural logarithm of the pension's assets, as the size of the pension is likely to play a role in accessing PE investments and subsequently returns. In model (2), I control for the current PE allocation of the plan in addition to the size, as the amount allocated to PE is likely to affect total returns. Similarly, in model (3), I control for the plan's current allocation to all alternative asset classes. In model (4), I control for past average asset allocations and investment consultants, reflecting public pension mandates and the importance of investment committees in investment decisions (Andonov et al. (2022)). This allows me to study the relationship between PE returns and underfunding ratios among plans using the same investment consultant. The number of observations in the last column is marginally lower than in the first three columns, as investment consultant information is not populated for all pension-years.

Table 3: Correlation Between Public Pension Underfunding Positions and Average Private Equity Returns for 2001-2021

| | PE Return (%) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.104*** | -0.103*** | -0.102*** | -0.136*** |
| | (0.031) | (0.031) | (0.031) | (0.036) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1988 | 1988 | 1988 | 1786 |
| Adjusted R squared | 0.650 | 0.650 | 0.650 | 0.654 |
| Dependent Variable Mean | 0.115 | 0.115 | 0.115 | 0.119 |
| Dependent Variable Std | 0.153 | 0.153 | 0.153 | 0.148 |

*Notes:* The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021. The PE performance is measured using the net internal rate of return (IRR). Column (1) controls for LP size, Column (2) controls for LP size and includes current PE allocation, Column (3) controls for LP size and includes current alternative asset allocation, and Column (4) controls for LP size, includes the average past three-year PE, equity, fixed income allocations, and investment consultant dummies. Robust standard errors in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

A clear pattern emerges from the estimates in the table. Public pensions with more underfunded

positions consistently exhibit, on average, lower total PE returns than the less underfunded public pensions, even after controlling for public pension characteristics of size, past average asset allocations, and investment consultants reflecting public pension mandates. That is, across their portfolio of PE investments, more severely underfunded pensions perform worse than those with smaller funding shortfalls. Specifically, focusing on column (4), I find that a 10 percentage point increase in the underfunding ratio of the pension plan is associated with a 1.36 percentage point decrease in average annual net IRR. With an average net IRR of 11.9 points, this corresponds to an 11.4% decline. The coefficient on underfunding ratio remains consistently negative and statistically significant across specifications. For instance, in columns (1)-(3), a 10 percentage point increase in the underfunding ratio of the pension plan is associated with a 1.02 to 1.04 percentage point decrease in average annual net IRR, which corresponds to an 8.9% to 9.0% decline. In comparison with the existing related literature which studies a different question, the results are reasonable in economic magnitude. For instance, Andonov et al. (2018) finds that pension plans with a 10 percentage point increase in the proportion of board with state-appointed members exhibit a 0.91 percentage point decrease in annual net IRR.[19]

Next, I analyze the investments in the PE portfolio of public pensions.

### 3.2 Matching of Underfunded Pensions and GPs

Having shown that more underfunded public pensions exhibit lower returns from PE, I next analyze the investments of public pensions' PE portfolio. PE investments are varied, having different strategies, region focus, and target companies, which generate returns for GPs, and ultimately the LPs. The focus is on PE buyout funds, aggregated at the GP level, and ultimately at the pension level, capturing a public pension's portfolio of PE assets. In addition to the average PE performance measure studied in Section 3.1, I use GP size aggregated to the pension level as the main variable of interest, following the literature on mutual funds which shows that managerial skill is reflected in the cross-sectional distribution of fund size and assets under management (Berk and van Binsbergen (2015)). This is indicative of GP quality, more broadly. I also find evidence for smaller GPs having less number of PE funds, which is also suggestive of having less connections with LPs. Small GPs also have fewer teams to manage the fund. All these measures are ultimately correlated with the type of deals financed by these funds. This is a useful measure, especially since majority of the GPs are non-traded.

[19]Reassuringly, the average annual net IRR in the regression sample is 11.9 points, which is similar in magnitude to the average net IRR of 10.4 points reported in Andonov et al. (2018)'s Table 1. The data here on returns is based on PPD and Andonov et al. (2018)'s data is based on Preqin, suggesting the two data sources yield consistent estimates.

I use two measures of GP size: (1) the book value measure, which is the sum of total size of existing PE funds within the GP for each year, and (2) the market based measure, which is the sum of book value of capital committed by investors, additional market value of PE investments, and the capital yet to be called ("dry powder"), covering all private asset classes.[20] I use the book value measure in this section, as this measure allows me to track the GP size within the private equity asset class over time. It is very hard to construct a reliable measure of market values of individual PE funds over time, given the well known difficulty in valuing private assets. Moreoever, it's hard to disentangle the appropriate GP market value stemming from PE versus from other private asset classes, as the reputational advantages of a GP span across asset classes. Nonetheless, I also use the market value GP size measure provided by Preqin in Section 4.2, which is a good robustness check for the book asset measure. Rankings of GPs across measures are generally consistent. To summarize, larger PE book assets represent bigger scale and better quality within the PE industry. This is also visible in a high percentage of deal volume via the top few GPs.

To construct the GP size measure, I use the year of inception and lifespan of the fund to determine the end year for each PE fund. When I do not observe the lifespan, I take the median value of 10 years (motivated by Kaplan and Strömberg (2009)). Next, I use the GP and PE fund linkages from Preqin and cumulate the sizes of constituent PE funds in each year for each GP to arrive at the GP size measure over time. Next, I use the investments made by public pensions (the LPs) in individual PE funds. When the pension invests in the PE fund, it is referred to as a capital commitment by the LP to the fund. Investments are made by the LPs at the beginning of a PE fund's lifespan, which is also referred to as the vintage year of the PE fund.[21] It is important to note that the missing data on capital commitment amounts in Preqin does not affect the analysis here, as I only use the public pension and PE fund linkages, and the vintage year of the fund, which is when the capital is committed by the LP to the PE fund. Lastly, using the public pension and PE fund linkages, and vintage year of the PE fund, I connect the GP size measure to the public pension over time. When a public pension invests in multiple GPs, I take the average size across GPs in which the public pensions invest in for each year, to arrive at a series of GP size at the public pension by year level. This measure is consistent with the PE performance measure at the pension by year level, capturing average returns across PE investments.

[20] This is reported directly by the GP. It is a complicated and opaque measure as it covers market value of non-traded private assets, which is hard to measure. This is only available as of the latest date reported by the GP ranging from 2019 to 2022, depending on the GP. Hence, I also use the book value measure in this section. Construction of the book value measure is described in detail in Appendix B.2.2.

[21] Supported by interviews with industry professionals and Preqin, the data provider.

To study the relation between the underfunding positions of public pensions and PE investments, I manually build a bridge between the public pension investor identifiers in Preqin and the PPD to connect pension plan characteristics with PE investment characteristics. This is tricky as in some cases Preqin records the entire retirement system as one investor, while the PPD records individual plans within the retirement system separately. In such cases, I check the public pension websites to find the individual constituent plans and make the link. Subsequently, for each Preqin public pension identifier and year, I take the average of underfunding ratios and plan characteristics across the constituent pension plans in the PPD, weighting them by the size of the pension plan. Hence, the number of observations in the sample differs from studying the relation between underfunding ratios and average PE returns to studying the relation between underfunding ratios and average GP size. Ultimately, I end up with a public pension by year panel with underfunding ratios, GP size, and other pension and GP time varying and time invariant characteristics.

I carry out the empirical analysis in Table 4. As in Section 3.1, the time period is 2001-2021, when public pensions increased their allocations towards alternative assets (Figure 2), and the observation level is public pension-year. The main independent variable is the reported underfunding ratio by public pensions, and Section 3.4 shows additional robustness tests with the market value of underfunding positions. I estimate the regression specification 1 of GP size in the PE industry on underfunding positions of public pensions in the year of capital commitment. $y_{pst}$ is the average size of the GP (in logarithmic terms) which the public pension $p$ in state $s$ and year $t$ invests in, where the GP size is the sum of the sizes of all its component PE funds existing in that year, as described above.

The control variables increase gradually from models (1) to (4) in Table 4. Vintage year fixed effects are included in all models to account for changes over time. I do not include pension fixed effects as the matching is across public pensions and GPs. PE investments are long-term, slow moving, sticky, and relationship based, with the pension capital locked into the investment for a minimum of 5-7 years according to the limited partner agreements, making it infeasible to show changes in GP allocations over a short horizon.[22] Model (1) controls for the natural logarithm of the LP's AUM as larger pension systems will have a higher bargaining power with the larger GPs.[23] After controlling for LP AUM, I find a negative and statistically significant relation between underfunding ratios of public pensions and GP size. Since the goal is to study the quality of public pensions' investments within PE, model (2)

[22] Appendix Figure B.5 shows the changes over time.

[23] For instance, there are situations where GPs require capital with immediacy, and they rely on the larger LPs to commit extra capital.

conditions for the pension to have a positive allocation to PE in that year. Note that there are very few observations in the which the PE allocation variable is missing in the PPD database but the Preqin data shows a positive GP size connection with the plan in that year. Further, upon manual investigation I find that most of these plans have positive PE allocations in the years before and after the missing allocation. Both these facts confirm that these observations indeed represent a positive PE allocation, and hence are part of the analysis. Model (3) adds controls for average past three-year asset allocations of the pension. Model (4) is the most saturated. To account for concerns of more underfunded pensions matching with different types of PE funds, I control for multiple fund characteristics which are stable over time, such as, industry focus, additional granular strategies, investment region focus, and domicile of the fund. Additionally, I control for public pension characteristics – assets, average past three-year allocations to different asset classes, fund benchmark returns to account for fundamentals other than underfunding ratios of pensions.

Table 4: More Underfunded Pensions Match with Smaller GPs on Average, 2001-2021

| | GP Size (Log) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.430** | -0.608*** | -0.482** | -0.517*** |
| | (0.177) | (0.184) | (0.188) | (0.167) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1504 | 1275 | 1202 | 1028 |
| Adjusted R squared | 0.299 | 0.304 | 0.292 | 0.486 |
| Dependent Variable Mean | 8.875 | 8.902 | 8.963 | 8.999 |
| Dependent Variable Std | 1.216 | 1.207 | 1.137 | 1.118 |

*Notes:* The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Column (1) controls for LP AUM, Column (2) controls for LP AUM and condition of having a positive allocation to PE, Column (3) also add LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Column (2), and Column (4) adds GP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Column (3). For Columns (2)-(4), note that there are very few observations in the which the PE allocation variable is missing in the PPD database but the Preqin data shows a positive GP size connection with the plan in that year. Further, upon manual investigation I find that most of these plans have positive PE allocation in the years before and after the "missing". Both these facts confirm that these observations indeed represent a positive PE allocation and hence are part of the analysis. Robust standard errors in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

A clear pattern is seen from the estimates across the four models. There is a negative and statistically significant relation between public pensions' underfunding positions and PE investment size, as measured by average GP size within pensions' PE allocations. That is, more severely underfunded pensions allocate capital to, on average, smaller GPs. Column (4) shows the estimates for the most saturated specification with pension and PE fund controls. Coefficient estimate for underfunding ratio is $-0.52$ $(t = -3.09)$, statistically significant at 1% level. The estimate is also economically significant. For a one-standard-deviation increase in underfunding ratio (19.22%), logarithmic size decreases by 0.10 log points. In levels, the average GP size is \$8,022 mn., and a 0.10 log point change corresponds to a $-9.5\%$ change.[24]

Overall, all four sets of models on PE returns estimation in Table 3 and on GP size estimation in Table 4 together suggest that more underfunded public pensions are allocating capital to smaller GPs on average, and receive lower returns from their portfolio of PE investments, relative to the least underfunded public pensions. These relationships exist after accounting for public pension and GP characteristics. Lower total PE returns earned by the more underfunded pensions provides suggestive evidence in support of the narrative of smaller funds getting worse deals, or being of lower quality, interpreted more broadly. This finding is not universal to all PE funds, but instead specific to PE investments by public pensions. Further, this result is consistent with the evidence in Andonov et al. (2018), which finds that the larger funds invested in by public pensions exhibit a higher net IRR. The positive association between average GP size and PE returns also does not invalidate the literature finding a negative relation between a PE fund size and its returns, as the returns exhibited by public pensions are an average across a portfolio of PE investments. It is also possible that the investments by pensions in small GPs, or small PE funds within a GP, are performing well, but not enough to overturn an under performing investment in a large GP. It is also worth noting that this is an inference about the cross-section of funds, and not an inference about the performance within a fund over time.

### 3.3 Why does the matching exist?

The results encompass a couple of possible explanations. First, the existing large funds have established relationships with investors, while the small funds are new. I observe this as the biggest funds

[24] Average logarithmic size is 8.99 ($\equiv$ \$8,022 mn). Table 1 shows the standard deviation of public pension underfunding ratios is 19.22%. With a coefficient of $-0.52$, change in log points is $-0.52 \times 19.22\% = 0.10$ log points change in the dependent variable. The average dependent variable in log terms along with the effect of underfunded positions is $8.99 - 0.10 = 8.89$ ($\equiv$ \$7,259 mn.). In level terms, the change in average size of the GP is -\$763 mn., which is a $-9.5\%$ change.

have consistent relationships with their limited partners over time, contributing to the surge of continuation funds in private equity (Kastiel and Nili (2024)). Second, larger and reputable GPs get the best deals, while smaller funds get the remaining, possibly worse deals (Demiroglu and James (2010)). This is plausible with larger GPs occupying a concentrated market share (With Intelligence PE Outlook (2025), McKinsey & Company PE Report (2026)), for instance, the top 30 GPs have greater than 50% market share. A third possibility is that in the sample of PE funds which public pensions invest in, small funds have smaller teams to manage target firms, which may factor in their inability to get high returns from investments. Small funds may require appropriate management attention to manage more deals (Lopez-de Silanes et al. (2015)). Lastly, small funds might be showing high return expectations but are not able to meet them. The most underfunded pensions prefer higher expected returns to cover up their extreme funding shortfall positions. I do not observe return expectations at the fund level as these would be included in confidential limited partnership agreements between investors and funds.

### 3.4 Market Value of Public Pension Liabilities, Varying Fiscal Year End Dates, and Varying Actuarial Cost Methods

In this section, I conduct a number of additional robustness checks. In the tables here, I use the most saturated specifications, which control for public pension and GP characteristics. In Appendix A, I describe the methodologies in detail, and in Appendix Tables A.4-A.9, I show results with other regression specifications.

First, according to the Governmental Accounting Standards Board (GASB) rulings, the U.S. public pensions use the average assumed return on their investments to discount their unfunded liabilities. According to the logic of financial economics, financial streams of payments should be discounted at a rate that reflects their risk (Modigliani and Miller (1958)). This creates a major distortion between the reported and the actual unfunded liabilities. The reported underfunding positions might obscure their true need for risk taking. Since the ranking of public pensions based on their underfunded positions is likely consistent irrespective of the discount rate used, the accounting measure or the market value measure should not likely drive the estimates.[25] However, as robustness, I calculate the market value of underfunding positions of pensions and perform the analyses.

Novy-Marx and Rauh (2011) takes a substantial step in correcting the bias by estimating the future

[25]Refer to Figure A.1 of Giesecke and Rauh (2022), which shows the rankings of public pensions based on the reported underfunding positions and market value of underfunding positions calculated using the U.S. Treasury Yield Curve, are consistent.

stream of payments for each pension plan based on plan characteristics, and discounting them to the present with an appropriate discount rate, common across pension plans.[26] The Bureau of Economic Analysis (BEA) adapts this methodology to publish the market value of unfunded liabilities and funding ratios (Lenze (2013), Lenze (2020)), which is subsequently used by the Federal Reserve Board Financial Accounts. To construct the market value of funding ratios, I follow step-by-step the approach used by the BEA.[27] I collect data on total pension liabilities at plan discount rate plus/minus one percentage point from the Comprehensive Annual Financial Reports (CAFRs), which is used to construct duration estimates, and subsequently calculate the market value of total pension liabilities using BEA discount rates. I divide the pension reported assets by the market value of total pension liabilities to calculate the market value of funding ratios. Appendices A.1 and A.2 provide the formulae and describe the data collection from the CAFRs. Lenze (2020) provides state-level funding ratios only for 2018 and duration estimates only for 2017. Aggregating pension level BEA-Adjusted funding ratio estimates constructed in this paper to the state level, and comparing them to Lenze (2020)'s 2018 estimates yields a correlation of 94.2%. The analogous correlation for 2017 duration estimates is 88.0%. These high correlations support the validity of the pension-level estimates constructed in this paper, with the small remaining differences likely attributable to differences in plan coverage. Appendix Tables A.2 and A.3 provide a state by state comparison.[28]

Table 5 column (1) shows estimates from model (4) of Table 3 using BEA-Adjusted underfunding ratios calculated at the pension plan-year level, based on the data collected from the pensions' CAFRs. As a sanity check, column (2) uses BEA-Adjusted underfunding ratios, constructed using state-level duration estimates from Lenze (2020), where pension plans are attributed their state's duration estimate. A 10 percentage point increase in the underfunding ratio of public pensions is associated with a 1.50 percentage point (12.6%) to 1.57 percentage point (13.2%) decrease in average annual net IRR, which is

[26]The PPD or the pension plan Comprehensive Annual Financial Reports (CAFRs) do not provide individual stream of pension benefit payments. Novy-Marx and Rauh (2011) model the prospective stream of payments from state pension promises using each state's stated liability, discount rate, actuarial cost method, benefit formulas, the number and average wage of state employees by age and service, salary growth assumptions by age, mortality assumptions, cost of living adjustments (COLAs), and separation (job leaving) probabilities by age. After constructing the cash flow streams for each plan, they use the discount rate stated by the plan and reconstruct the plan's reported liabilities. Next, they recalculate the liabilities for each plan-year using the approprate discount rate. I use the discount rate used by the Bureau of Economic Analysis (BEA).

[27]For 2000-2003, the BEA has used a 6% discount rate, from 2004-2009 a 5.5% discount rate, from 2010-2012 a 5% discount rate, and from 2013-2018 a 4% discount rate. There has not been an update on the discount rate for 2019 onwards. I assume the continuation of the 4% discount rate till the end of the sample period (refer to Appendix Table A.1). This also makes studying the time period until 2021 and not until more recent, appropriate, as liability recording, estimation, and adjustment runs with a lag.

[28]The BEA only provides funding ratios at the state level for 2018, and duration estimates at the state level for 2017. Refer to Lenze (2020) Tables 4 and 14, respectively.

Table 5: Correlation Between Public Pensions Underfunding Positions and Average Private Equity Returns for 2001-2021 (robustness)

| | BEA-Adjusted underfunding ratio using CAFR collected data (1) | BEA-Adjusted underfunding ratio using Lenze (2020) duration (2) | All plans (3) | June or December FY end (4) | June FY end (5) | EAN in $\geq$ 5 years (6) |
|---|---|---|---|---|---|---|
| BEA-Adjusted Underfunding Ratio | -0.150*** | | | | | |
| | (0.047) | | | | | |
| BEA-Adjusted Underfunding Ratio | | -0.157*** | | | | |
| | | (0.047) | | | | |
| Underfunding Ratio | | | -0.090** | -0.096** | -0.103** | -0.185*** |
| | | | (0.036) | (0.038) | (0.041) | (0.067) |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocation Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Investment Consultant Dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Public Pension FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | | | Yes | Yes |
| Pension FY End Month-Year FE | | | Yes | Yes | | |
| Observations | 1771 | 1771 | 1782 | 1684 | 1323 | 1563 |
| Adjusted R squared | 0.652 | 0.653 | 0.706 | 0.707 | 0.737 | 0.633 |
| Dependent Variable Mean | 0.119 | 0.119 | 0.118 | 0.119 | 0.122 | 0.119 |
| Dependent Variable Std | 0.148 | 0.148 | 0.148 | 0.148 | 0.154 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021. This table shows robustness for Table 3. The PE performance is measured using the net internal rate of return (IRR) (%). All columns control for LP size, the average past three-year PE, equity, fixed income allocations, investment consultant dummies, and public pension fixed effects. Columns (1)-(2) and (5)-(6) include year fixed effects. Columns (3)-(4) include fiscal year end month by year fixed effects. Column (1) uses the BEA-Adjusted underfunding ratio constructed using the data collected from Comprehensive Annual Financial Reports (CAFRs). Column (2) uses the BEA-Adjusted underfunding ratio constructed using state-level liability-weighted average duration estimates from Lenze (2020). Column (3)-(6) use the reported underfunding ratios. Column (3) considers all plans, but instead of year fixed effects, it includes fiscal year end month by year fixed effects. Column (4) considers plans ending in June or December, but instead of year fixed effects, it includes fiscal year end month by year fixed effects. Column (5) considers plans ending in June and has year fixed effects. Column (6) considers plans using the Entry Age Normal (EAN) actuarial cost method in at least 5 years.

Table 6: More Underfunded Pensions Match with Smaller GPs on Average, 2001-2021 (robustness)

| | BEA-Adjusted underfunding ratio using CAFR collected data | BEA-Adjusted underfunding ratio using Lenze (2020) duration | All plans | June or December FY end | June FY end | EAN in $\geq$ 5 years |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| BEA-Adjusted Underfunding Ratio | -0.619** | | | | | |
| | (0.244) | | | | | |
| BEA-Adjusted Underfunding Ratio | | -0.661*** | | | | |
| | | (0.231) | | | | |
| Underfunding Ratio | | | -0.590*** | -0.592*** | -0.614*** | -0.566** |
| | | | (0.177) | (0.176) | (0.192) | (0.232) |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocations | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Strategy | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Industry Focus | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Region Focus | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Benchmark Returns | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Domicile | Yes | Yes | Yes | Yes | Yes | Yes |
| Vintage Year FE | Yes | Yes | | | Yes | Yes |
| Pension FY End Month-Year FE | | | Yes | Yes | | |
| Positive PE Allocation | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 1027 | 1027 | 1002 | 994 | 837 | 904 |
| Adjusted R squared | 0.485 | 0.486 | 0.480 | 0.470 | 0.457 | 0.480 |
| Dependent Variable Mean | 8.998 | 8.998 | 8.993 | 8.997 | 9.033 | 9.002 |
| Dependent Variable Std | 1.118 | 1.118 | 1.123 | 1.112 | 1.114 | 1.125 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021. This table shows robustness for Table 4. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. All columns control for LP AUM, LP's past allocations (average past three-year PE, equity, fixed income allocations), GP and pension characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile), and condition on having a positive allocation to PE. There are very few observations in the which the PE allocation variable is missing in the PPD database but the Preqin data shows a positive GP size connection with the plan in that year. Upon manual investigation, I find that most of these plans have positive PE allocations in the years before and after the "missing". Both these facts confirm that these observations indeed represent a positive PE allocation and hence are part of the analysis. Columns (1)-(2) and (5)-(6) include vintage year fixed effects. Columns (3)-(4) include pension fiscal year end month by year fixed effects. Column (1) uses the BEA-Adjusted underfunding ratio constructed using the data collected from Comprehensive Annual Financial Reports (CAFRs). Column (2) uses the BEA-Adjusted underfunding ratio constructed using state-level liability-weighted average duration estimates from Lenze (2020). Column (3)-(6) use the reported underfunding ratios. Column (3) considers all plans, but instead of year fixed effects, it includes pension fiscal year end month by year fixed effects. Column (4) considers plans ending in June or December, but instead of year fixed effects, it includes pension fiscal year end month by year fixed effects. Column (5) considers plans ending in June and has year fixed effects. Column (6) considers plans using the Entry Age Normal (EAN) actuarial cost method in at least 5 years.

larger than the 1.36 percentage point (11.4%) decrease observed using reported underfunding ratios in Table 3. Similarly, Table 6 columns (1) and (2) show coefficient estimates of model (4) Table 4 using BEA-Adjusted underfunding ratios. The coefficient estimates depict a 11.3% to 12.2% decrease in average GP size for a one standard deviation (19.22%) increase in underfunding positions of pensions. Appendix Tables A.4 and A.7 show consistent estimates for other model specifications. The strengthening of the magnitudes relative to the reported underfunding ratios (in Tables 3 and 4) highlight that the reported unfunded liabilities understate the true unfunded positions and incentives of public pensions, and its subsequent impact on average returns.

A second concern is that pension plans have different fiscal year end dates.[29] While the BEA-Adjusted underfunding methodology already adjusts for varying fiscal year ends, to be further conservative, I do three more tests. First, in column (3) of Tables 5 and 6, instead of year fixed effects, I include month by year fixed effects for all fiscal year end month-years.[30] Second, to ensure that the fiscal year end month-years with few observations are not driving the estimates, in column (4) of Tables 5 and 6, I only consider plans with fiscal year end months of June and December, which together comprise 91% of observations. I include month by year fixed effects for varying fiscal end periods. Third, to be even more restrictive, in column (5) of Tables 5 and 6, I limit attention to plans with only June ending periods, and include year fixed effects. A 10 percentage point increase in the underfunding ratio of public pensions is associated with a 0.9 percentage point to 1.03 percentage point lower average net IRR. Similarly, a one standard deviation (19.22%) increase in underfunding ratio of pensions, corresponds to a 10.4% to 11.3% decrease in average GP size. The coefficient estimates are economically large, consistent, and not statistically different from the estimates in main Tables 3 and 4.[31]

A third concern is that public pensions employ various actuarial cost methods for liability accounting. The most popular one is the Entry Age Normal (EAN) method. Other methods used are: Projected Unit Credit (PUC), Frozen Initial Liability (FIL), Aggregate Cost Method (AGG). Sometimes plans may also use a variation of these methods. The BEA-Adjusted underfunding methodology already accounts for varying actuarial cost methods across plans, however, to be further conservative, I do another test. In column (6), I limit attention to plans which have used the EAN method for at least five years, which is approximately 90% of the plans. Once again, I observe patterns similar to those in Tables 3 and 4.

[29] For instance, in the PE return sample, 66.4% of the observations have fiscal year end date of June 30, 24.8% of observations have fiscal year end date of December 30/31, 5.8% of observations have a fiscal year end date of September 30 (one observation being September 20), and 1.5% and 1.6% of observations have fiscal year end dates of April 30 and August 31 respectively.

[30] Special thanks to Aleks Andonov for this suggestion.

[31] Appendix Tables A.5 and A.8 report results with other model specifications for average PE returns and GP size respectively.

The coefficient on underfunding ratio for both, average PE return and GP size estimation is negative, statistically, and economically significant. Appendix Tables A.6 and A.9 observe consistent estimates for all model specifications.

Overall, I find that estimates exhibiting how average PE returns and the average size of PE investments within the PE portfolio of public pensions respond to their underfunded positions become stronger in economic magnitude, when I use the true market value instead of the accounting based measures of underfunding. This implies that the accounting underfunding measure under represents public pensions' incentives within PE asset allocation. The estimates are robust to varying fiscal year end dates and actuarial cost methods used by public pensions.

# 4 Evaluating Real Outcomes at PE Target Firms

In this section, I examine real outcomes at the PE target firms obtaining funding from public pensions. Public pensions commit capital to PE funds, and receive returns after a period of 5-7 years of their investments. PE funds invest the capital in target firms to generate returns and pay out the LPs. As Section 3.1 shows that the more underfunded public pensions exhibit, on average, a lower PE performance than the less underfunded ones, it is helpful to examine outcomes at the underlying target firms when they receive PE capital.

Majority of the firms are privately held, and studying private firms encounters several limitations. First, basic balance sheet and income statement statistics are scarcely available. Second, studies have relied on surveys which represent an incomplete and a potentially selective image. Third, firms do not have a reliable market valuation, which is present in the public firm space. Fourth, and importantly, most of the firms undergo structural changes during private equity transactions, which often times, involves a name, ownership, or identifier change, making it hard to study a firm over time. To deal with these central concerns, I turn to the Longitudinal Business Database (LBD) of the U.S. Census Bureau, which has the most comprehensive information on the universe of private firms in the U.S. along with time consistent firm identifiers, allowing me to study the firm over time. I study the evolution of target firms pre- and post-buyout transactions, building on the earlier literature (Davis et al. (2014), Davis et al. (2019)).

To execute this, I supplement the data on PE target firms financed by PE funds with the public pension (LP) and the PE fund linkages examined in Section 3. Preqin provides information on PE fund-

target firm linkages, along with the target firm's identifying characteristics, such as, the name, address, zipcode, and the date of the deal. Now, I have the connection between the LP and the PE fund based on the vintage year of the fund, and the connection between the PE fund and the target firm based on the deal date. I use the Standard Statistical Establishment List (SSEL), also known as the Business Register (BR), in the U.S. Census micro-data to match the PE target firms from Preqin via firm name, address, state, based on the year before the buyout deal.[32] I use the linking table between the SSEL and the LBD, to track employment, pay, revenue, and labor productivity measures of target firms before and after the buyout deal. The matched target firms represent close to 8% of the total U.S. non-farm payroll employment in 2018 (Figure F.1). Appendix F has detailed description on the matching methodology and sample construction.[33] Naturally, since these are target firms from the Preqin database, the firms have substantial presence of public pension LP capital, which is the focus of this study.

As a first step, I estimate the following regression specification across all the matched PE target firms in the sample. The model estimates the response of growth rates of real outcomes at the firm level over time after the buyout transaction, following Davis et al. (2014). This model serves as a benchmark to study variations by LP and GP characteristics in the follow on sections, which is the main focus of the study.

$$y_{it} = \alpha_t + \alpha_0 \text{Post Buyout}_{it} + \gamma \text{LFIRM}_i + \text{Fixed Effects} + \epsilon_{it} \tag{2}$$

$y_{it}$ is the outcome variable in growth rates for firm $i$ at time $t$. These are one-year "DHS" growth rates (Davis et al. (1996)) from $t-1$ to $t$ for firm $i$, where $y_{it} = \left(Y_{it} - Y_{it-1}\right)/\left(X_{it}\right)$ and $X_{it} = 0.5 \times \left(Y_{it-1} + Y_{it}\right)$.[34] Post Buyout is a dummy which takes the value 1 for the year corresponding to the buyout and after. $\alpha_0$ is the coefficient of interest measuring the responses of outcome variables following the buyout activity. Following earlier literature, I include year fixed effects to account for time varying changes, granular 22 two-digit NAICS industry categories to account for sectoral changes, 12 firm size and 5 firm

[32] Using the year before the buyout deal avoids mismatches in the merge between PE target firms from Preqin and firms in the U.S. Census Bureau micro data, due to potential firm name changes during the deal. This methodology is followed in the existing literature.

[33] Particularly, Appendix F describes the merging of LP, PE fund, and firm information (Section F.1), merging with the SSEL (Section F.2) and the Revenue Enhanced LBD (Section F.3), which is the updated version of the original LBD and includes the revenue measures at the firm level. Figure F.1 shows the employment and revenue of the matched firms as a percentage of the total U.S. non-farm payroll employment over time, and Figure F.2 shows the same in the year of the buyout. Figure F.3 shows the match rates over time, and Figure F.4 shows the match rates by industry and state.

[34] I calculate the DHS growth rates of revenue and total payroll in 2020 U.S. dollars. Revenue is deflated by the U.S. GDP Price Deflator Series, link: https://fred.stlouisfed.org/series/USAGDPDEFQISMEI. Pay is deflated by the Consumer Price Index for All Urban Consumers (CPI-U).

age buckets, and type of firm (multi or single unit) fixed effects to control for potential differences across entities. $\text{LFIRM}_i$ represents lagged firm growth rates from $t-3$ to $t-1$ relative to buyout, to control for firm specific pre-deal organic growth. Regressions are weighted by employment in the year of buyout.

Table 7 shows estimated coefficients $\alpha_0$ for Equation 2. Employment changes by $-8.4\%$ per year, revenue by $-8.6\%$, with an insignificant $-0.2\%$ change in labor productivity.[35] The coefficient estimates are larger, but in reasonable magnitude, when compared to the existing literature. Davis et al. (2019) studies firms across industries, and highlights substantial heterogeneity in employment changes ranging from -12.6% to +12.8% based on the type of deal. The differences are not surprising as the sample of firms is much smaller in size, a substantial proportion of firms are single-unit establishments, which tend to exhibit large changes, at times driven by firm closures. The sample is focused on public pension LP capital financing, not necessarily representative of an average PE transaction in the U.S. Moreover, comparison with earlier literature should be with caution as the estimates in this section are not based on a difference in differences setting (as in Davis et al. (2014), Davis et al. (2019)), but rather based on an event study setting, and study public pension financed target firms. This event study specification serves as a point of comparison for studying the target firms based on the LP and GP heterogeneity in the next sections.

Table 7: Event Study Estimated Coefficients of Post Buyout, Private Equity Deals 1997-2018

| Dependent Variable: | Emp $g$ (1) | Pay $g$ (2) | Rev $g$ (3) | Rev $g$ -Emp $g$ (4) |
|---|---|---|---|---|
| Post Buyout | $-0.084^{***}$ | $-0.075^{***}$ | $-0.086^{***}$ | $-0.002$ |
| | (0.016) | (0.016) | (0.011) | (0.013) |
| Year FE | Y | Y | Y | Y |
| Firm Size FE | Y | Y | Y | Y |
| Firm Age FE | Y | Y | Y | Y |
| Industry FE | Y | Y | Y | Y |
| Type of Unit FE | Y | Y | Y | Y |
| Lagged Firm $g$ | Y | Y | Y | Y |
| Weighted Emp $t_0$ | Y | Y | Y | Y |
| Observations | 70,000 | 70,000 | 70,000 | 70,000 |
| Adjusted $R^2$ | 0.183 | 0.193 | 0.132 | 0.015 |
| Dependent Variable Mean | 0.023 | 0.028 | 0.026 | 0.003 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The table displays coefficients $\alpha_0$ of the event study specification 2: $y_{it} = \alpha_t + \alpha_0 \text{Post Buyout}_{it} + \gamma \text{LFIRM}_i + \text{Fixed Effects} + \epsilon_{it}$. $y_{i,t}$ are DHS growth rates. Post Buyout is a dummy which equals to 1 for the year corresponding to the buyout and after. All columns include year fixed effects, granular 22 two-digit NAICS industry categories, 12 firm size and 5 firm age buckets, type of firm (multi or single unit) fixed effects, and $\text{LFIRM}_i$ representing lagged firm growth rates. Column (1) shows employment growth rates, Column (2) and (3) show pay and revenue growth rates respectively (in 2020 U.S. dollars). Column (4) shows labor productivity estimates. Standard errors are in parentheses, and clustered at the firm level. Regressions are weighted by employment in the year of buyout.

[35] Additional robustness checks (not reported) include specifications with year and industry fixed effects; year and firm size fixed effects; year and firm age fixed effects; year and type of unit fixed effects; year, industry and firm size fixed effects; industry $\times$ year, firm size, age, and type of unit fixed effects. Results are robust to these alternate specifications.

A few caveats are worth noting when interpreting these estimates. First, labor productivity is not the ideal measure as it does not consider costs incurred by the firm. However, the U.S. Census Bureau does not provide cost measures *across* all industries. The Census of Manufactures (CMF) and the Annual Survey of Manufactures (ASM) provide total factor productivity (TFP) for only manufacturing firms — an unrepresentative and small subset of PE transactions — making it infeasible to study TFP for the target firm splits based on LP and GP characteristics in later sections. Especially in the 2000s, the growth of PE has dominated industries, such as, technology, healthcare, and services. Hence, recent literature has mainly studied labor productivity measures. Despite these concerns, in Appendix D.5, I show no substantial increases in TFP and labor productivity following a PE deal, for the small sample of manufacturing firms (Figures D.4, D.5). Second, these estimates suggest the role of a PE buyout transaction in employment, revenue, and productivity changes, after controlling for all observable characteristics. Because buyout targets are not randomly selected, I cannot rule out that these relationships partly reflect selection on unobservables.

### 4.1 PE Target Firms by Public Pension Underfunding Positions

Figure 2 shows that allocation to alternatives by public pensions has increased from 8.4% in 2001 to 27.6% in 2021, and PE has acquired a significant share within alternative assets. Further, within public pensions, the share of capital committed by the most underfunded public pension tercile, defined in the year prior to commitment, has increased over time (Figure E.2). Out of the total capital committed by public pensions to PE, the most underfunded tercile contributed 9.6% in 2001 and 29.0% in 2018.[36] These facts motivate the study of target firms receiving the underfunded public pension LP capital.

U.S. public pensions are one of the largest groups amongst the LPs in Preqin. This is not surprising, as Preqin is representative of U.S. public pension LP capital, given that most of its data is collected from FOIA requests and public reports. Begenau et al. (2020) confirms this by comparing Preqin data against self-filed FOIA requests, and recent academic literature (Andonov et al. (2018), Begenau and Siriwardane (2020), Korteweg and Nagel (2022)) has used Preqin data to study U.S. public pensions. On average, 20 LPs are involved in financing a deal though commingled PE funds. Targets are bought by commingled PE funds, where capital from multiple investors is pooled together. To categorize firms financed by the most underfunded public pension LP capital, I calculate underfunded ratios $\phi_i$ at the PE target firm level $i$, weighted by the capital committed $w_{pji}$ by the individual public pension $p$ to firm $i$

[36] This has been accompanied by subsequent new investments in private equity (Figure B.6).

via PE fund $j$ within a GP,

$$\phi_i = \frac{\sum_{pji} w_{pji} \cdot \text{Underfunded Ratio}_{p,p \in ji}}{\sum_{pji} w_{pji}} \tag{3}$$

This represents the public pension presence in the deal based on the degree of underfunding of the pension. If the capital commitment amount is missing, I weight by the size of the pension, as it is reasonable to expect that a larger pension is likely to commit more capital. If a firm does not receive any U.S. public pension capital, they will not appear. This makes the sample consistent in studying firms receiving U.S. public pension capital. I split $\phi_i$ into terciles to estimate the following specification,

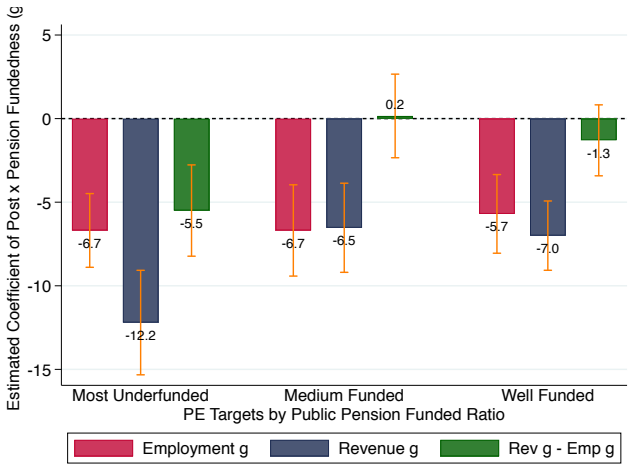
$$y_{it} = \alpha_t + \alpha_0 \text{Post}_{it} + \sum_{r=1,2} \beta^r \left( \text{Post}_{it} \times \mathbb{I}_i^{UF^r} \right) + \gamma \text{LFIRM}_i + \text{Fixed Effects} + \epsilon_{it} \tag{4}$$

This is similar to specification 2 with an additional interaction term of $\text{Post}_{it}$ with $\mathbb{I}_i^{UF^r}$, where $\mathbb{I}_i^{UF^r}$ is a dummy which takes the value 1 for target firms financed by public pensions in underfunded tercile $r$. $r = 1$ represents the most underfunded tercile, and $r = 2$ represents the medium underfunded tercile. $\text{Post}_{it}$ captures the least underfunded tercile category. I use the fully saturated specification as in equation 2, controlling for industry, size, age, and type of the firm in addition to year fixed effects, to compare very similar target firms but differing by the type of investor in the deal. I also control for pre-buyout growth to account for pre-buyout firm trends.

Figure 3 shows the post buyout coefficient estimates from regression 4 for employment, revenue, and labor productivity yearly growth rates for firms split based on the underfunding ratios of public pensions financing the firms. These are total coefficient estimates. Deals financed by the most underfunded public pension tercile experience a $-5.5\%$ significant change in labor productivity per year, whereas, those financed by the least underfunded public pension tercile face a $-1.3\%$ insignificant yearly productivity change. There are many exits in short time periods, and substantial firms are single establishment units, which results in large losses on firm closure. Table 8 shows that the incremental differences for the most underfunded public pension category relative to the least underfunded category are also large and significant. In relative terms, firms financed by the most underfunded tercile of pensions experience a $-4.2\%$ significant productivity change relative to the least underfunded tercile category. These results suggest that there is substantial heterogeneity in target firms by public pension underfunding. It is important to note that this result should *not* be interpreted as public pensions causing target firms to perform poorly as they do not have direct contact with firms. The target firms evolve differently based

on the type of investor, given the GPs that underfunded pensions match with.

Figure 3: Estimates of Post Buyout × Tercile Split Based on Underfunding Ratios of Public Pensions, Private Equity Deals 1997-2018 (Total Estimates)




*Notes:* The above figure plots estimated (total) coefficients for employment (red), revenue (blue), and labor productivity (green) yearly DHS growth rates of Post × Pension Underfunding Split in Equation 4 for each tercile. Firms are split based on the weighted average of underfunded positions of public pensions financing the firms. Weights are capital commitments by each pension in the firm via PE funds. All specifications control for industry, size, age, and type of the firm in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level to account for potential heterogeneity. Orange lines represent 90% confidence intervals.

Table 8: Estimated Coefficients for Post Buyout × Tercile Split Based on Underfunding Ratios of Public Pensions, Private Equity Deals 1997-2018 (Differential Estimates Relative to Base: Least Underfunded)

| Dependent Variable: | Emp $g$ (1) | Rev $g$ (2) | Rev $g$-Emp $g$ (3) |
|---|---|---|---|
| Post Buyout (Base: Least Underfunded) | $-0.0570^{***}$ | $-0.0700^{***}$ | $-0.0130$ |
| | (0.0143) | (0.0126) | (0.0129) |
| Post Buyout × Most Underfunded | $-0.0099$ | $-0.0519^{***}$ | $-0.0420^{**}$ |
| | (0.0140) | (0.0200) | (0.0188) |
| Post Buyout × Medium Underfunded | $-0.0099$ | 0.0047 | 0.0146 |
| | (0.0159) | (0.0161) | (0.0156) |
| Observations | 44,500 | 44,500 | 44,500 |
| Adjusted $R^2$ | 0.2020 | 0.1460 | 0.0197 |
| Dependent Variable Mean | 0.0237 | 0.0223 | $-0.0014$ |
| Year FE | Y | Y | Y |
| Firm Size FE | Y | Y | Y |
| Firm Age FE | Y | Y | Y |
| Industry FE | Y | Y | Y |
| Type of Unit FE | Y | Y | Y |
| Lagged Firm $g$ | Y | Y | Y |
| Weighted Emp $t_0$ | Y | Y | Y |

*Notes:* The table displays coefficients $\alpha_0$ and $\beta^r$ from Equation 4. The regression consists of three categories: most underfunded, medium underfunded, and least underfunded public pension financed firms. Firms are split based on the weighted average of underfunded positions of public pensions financing the firms. Weights are capital commitments by each pension in the firm via PE funds. All specifications control for industry, size, age, and type of the firm in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are in parentheses, clustered at the firm level to account for potential heterogeneity. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

This evidence is robust to alternative splits of the data. For example, quartile splits based on pen-

sions' underfunding positions show that firms financed by the most underfunded quartile of public pensions experience a $-5.3\%$ change in labor productivity per year (Appendix Figure B.2). Appendix Table B.1 also shows that the firms financed by the most underfunded quartile category experience statistically significant negative changes relative to the least underfunded quartile category. Moreover, to account for macroeconomic conditions, I residualize underfunding ratios with local region fixed effects and 10 year interest rates, and find similar results (Appendix Figure B.1).

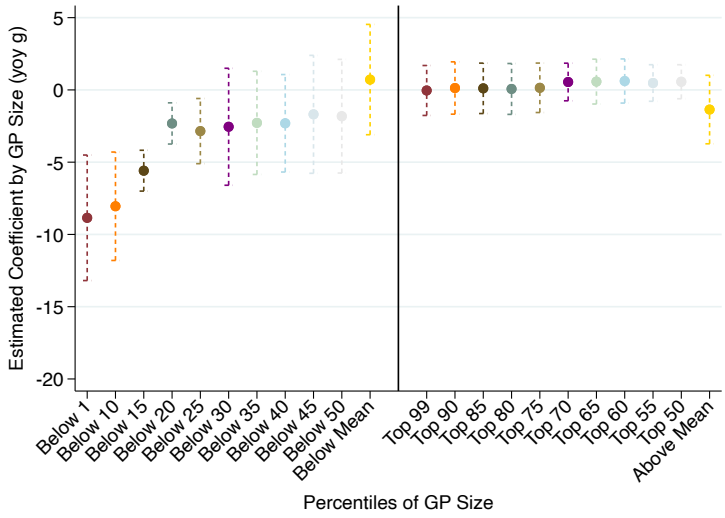
### 4.2 PE Target Firms by GP Size

LPs provide capital to PE funds, which ultimately invest in target firms. PE funds are active managers directly engaging in operations of target firms, while LPs are passive investors and do not manage target firms directly. Having documented that the most underfunded public pensions match with the smallest GPs on average (Section 3.2), I turn next to the ultimate allocation and performance of this capital. To substantiate this further, I examine differences in target firms based on GP size. I aggregate the average GP size at the target firm level. I weight GP characteristics by the number of funds within a GP involved in a deal. Rankings across GPs are persistent over time and across the two size measures.

I estimate Equation 4 with varying splits of target firms based on the GP size distribution. Instead of $\text{Post}_{it}$ interacted with $\mathbb{I}_i^{UF^r}$, I now estimate coefficients for $\text{Post}_{it}$ interacted with $\mathbb{I}_i^{GP^x}$. $\mathbb{I}_i^{GP^x}$ is an indicator variable which takes the value 1 for firms financed by GPs of size less than or equal to $x$th percentile. Studying multiple size cutoffs establishes that the estimates are not purely driven by one cut of the data. Similar to above, I am comparing outcomes at target firms post buyout within granular 22 two-digit NAICS industry codes, 5 firm age and 12 firm size buckets, and same type of firm – multi or single establishment unit, and the year of buyout, but differing by the average GP size financing the deal. Inclusion of granular characteristics of target firms allows me to get close to comparing similar firms undergoing a private equity transaction.

Figure 4 plots the estimated coefficients. Each regression has two estimates (represented in the same color), one for less than or equal to $x$th percentile, and the other for greater than $x$th percentile. The figure documents that the target firms financed by the smallest GPs on average (extreme left of the distribution) experience the greatest decreases in labor productivity. For instance, firms financed by the bottom 20th percentile experience a $-2.3\%$ year over year labor productivity change, those financed by the bottom 15th percentile experience a $-5.6\%$ yearly change, and those financed by the bottom 10th percentile undergo a $-8.1\%$ yearly change. This trend holds across different size percentile distribu-

Figure 4: Labor Productivity $g$ Estimates of Post Buyout $\times$ GP Size Percentile, PE Deals 1997-2018




*Notes:* The figure shows labor productivity yoy growth rate estimates from Equation 4, where $(\text{Post}_{it} \times \mathbb{I}_i^{LIF^r})$ is substituted with $(\text{Post}_{it} \times \mathbb{I}_i^{GP^x})$. $\mathbb{I}_i^{GP^x}$ is an indicator variable which takes the value 1 for target firms financed by GPs of size less than or equal to $x$th percentile. I estimate the regression for different percentiles: $x = 1, 10, 15, 20, 25, 30, 35, 40, 45, 50$, mean, where each color shows estimates from a different percentile cut. I use the market-based size measure. The ranking of GP in terms of size remains consistent over time. The figure considers deals from 1997 to 2018. These are total coefficient estimates for each category. All specifications control for industry, size, age, and type of the firm in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level. Bars show 90% confidence intervals.

tions.[37] Appendix Figure B.3 shows labor and revenue separately. A large proportion of firms are single establishment firms, which close down in short periods driving down revenue. Appendix Figure B.7 studies firms by granular strategy types. Majority of the funds are balanced buyout funds, hence strategy doesn't appear to be driving the difference.[38] Appendix Figure B.4 provides further detail, showing that the cross-sectional differences in labor productivity estimates for firms based on average GP size are stronger in the later half of the 2000s. This is supported by the fact that, on average, the smaller GPs observe a greater increase in matches with the more underfunded public pensions in the second half of the 2000s, than the larger GPs (Appendix Figure B.5).[39]

[37] These are total coefficient estimates for each category. The incremental estimates are statistically and significantly different across categories in the same regression, i.e., the coefficient estimate of below 1 percentile is statistically and significantly different than the coefficient estimate of top 99th percentile.

[38] Granular fund strategies are also formally controlled for in the regression estimation of average GP size and public pension underfunding positions in Section 3.2.

[39] Appendix Figure B.5 shows the percentage of investment linkages, or capital commitments, for the most underfunded public pension tercile (as a share of all public pension commitments) to GPs, split into deciles based on AUM, over time. Panel A shows the first and second half of 2000s separately, and Panel B depicts the change over time. The patterns are robust to alternative size cutoffs. The underfunding ratio of pension is considered at the time of capital commitment to the PE fund. The smallest GPs had 44.7% links with the most underfunded category (as a percent of all public pension commitments) in 2011-2018, which is 7.7% higher than in 1999-2010. This increase is substantial as PE investments are long-term, sticky, and relationship based. As an example, the large GPs, such as, Blackstone Group, Kohlberg Kravis Roberts & Co. (KKR), and Goldman Sachs Alternatives (AIMS) Group have connections with all types of investors, and maintain consistent matches over time. Small GPs, such as, Wicks Group had $15 mn. capital commitments from Philadelphia Board of Pensions and Retirement in 2005, and combined $65 mn. capital from Philadelphia Board of Pensions, Illinois State Board of Investment and Oklahoma Teachers Retirement System in its 2012 fund. Many of the funds in the later half of 2000s are new funds – either

As PE funds have an active role in determining capital allocation towards deals. It is possible that small funds are not good at managing target firms, because they either don't have the expertise or need more management attention (Lopez-de Silanes et al. (2015)). I account for observable characteristics of target firms to compare very similar firms differing by the average GP size financing the deal. On the other hand, it is also possible that smaller funds, with less developed networks, get worse deals (Hochberg et al. (2007)). With the influx of capital from underfunded pensions, more small PE funds, on average, are being supported at the margin with only limited good projects. I do not observe the information set of PE funds during deal selection. Both, selection and treatment are possible between the PE fund and the target firm. The goal is not to disentangle these two possibilities beyond controlling for observable characteristics.

Taken together, these results on real outcomes further substantiate the results documented in Section 3. First, they corroborate the matching pattern between the most underfunded pensions and on average smaller GPs documented in Section 3.2 – target firms financed by the most underfunded pensions and target firms financed by the smallest GPs, each separately exhibit large declines in labor productivity. Second, although labor productivity is at best an imperfect reflection of firm performance, observing labor productivity losses at the firm level are broadly consistent with these losses being reflected in the lower average returns exhibited by the most underfunded pensions (Section 3.1).

## 5 Evidence for Underfunding Positions of Pensions

In this section, I provide support for the channel of underfunding positions of public pensions, being the driver for pensions realizing lower average returns from PE and matching with smaller GPs on average. In Section 5.1, I use public unionization rates as an instrument for underfunding positions. I also address potential threats to the exogeneity condition, such as union members serving on pension boards. In Section 5.2, I discuss and evaluate alternate mechanisms, such as, political motives, local incentives, and politically diverse and large states with major pension systems.

new GPs or small old GPs which had gone dormant and started raising new funds. While this is mostly a post-financial crisis phenomenon due to gradual increase in underfunding positions of pensions, it takes couple of years for PE funds to raise capital from investors, "call" the committed capital, and deploy into investments (Appendix Figure B.6 shows the rise of new PE AUM from around 2011 onwards).

## 5.1 Instrument for Underfunding Positions

### 5.1.1 Instrument Description

Following the Great Financial Crisis (GFC) in 2008, the funded ratio of public pensions did not recover even though the stock market bounced back. For instance, public pensions were funded at 72.4% in 2020, i.e., for every $100 of liabilities, a public pension only had $72.4 in assets. To help disentangle the channel of underfunding positions of pensions, I aim to rule out unobservable investor characteristics that might be correlated with both, underfunding positions and private equity fund size. One possible confounder is investor skill. Underfunded pensions might also be low skilled, which may lead them to mismanage capital resulting in higher underfunding ratios, and on average more matches with smaller and lower quality PE funds. To provide support for underfunding positions, I use exogenous variation in underfunding ratios which only affects the liability side. Further, I control for investment consultants, who are incharge of making investment decisions for pensions (Andonov et al. (2022)).

I introduce a novel instrument for public pension underfunding positions, by exploiting cross-sectional variation in unionization amongst public employees in a state-year. Public unionization rate, also known as union density is reported by the Current Population Survey (CPS). As part of the CPS conducted by the U.S. Bureau of Labor Statistics (BLS), survey respondents are asked: 1. "Are you a member of a union?". Empirically,

$$\text{Union Density } (\%)_{st} = \frac{\text{Number of members}_{st}}{\text{Number of Government Employees}_{st}} \tag{5}$$

There is a wide cross-sectional variation in public unionization rates across states. Figure C.1 shows variation in public union density across all U.S. states over time. While North Carolina had a union density of 6.6% in 2018, New York had 66.6% of its public workers as part of a union.[40]

This instrument is valid under two identifying assumptions. First, the relevance condition, i.e., public unionization affects underfunding ratios of public pensions. Intuitively, this makes sense as public workers, such as, teachers, firemen, and state employees heavily rely on public pensions for their pay, and higher unionization amongst public workers leads to higher monetary and non-monetary benefits, which strains funded ratios of public pensions. Freeman (1983) shows unions increase pension coverage.[41] A plausible concern related to instrument relevance is that the union density measure is available

[40] I did not find pension plan level union numbers, and hence use state-level union numbers. This is not a major concern as there are few large plans within each state.

[41] Also refer to: https://uniontrack.com/blog/unions-retirement-benefits, which mentions ways unions impact pen-

at the state level. However, the sample of public pensions generally covers the large plans, which likely eases this concern. Another concern is that there might be substantial heterogeneity in unionization and political influence in large and politically diverse states, such as California and Texas. To ensure that the instrument's strength and validity are not limited to a subset of the sample, I conduct empirical tests that exclude large states one at a time (California, Texas, Massachusetts) and that exclude a couple of large states simultaneously (California, Illinois, Massachusetts, New York), reported in Section 5.2, Tables 11 and 12. I will discuss these in detail in the next section.

Second, the exogeneity condition should hold, i.e., public unionization rates affect investments by pensions to specific private equity funds and ultimately private equity returns only through pensions' underfunding positions. This is plausible as portfolio allocation decisions are made by an investment committee, which is operationally independent from broader pension governance operations. Further, the instrument of unionization rate is at the state-year level, and not at the pension-year level. Hence, it is reasonable to assume that the unionization rate is taken as given by the public pension. Additionally, to be further conservative about concerns related to reverse causality, i.e., higher underfunding might lead to higher union representation, I use union density from one year prior to the underfunding ratio.

A potential threat to the exclusion restriction might still exist if union representatives serve directly on pension boards and exert direct influence over investment decisions. For instance, the board of trustees of the New York City Employees' Retirement System (NYCERS) includes heads of three unions. In such scenarios, unionization might plausibly affect both asset allocation and the downstream firm-level labor productivity outcomes. To address this concern, I use data on board composition of public pensions from Andonov et al. (2018), and merge it to the sample using Preqin investor identifiers. Andonov et al. (2018) finds that public pensions having ex officio participant board members are usually union representatives. In addition to studying all plans, I exclude plans with potential union members on board by carrying out the instrumental variable analysis for the sample excluding plans which have at least one ex officio participant board member — Illinois SERS, Louisiana Schools, New York City ERS, New York City Fire, New York City Police, and Oklahoma Fire.[42] Tables 9 and 10 present results for both samples. Appendix Tables C.1 and C.2 show OLS estimates for average PE returns and average GP size,

sions. Further, Appendix Figure C.2 shows empirical evidence of a +17.3% significant correlation between underfunded public pensions and one year lagged public union density for 2011 to 2018.

[42] I find similar estimates when limiting the sample to pension plans that have elected participant members or appointed participant members as board of trustees in less than 50% of cases (and hence, cannot override the board decision), as these groups might also represent union members (Andonov et al. (2018)). I do not have person-level history of pension plan board members, which would be ideal to rule out union representatives. However, these exercises get very close to the ideal scenario.

for all model specifications considering plans excluding potential union board members.

### 5.1.2 Empirical Methodology and Results

The first and second stages of the estimation are shown in Equations 6 and 7.

$$\text{Underfunded Ratio}_{pst} = \alpha_t + \beta \cdot \text{Union Density (\%)}_{st-1} + \text{Controls} + \epsilon_{pst} \tag{6}$$

$$y_{pst|p\in j} = \gamma_t + \beta_{IV} \cdot \widehat{\text{Underfunded Ratio}}_{pst} + \text{Controls} + \epsilon_{pst} \tag{7}$$

As before, $p$ stands for public pension, $s$ is state, $j$ is private equity fund, and $t$ is year. $y_{pst|p\in j}$ is the average size of a private equity fund a public pension commits capital to, in time $t$, and aggregated to the GP level. $y_{pst}$ will also measure the average total realized PE returns for a public pension $p$ in time $t$. The controls follow the most saturated specification of the OLS for the respective dependent variables.

Tables 9 and 10 reproduce the OLS estimates in Column (1) from main model (4) of Tables 3 and 4 respectively, and present the first and second stage IV results. Table 9 shows estimates for average realized PE returns, and Table 10 shows estimates for average GP size. The first three columns in both tables correspond to all plans, and the last three columns consider plans excluding those with potential union members on the board, i.e., excluding plans with at least one ex officio participant board member, which are generally union representatives (Andonov et al. (2018)).

The first stage coefficient of interest is $\beta$, and expected to be positive, i.e., a higher unionization rate corresponds with a higher underfunding ratio. The coefficient estimates for one-year lag union density are not very different in the first stage estimation for average PE returns, when considering all plans and when limiting attention to plans excluding potential union members as board trustees (Columns (2) and (5) of Table 9). The one-year lag union coefficient estimates in the average GP size estimation exercise are marginally smaller for plans excluding potential union board members as compared to all plans. This is plausible as board members might directly engage with PE investment selection and the regression is saturated with PE fund characteristics, while returns take substantial time following the investments. For average GP size estimation in Table 10, the coefficient on "Lag 1 Year Union Density" is positive and highly significant (Column (2): $\beta = 0.181$, $t = 6.73$). The effect is economically significant, as a one-standard-deviation (18.5%) increase in public unionization rates, increases underfunding positions by $0.181 \times 18.5\% = 3.3$ percentage points. With an average underfunding ratio of 23.3%, this corresponds to a 14.2% change. For the sample excluding plans with potential union affiliates as board

trustees, the coefficient on "Lag 1 Year Union Density" is also positive and statistically significant (Column (5): $\beta = 0.141$, $t = 5.13$), although marginally smaller in magnitude than when considering all plans. The coefficient is economically significant, as a one-standard-deviation (18.0%) increase a public unionization rates, corresponds to a $0.141 \times 18.0\% = 2.5$ percentage point change or a 11.1% change. Accordingly, higher unionized states have pension plans with higher underfunded ratios.

It is important for the IV to be "strong", i.e., the exogenous variable – one year lagged public union density to be strongly correlated with the endogenous variable – underfunding positions of public pensions, especially for IV estimation in finite samples. In Column (2) of Table 9, the $F$ statistic for the null that $\beta = 0$ is 24.0, which is greater than the rule of thumb ($F \geq 10$) proposed by Staiger and Stock (1997), and the 10% critical value in Table 5.2 of Stock and Yogo (2005). Similarly, in Column (5), the $F$ statistic is 23.9, which satisfies both conditions of a strong IV. Thus, weak instrument is unlikely to be a concern. The 2SLS coefficients are in the same direction as the OLS and statistically significant. The OLS is biased downward as the 2SLS coefficient $(-0.497)$ is higher in magnitude than the OLS coefficient $(-0.136)$. The coefficients are not statistically significantly different from each other at 10% level. This is true for both, the average size and return regressions. The standard errors are bound to be large in small samples with multiple dummies and controls. Overall, the estimates lend support to the fact that underfunding ratios is a key driver behind these results.

Overall, this section provides support for underfunding positions of pensions being a driver for public pensions ending up with, on average, smaller GPs, and receiving lower average private equity returns than the less underfunded pensions. The results also show that unionization rates capture a relevant variation of underfunding positions of public pensions, even after excluding plans with potential union members as trustees. To further substantiate the role of underfunding positions of pensions, in the next section I examine a set of alternative explanations, such as, the inclusion of political and large diverse states, political motives of board members, and local investments by pensions, and show that these channels do not account for the observed relationship.

### 5.2 Evaluating Alternative Explanations: Political Motives, Large States, Local Investments

In this section, I evaluate alternative explanations and a conduct a number of additional robustness checks to ensure that other mechanisms are not driving the estimates. The main model of comparison is model (4) in Table 3 for average PE returns and in Table 4 for average GP size. Tables 11 and 12 present estimates from robustness checks for average PE returns and average GP size, respectively. I consider

Table 9: Instrumental Variable Results for Underfunding Ratios and Average PE Returns, 2001-2021 for 1) All plans and 2) Plans excluding ex officio participant board members, generally union representatives

| | All plans | | | Plans without ex officio participant board members (generally union representatives) | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Dependent Variable: | PE Return | Underfunding Ratio | PE Return | PE Return | Underfunding Ratio | PE Return |
| Underfunding Ratio | -0.136*** | | -0.497** | -0.134*** | | -0.510** |
| | (0.036) | | (0.240) | (0.036) | | (0.241) |
| Lag 1 Year Union Density | | 0.367*** | | | 0.372*** | |
| | | (0.075) | | | (0.076) | |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocation Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Investment Consultant Dummies | Yes | Yes | Yes | Yes | Yes | Yes |
| Public Pension FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Regression Type | OLS | First Stage | Second Stage | OLS | First Stage | Second Stage |
| Observations | 1786 | 1786 | 1786 | 1750 | 1750 | 1750 |
| Adjusted R squared | 0.654 | 0.899 | -0.142 | 0.654 | 0.896 | -0.148 |
| Dependent Variable Mean | 0.119 | 0.233 | 0.119 | 0.118 | 0.230 | 0.118 |
| Dependent Variable Std | 0.148 | 0.199 | 0.148 | 0.149 | 0.197 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the OLS (Columns (1), (4)) and instrumental variable (IV) estimates (Columns (2) and (5) are first stage estimates, Columns (3) and (6) are second stage estimates) for the relation between underfunding ratios of public pensions and average PE returns for 2001-2021. Column (1) reproduces main model (4) of Table 3. Columns (1)-(3) correspond to all plans. Columns (4)-(6) correspond to the sample *excluding* plans with ex officio participant board members, which are generally union representatives (Andonov et al. (2018)). The plans excluded are: Illinois SERS, Louisiana Schools, New York City ERS, New York City Fire, New York City Police, and Oklahoma Fire. The PE performance is measured using the net internal rate of return (IRR) (%). All columns control for LP size, the average past three-year PE, equity, fixed income allocations, investment consultant dummies, public pension and year fixed effects.

Table 10: Instrumental Variable Results for Underfunding Ratios and Average GP Size (logarithmic), 2001-2021 for 1) All plans and 2) Plans excluding ex officio participant board members, generally union representatives

| | All plans | | | Plans without ex officio participant board members (generally union representatives) | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Dependent Variable: | GP Size | Underfunding ratio | GP Size | GP Size | Underfunding ratio | GP Size |
| Underfunding Ratio | -0.517*** | | -1.900** | -0.578*** | | -2.891** |
| | (0.167) | | (0.839) | (0.176) | | (1.183) |
| Lag 1 Year Union Density | | 0.181*** | | | 0.141*** | |
| | | (0.027) | | | (0.027) | |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocations | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Strategy | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Industry Focus | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Region Focus | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Benchmark Returns | Yes | Yes | Yes | Yes | Yes | Yes |
| Fund Domicile | Yes | Yes | Yes | Yes | Yes | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Positive PE Allocation | Yes | Yes | Yes | Yes | Yes | Yes |
| Regression Type | OLS | First Stage | Second Stage | OLS | First Stage | Second Stage |
| Observations | 1028 | 1028 | 1028 | 991 | 991 | 991 |
| Adjusted R squared | 0.486 | 0.225 | 0.287 | 0.481 | 0.211 | 0.207 |
| Dependent Variable Mean | 8.999 | 0.233 | 8.999 | 8.985 | 0.226 | 8.985 |
| Dependent Variable Std | 1.118 | 0.175 | 1.118 | 1.125 | 0.171 | 1.125 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the OLS (Columns (1), (4)) and instrumental variable (IV) estimates (Columns (2) and (5) are first stage estimates, Columns (3) and (6) are second stage estimates) for the relation between underfunding ratios of public pensions and average GP size for 2001-2021. Column (1) reproduces main model (4) of Table 4. Columns (1)-(3) correspond to all plans. Columns (4)-(6) correspond to the sample *excluding* plans with ex officio participant board members, which are generally union representatives (Andonov et al. (2018)). The plans excluded are: Illinois SERS, Louisiana Schools, New York City ERS, New York City Fire, New York City Police, and Oklahoma Fire. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. All columns control for LP AUM, LP's past allocations (average past three-year PE, equity, fixed income allocations), GP and pension characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile), condition on having a positive allocation to PE, and control for vintage year fixed effects. There are very few observations in the which the PE allocation variable is missing in the PPD database but the Preqin data shows a positive GP size connection with the plan in that year. Upon manual investigation, I find that most of these plans have positive PE allocation in the years before and after the "missing". Both these facts confirm that these observations indeed represent a positive PE allocation and hence are part of the analysis.

the most saturated models in the main text and present estimates from other model specifications in Appendix Tables C.3-C.8.

A first concern is that few states with large public pension plan coverage might be driving the coefficient estimates. I evaluate California, Texas, and Massachusetts separately, where Texas and California might have a slightly greater political heterogeneity than Massachusetts. It is useful to evaluate the results leaving these states out one at a time. Column (1) of Table 11 shows that excluding California-based pension plans, a 10 percentage point increase in underfunding ratio of pension plans is associated with a $-1.43$ percentage point change in average annual net IRR, which is approximately a 5.1% change relative to the $-1.36$ percentage point change for all plans (Column (4), Table 3). Column (2) shows that excluding Texas-based plans, a 10 percentage point increase in underfunding ratio corresponds to a $-1.24$ percentage point change in average annual net IRR, which is an 8.8% change relative to the overall $-1.36$ percentage point change. On the other hand, Column (3) shows that excluding Massachusetts-based plans gives a very similar change ($-1.37$ percentage point change) relative to the $-1.36$ percentage point change for all plans. A similar trend is visible in the coefficient estimates on underfunding ratio, when estimating average GP size (Table 12). When excluding Massachusetts-based plans (Column (3)), the coefficient estimate $-0.517$ is the same as that in all plans (Column (4), Table 4). When excluding California-based and Texas-based plans, the coefficient estimates change by 3.1% to 4.8% relative to the baseline estimate (measured in logarithmic terms).[43] While these coefficients are not substantially different from the main models, this analysis highlights the greater heterogeneity in politically diverse states, such as, California and Texas. These results suggest that while political diversity has a role to play, as expected, it does not eliminate the main results.

A second concern is that variation in PE performance may be driven due to political incentives of state board members. Andonov et al. (2018) show that while existence of a higher percentage of state-appointed or state-ex officio board members reduces pensions' performance in venture capital and real estate investments, they do not find this to be the case for buyout funds, which is the focus of this study.[44] I test this alternative mechanism using data on board composition of public pensions. In Column (4), I limit attention to plan-year observations where the percentage of state political board members, defined as the total of state appointed and state ex-officio, is less than 50% of all board members. It is reasonable

[43] Appendix Tables C.3 and C.6 provide consistent estimates across other model specifications.

[44] Table VI of Andonov et al. (2018) shows that a higher percentage of state-appointed board members has a positive and insignificant impact on PE buyout performance, and having more state ex-officio board members has a negative and insignificant impact on PE buyout performance.

Table 11: Evaluating Alternate Mechanisms for More Underfunded Pensions Realizing Lower Average PE Returns, 2001-2021 (robustness)

| | Excluding California | Excluding Texas | Excluding Massachusetts | State Political Board Members $< 50\%$ | Excluding States with In-State Investments $> 20\%$ |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Underfunding Ratio | -0.143*** | -0.124*** | -0.137*** | -0.156** | -0.148*** |
| | (0.040) | (0.036) | (0.036) | (0.069) | (0.042) |
| LP AUM | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocation Controls | Yes | Yes | Yes | Yes | Yes |
| Investment Consultant Dummies | Yes | Yes | Yes | Yes | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes |
| Observations | 1531 | 1715 | 1753 | 1366 | 1347 |
| Adjusted R squared | 0.626 | 0.666 | 0.650 | 0.538 | 0.615 |
| Dependent Variable Mean | 0.119 | 0.119 | 0.117 | 0.102 | 0.117 |
| Dependent Variable Std | 0.147 | 0.147 | 0.149 | 0.133 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021. This table evaluates alternate mechanisms and shows robustness for Table 3. The PE performance is measured using the net internal rate of return (IRR) (%). All columns control for LP size, the average past three-year PE, equity, fixed income allocations, investment consultant dummies, public pension and year fixed effects. Column (1) excludes California. Column (2) excludes Texas. Column (3) excludes Massachusetts. Column (4) considers plan-year observations where state political board members are $< 50\%$. State political board members comprise of state appointed and state ex-officio board members. This data is obtained from Andonov et al. (2018). Column (5) excludes states with $> 20\%$ in-state investments across alternative assets: California, Illinois, Massachusetts, New York, as defined by Andonov et al. (2018), Online Appendix Table IAII corresponding to the 1990-2011 period. As investments are sticky and long dated, it is reasonable to assume that the cross-sectional trend across states persists beyond 2011 as well.

Table 12: Evaluating Alternate Mechanisms for More Underfunded Pensions Matching with Average Smaller GPs, 2001-2021 (robustness)

| | Excluding California | Excluding Texas | Excluding Massachusetts | State Political Board Members $< 50\%$ | Excluding States with In-State Investments $> 20\%$ |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Underfunding Ratio | -0.533*** | -0.542*** | -0.517*** | -0.445** | -0.576** |
| | (0.183) | (0.171) | (0.167) | (0.204) | (0.225) |
| LP AUM | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocations | Yes | Yes | Yes | Yes | Yes |
| Fund Strategy | Yes | Yes | Yes | Yes | Yes |
| Fund Industry Focus | Yes | Yes | Yes | Yes | Yes |
| Fund Region Focus | Yes | Yes | Yes | Yes | Yes |
| Fund Benchmark Returns | Yes | Yes | Yes | Yes | Yes |
| Fund Domicile | Yes | Yes | Yes | Yes | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes | Yes |
| Positive PE Allocation | Yes | Yes | Yes | Yes | Yes |
| Observations | 844 | 949 | 1028 | 843 | 717 |
| Adjusted R squared | 0.488 | 0.480 | 0.486 | 0.490 | 0.484 |
| Dependent Variable Mean | 8.978 | 8.996 | 8.999 | 8.988 | 9.000 |
| Dependent Variable Std | 1.159 | 1.110 | 1.118 | 1.108 | 1.165 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021. This table evaluates alternate mechanisms and shows robustness for Table 4. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. All columns control for LP AUM, LP's past allocations (average past three-year PE, equity, fixed income allocations), GP and pension characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile), condition on having a positive allocation to PE, and control for vintage year fixed effects. There are very few observations in the which the PE allocation variable is missing in the PPD database but the Preqin data shows a positive GP size connection with the plan in that year. Upon manual investigation, I find that most of these plans have positive PE allocation in the years before and after the "missing". Both these facts confirm that these observations indeed represent a positive PE allocation and hence are part of the analysis. Column (1) excludes California. Column (2) excludes Texas. Column (3) excludes Massachusetts. Column (4) considers plan-year observations where state political board members are $< 50\%$. State political board members comprise of state appointed and state ex-officio board members. This data is obtained from Andonov et al. (2018). Column (5) excludes states with $> 20\%$ in-state investments across alternative assets: California, Illinois, Massachusetts, New York, as defined by Andonov et al. (2018), Online Appendix Table IAII corresponding to the 1990-2011 period. As investments are sticky and long dated, it is reasonable to assume that the cross-sectional trend across states persists beyond 2011 as well.

to expect that if political board members do not hold the majority seats, there is a greater chance that pension plans do not make politically oriented investment decisions. The coefficient estimates on underfunding ratio of pension plans for average PE return estimation (Column (4), Table 11) and average GP size estimation (Column (4), Table 12) are not statistically different from their counterpart main model (4) in Tables 3 and 4 respectively.[45] This confirms the existing results in the literature, that existence of politically oriented state board members do not appear to differentially drive performance outcomes in PE buyout funds.

A third concern is that pension plans may want to invest locally, given their objectives for local development, and the corresponding PE investments might not perform well (Hochberg and Rauh (2013)). First, the literature finds evidence for overweighting of in-state investments for state and political goals in venture capital and real estate, but not in PE buyout investments (Andonov et al. (2018), Hochberg and Rauh (2013)).[46] Venture capital investments might benefit the state via innovation, and real estate investments might benefit the state via infrastructure development, both of which are more directly aligned with local development goals than buyout funds. Second, in order to test this hypothesis, I remove states which have at least 20% of investments by their pension plans in in-state investments across various type of alternative asset classes.[47] It is important to note that this is a more restrictive test than dropping plans with a high percentage of in-state investments within the buyout category only. Column (5) of Tables 11 and 12 present coefficient estimates on underfunding ratio for average PE return and average GP size estimation, respectively. Once again, the coefficient estimates are not statistically and significantly different than the main models in Tables 3 and 4, which is consistent with the existing literature and economic intuition discussed above.[48] Lastly, when I limit the LP-PE target firm sample to out-of-state investments relative to the investor's states, most of the sample is retained, further supporting the consistency of the results above.

Overall, this section demonstrates that while states with major pension plans and politically diverse states can marginally account for part of the changes in average PE returns and average GP size driven

[45] Appendix Tables C.4 and C.7 confirm consistent inferences from other model specifications. Additionally, the estimates for all model specifications are similar when I limit the sample to plan-year observations, where percentage of *all state* board members is less than 50% instead of limiting attention to observations, where percentage of all *state political* board members is less than 50%. This is because the share of board members which are state elected — the difference between all state members and state political members — is close to negligible.

[46] For reference, see Table VII of Andonov et al. (2018).

[47] These states are California, Illinois, Massachusetts, and New York, as defined in Online Appendix Table IAII of Andonov et al. (2018) corresponding to the 1990-2011 period. As investments are sticky and long dated, for this robustness exercise, it is reasonable to assume that the cross-sectional trend across states persists beyond 2011 as well.

[48] Appendix Tables C.5 and C.8 demonstrate the same across model specifications.

by underfunding ratios of pensions, this does not substantially affect the main model coefficient estimates. Further, political and local incentives of pensions do not seem to significantly affect their PE buyout investments.

## 6 Policy and Broader Economic Implications

The first implication of this study is on the side of public pension policy. According to the Governmental Accounting Standards Board (GASB) rulings, U.S. public pensions' reported funding ratios, which is assets divided by liabilities, use individual plans' assumed future return on assets to discount future obligations when calculating the present value of liabilities. There is no single defined discount rate for U.S. state pensions to value liabilities, as in Europe (Greenwood and Vissing-Jorgensen (2018)). The median pension plan discount rate in the sample is 7.75%. An increase in assumed returns mechanically decreases the present value of liabilities, and increases funding ratios. This obscures the true extent of public pension liabilities and furthers their incentives toward high return assets, like private equity. Investments in private assets also provide less volatility to their portfolios.

Public pensions invest a substantial portion of their assets in PE, and investment returns are one of their major sources of financing.[49] This paper shows that once public pensions invest in PE, ex-post more severely underfunded public pensions invest in smaller GPs on average, and receive lower average returns, than the less underfunded pensions. A decrease in returns might further exacerbate pensions' underfunding for the following years via a decrease in asset values. This may provide the more underfunded pensions a greater incentive to swing for high returns, but if they further realize lower average returns from their PE investments, it might push them into an underfunding spiral. This further supports the argument for valuing liabilities using a common discount rate that reflects the true risk of cash flows, which are fixed obligations owed to retirees (Novy-Marx and Rauh (2011)).

The second implication of this study is for local public finances. U.S. public pensions had $4.1 tn. in assets in 2021, and supported 14.7 mn. active members and 11.2 mn. retirees. Public pension plans generally rely on state government support if pension obligations are not met. Hence, underfunded positions of public pensions also have broader implications for municipal and state finances, and the potential stability of retirement systems. Understanding public pensions' investment allocation decisions and incentives in private markets, which are opaque and transaction based, is key for policy making.

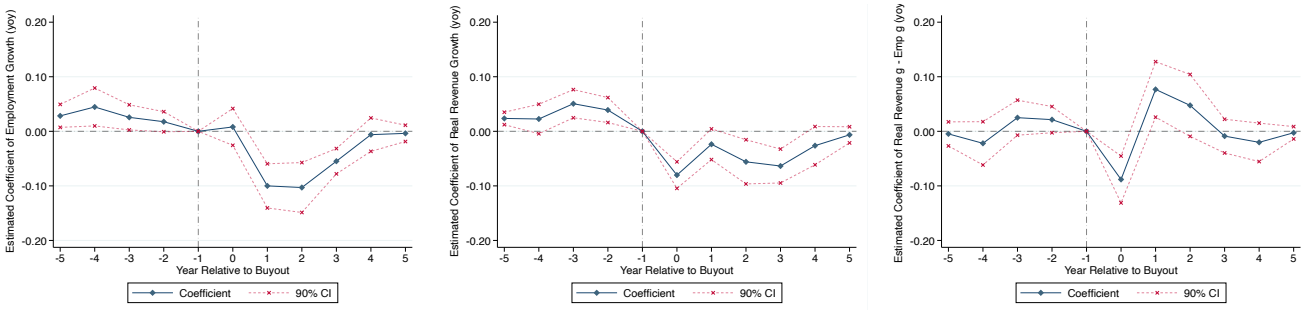
[49] Public pension assets were $4.1 tn. in 2021, and on average, they invested 10.6% of their total assets in PE. This corresponds to $435 bn.

The third implication relates to PE capital supply. Not only are the more underfunded public pensions understating their true costs to pay public sector employees, but their investment allocations also flows to projects funded by smaller GPs, on average, which suffer on the real measures at the firm level. How is an average firm that is receiving capital from public pension LPs performing relative to a comparable U.S. firm, which has never received any PE capital? Figure 5 shows the difference in differences coefficient estimates of employment growth, revenue growth, and labor productivity for target firms relative to control firms, and finds no significant increases in labor productivity at targets relative to control firms five years post buyout. The control firms are active firms in the buyout transaction year, in the same industry, firm size, firm age, and single-/multi-unit group as target firms, but never bought by PE.[50] These are consistent with the control groups used in the event study specifications in Section 4. Even within the small, unrepresentative subsample of manufacturing target firms, which have detailed cost measures, I do not find substantial increases in total factor productivity (TFP) (Appendix D.5). Combined with results in Section 4.2, this suggests that small GPs, whose target firms face productivity losses, are pulling down the average performance in the sample. It is possible that fund managers either are left with adversely selected deals within the granular matched groups, or lack the management expertise and experience.

These results provide evidence on a broader policy question: whether excess capital is flowing into PE such that, at the margin, lower quality deals are being financed. A related and policy-relevant question is whether capital from U.S. public pension LPs is a contributing source of this excess supply. Further research is needed to answer this question. Lastly, this phenomenon of desperate capital and ending up with worse returns on average has applications beyond PE. For instance, the study helps us think more broadly about investor-type and investment matches in other asset classes, such as the mortgage market, where lenders facing funding pressure or capital constraints may originate lower quality loans, leading to worse ex-post performance.

[50] It would be ideal to have an exogenous shock to capital flow in private equity funds across all industries. In the absence of that, the literature has relied on constructing control firms using a matched sample. The control firms are constructed based on the cross product of 22 industries (based on two-digit NAICS codes) $\times$ 12 firm size categories $\times$ 5 firm age categories $\times$ multi/single unit firm $\times$ buyout year (following Davis et al. (2014)). The regression accounts for lagged firm growth from $t_0 - 3$ to $t_0 - 1$ to control for pre-buyout organic growth, and is weighted by employment at the time of buyout. Appendix D.1 describes the specification in detail. Appendix D.2 compares raw data patterns in treated and control firms with prior research as a sanity check. As the control group, comprising of most firms in the U.S. economy, exceeds the treated group, I run into computing constraints. Following standard approach while using U.S. Census micro data, I select a 10% random sample from the universe of control firms within each group. Appendix D.3 shows robustness to different random samples. The parallel trends assumption is near satisfied. With additional controls (Appendix D.4), parallel trend assumptions are satisfied.

Figure 5: Difference in Differences Estimated Coefficients Over Time Relative to Buyout Year, PE Deals 1997-2018




**(A)** Employment $g$ **(B)** Revenue $g$ **(C)** Rev $g$ - Emp $g$

*Notes:* The figure plots difference in differences coefficients of interest $\gamma_j$ in equation: $y_{it} = \alpha_t + \sum_{j=-5, j\neq -1}^{j=+5} \gamma_j(\text{PE}_i \times \text{Buyout Year}_{it_0+j}) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it}$ for years -5 to +5 relative to buyout for employment (panel A), revenue (panel B), and revenue minus employment (panel C) growth rates. $\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the controls. $\text{Buyout Year}_{i,t_0+j}$ is a dummy for each $j$ taking a value of 1 in the year $t_0 \pm j$ relative to buyout year, with $j = -5, \cdots, 5$. $\text{LFIRM}_i$ is the lagged firm growth from $t_0 - 3$ to $t_0 - 1$. The year before buyout $t_0 - 1$ is the omitted category, and years beyond 5 years pre- and post-buyout are binned with year +/-5 relative to buyout. Dotted red lines show 90% confidence intervals. Standard errors are clustered at the firm level, for potential heterogeneity. The transaction takes place between year -1 and 0.

# 7 Conclusion and Future Research

This paper studies U.S. public pensions' investment allocations in private equity, based on their degree of underfunding. I trace their investments to the ultimate micro assets—the target firms, in which the PE funds invest. I construct a comprehensive micro-dataset, combining detailed investments from public pension LPs to individual PE funds, GPs, and target firms, along with measuring real outcomes at target firms over time from the U.S. Census Bureau micro-data.

First, I show that U.S. public pensions have increased their allocations to alternative assets by three-fold post 2000. Even within public pensions, the most underfunded pensions have increased their capital commitments drastically. Motivated by these facts, I show that public pensions with larger underfunding positions exhibit a lower average PE performance, and end up matching with, on average, smaller GPs. Larger GPs represent greater scale and deal flow, broadly interpretable for better PE fund quality. I account for various measurement challenges, such as market value adjustments to underfunding positions, varying fiscal end periods, and varying actuarial cost methods, and find consistent results. In fact, I find that the accounting underfunding measure understates the true incentives of pensions relative to the market value underfunding measure.

Second, I further investigate the returns to underfunded public pensions by examining real outcomes at PE target firms underlying the deals, using the U.S. Census Longitudinal Business Database (LBD). Since majority of the target firms are private, and often undergo name changes, the Census data

provides the most comprehensive information on real outcomes at firms. I study performances at target firms based on public pension LP and GP characteristics. I show that PE targets financed by the most severely underfunded pensions (i.e., the bottom tercile of funded pensions) experience a large, statistically significant $-5.5\%$ labor productivity change. Separately, PE targets financed by small GPs, on average, show decreases in productivity. Observing labor productivity decreases in both splits of PE target firms, by pension and GP characteristics, confirm the matching discussed earlier. Further, labor productivity decreases are consistent (without implying a one for one relation) with lower average returns exhibited by the most underfunded pensions, and with their matching to smaller GPs on average.

Third, I provide evidence supporting the underfunding channel. I use variation in public unionization rates across state-years as a novel instrument for pension underfunding. I conduct various robustness tests, such as controlling for investment consultants, and limiting attention to plans without potential union members on board. Further, I test alternate hypotheses to ensure that a few large states, and political and local incentives of pensions are not driving the estimates, and I corroborate the findings with the existing literature.

Lastly, I discuss policy and broader economic implications of the study. The paper shows that the more underfunded pensions realize lower average returns from PE. However, the severely underfunded pensions also have a greater incentive to swing for high returns, in order to close funding gaps. The results in the paper support the use of a common discount rate for valuing pension liabilities that reflects the true risk of the underlying cash flows. A common, risk-based discount rate would reduce distortion of pensions' incentives, and help prevent plans from being pushed into a self-reinforcing underfunding loop. Additionally, I highlight the potential stability risk for retirement systems, and state and municipal finances. On the side of PE capital supply, an average public pension financed PE target does not experience labor productivity increases following a PE deal relative to a peer that never received PE capital. I also do not find increases in total factor productivity for the small subset of manufacturing targets. This provides evidence for a broader policy question: Is there excess capital flowing into PE, such that, at the margin, lower quality deals are being financed? This raises further questions for future research: Are U.S. public pension LPs a contributing source of this excess supply? Is there an optimal investment threshold that realizes high returns while limiting the financing of marginal deals? I leave these questions to future research.

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# Appendix for Desperate Capital Breeds Productivity Loss: Evidence from Public Pension Investments in Private Equity

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# A Public Pension Adjusted Underfunding Ratios with BEA Methodology, Varying Fiscal Year End Dates and Actuarial Cost Methods

## A.1 BEA Adjustment Methodology

According to the Governmental Accounting Standards Board (GASB) rulings, U.S. public pensions use the average assumed return on their investments to discount their unfunded liabilities. This obscures the true value of their unfunded positions, and creates a major distortion between the reported and the actual unfunded liabilities. Discounting liabilities at the expected return on the assets of the plan runs opposite to the logic of financial economics, according to which the financial streams of payments should be discounted at a rate that reflects their risk (Modigliani and Miller (1958)). Novy-Marx and Rauh (2011) takes a substantial step in correcting this bias.

**Adjustment of public pension liabilities by Novy-Marx and Rauh (2011).** In order to correct this bias, ideally we take the future stream of payments for each pension plan and discount them back to the present with an appropriate discount rate, which is common across pension plans. This facilitates a proper comparison of unfunded liabilities across public pensions in a given year. However, we do not observe individual cash flows by active, retired, or separated (who is not an active employee but has not yet started receiving pension payments) employee for each plan-year. Novy-Marx and Rauh (2011) models the prospective stream of payments from state pension promises using each state's stated liability, discount rate, actuarial cost method, benefit formulas, the number and average wage of state employees by age and service, salary growth assumptions by age, mortality assumptions, cost of living adjustments (COLAs), and separation (job leaving) probabilities by age. After constructing the cash flow streams $C_{p,t}$ for each plan $p$ with the above stated assumptions, and using the discount rate stated by the plan $r_{p,stated}$, they reconstruct the plan's reported liabilities $L_{p,stated}$. Pension plans discount cash flows using the following discounted formula:

$$L_{p,stated} = \sum_{t=1}^{T} \frac{C_{p,t}}{(1 + r_{p,stated})^t} \tag{8}$$

Next, the liabilities for each plan-year can be recalculated using the appropriate discount rates $r^*$ reflecting the risk of cash flows.

$$L_p^* = \sum_{t=1}^{T} \frac{C_{p,t}}{(1 + r_p^*)^t} \tag{9}$$

With the adjusted liabilities for each plan-year $L^*_{p,t}$, the adjusted underfunding ratios can be recalculated as,

$$\text{UF}^*_{p,t} = \left(1 - \frac{A_{p,t}}{L^*_{p,t}}\right), \tag{10}$$

where $A_{p,t}$ are the pensions assets for plan $p$ and time $t$. Further, the liabilities can be recalculated under different actuarial cost methods and adjusting for differing fiscal year end dates.

**Bureau of Economic Analysis (BEA) Implementation.** In order to standardize liability estimates prepared by different pension plans using different actuarial cost methods, discount rates, and fiscal year end dates, the BEA implements Novy-Marx and Rauh (2011) (Lenze (2013)), and estimates annual transactions from 2000 to 2018, consistently measured on an accrual basis, using a common discount rate, and benchmarked to the 2017 Census of Governments (Lenze (2020)). The BEA uses these estimates to produce the National Income and Product Accounts (NIPA) estimates. The Federal Reserve Board receives the BEA liability estimates at an aggregate level, and separately for the federal, state and local level for the Financial Accounts (FA) construction. Hence, the BEA adjustment is the widely accepted series, and I will reconstruct the underfunded liabilities of public pensions using this method.[51]

In Table A.1, I reproduce from Table 2 of Lenze (2020), the discount rates used by the BEA for state and local government defined benefit plans over years. For 2000-2003, the BEA has used a 6% rate, from 2004-2009 a 5.5.% rate, from 2010-2012 a 5% rate, and from 2013-2018 a 4% rate. There has not been an update on the discount rate from 2019 onwards. I assume the continuation of the 4% discount rate till the end of the sample period (2021). This also makes studying the time period until 2021 and not until more recent, appropriate, as liability recording, estimation, and adjustment runs with a lag.

Table A.1: BEA Discount Rates for State and Local Government Defined Benefit Pension Plans from Lenze (2020) Table 2

| **Years** | **Discount Rate** |
|---|---|
| 2000–2003 | 6% |
| 2004–2009 | 5.5% |
| 2010–2012 | 5% |
| 2013–2018 | 4% |
| 2019–2021 | 4% (assumed because no BEA update) |

Let $r$ be the discount rate used by the plan and $r^*$ be the BEA discount rate. I have removed the

[51] I am grateful to Irina Stefanescu from the Federal Reserve Board for helpful discussions in this understanding. More recently, Giesecke and Rauh (2022) use the U.S. Treasury Yield Curve to discount public pension liabilities. However, the ranking of public pensions based on their underfunded positions is consistent irrespective of the discount rate used (Figure A.1 of Giesecke and Rauh (2022)).

time and plan subscripts for ease of notation. $L_{stated}$ is the total pension liability (also known as "$TPL$") reported by the pension plan based on the plan discount rate $r$. $L^*$ (also denoted as "$TPL^*$" is the total pension liability based on the BEA discount rate $r^*$. The adjustment of pension liabilities is given by

$$TPL^* = TPL\Big(1 + \frac{d}{100}\Big)^{r-r^*} \tag{11}$$

The adjustment depends on duration $d$, which can be calculated as,

$$d = \Big(\frac{NPL_{r-1} + FNP}{TPL} - 1\Big)/0.01, \quad if \quad r > r^* \tag{12}$$

$$\Rightarrow d = \Big(\frac{TPL_{r-1} - TPL}{TPL}\Big)/0.01, \quad if \quad r > r^*$$

$$d = \Big(1 - \frac{NPL_{r+1} + FNP}{TPL}\Big)/0.01, \quad if \quad r < r^* \tag{13}$$

$$\Rightarrow d = \Big(\frac{TPL - TPL_{r+1}}{TPL}\Big)/0.01, \quad if \quad r < r^*$$

where, $FNP$ is the plan fiduciary net position (also known as pension plan assets, $A$) at the end of the fiscal year. $NPL_{r-1}$ is the net pension liability at the plan discount rate less 1 percentage point (as reported by the pension plan). $NPL_{r-1}$ is calculated as $TPL_{r-1} - FNP$. $NPL_{r+1}$ is the net pension liability at the plan discount rate plus 1 percentage point, and calculated as $TPL_{r+1} - FNP$. To summarize,

$$NPL = TPL - FNP$$

$$NPL_{r-1} = TPL_{r-1} - FNP$$

$$NPL_{r+1} = TPL_{r+1} - FNP$$

To calculate the adjusted pension liabilities $TPL^*$, we need $TPL$, $r$, $r^*$, $FNP$, $NPL_{r+1}$ or $TPL_{r+1}$, and $NPL_{r-1}$ or $TPL_{r-1}$. I get $TPL$, $r$, and $FNP$ from the Public Plans Data (PPD) produced by the Center for Retirement Research at Boston College. $r^*$ is defined in Table A.1. I collect $NPL_{r+1}$, $TPL_{r+1}$, $NPL_{r-1}$ and $TPL_{r-1}$ from Comprehensive Annual Financial Reports (CAFRs) of individual pensions. I describe the methodology in Section A.2.

### A.2 Data Collection from Pension Plan Comprehensive Annual Financial Reports (CAFRs)

Financial statements of public pension plans report $NPL$ and $TPL$ for plan rate plus/minus one percentage point only from 2014 onwards. These sensitivities are key to constructing the duration $d$, which

is needed to calculate adjusted pension liabilities $TPL^*$. I collect the $NPL$ and $TPL$ sensitivities for the 204 public pension plans in the sample obtained from the Public Plans Data (PPD) for 2014 to 2021, the end of the sample period. For each plan-year, I reviewed the Comprehensive Annual Financial Reports (CAFRs), the Annual Comprehensive Financial Reports (ACFRs), and the Governmental Accounting Standards Board (GASB) pension disclosures, and extracted the total pension liability, fiduciary net position, net pension liability, net pension liability for pension plan discount rate plus/minus one percentage point, total pension liability for pension plan discount rate plus/minus one percentage point, and the plan discount rate. I also record the source URL, page where the text can be found, and the text in which the information is found. In some instances, the information is found in the image instead of the text.

The extraction follows a conservative source hierarchy. Official CAFRs or pension system reports are preferred. If official links are unavailable, government archives, PPD reposts, or archived official copies are used, but source quality is tracked. PPD values are retained as cross-check fields for report-scraped variables.

When the report provides direct $TPL$ values for pension plan rate plus/minus one percentage point $TPL_{r\pm1}$, I use the reported $TPL$ values. When the report only provides $NPL$ values for pension plan rate plus/minus one percentage point $NPL_{r\pm1}$, I derive the corresponding $TPL_{r\pm1}$ values using $TPL_{r\pm1} = NPL_{r\pm1} + FNP$, when $FNP$ is available from the report. With these values, I derive duration $d$ using Equations 12 and 13, adjusted total pension liability $TPL^*$ using Equation 11, and adjusted underfunded ratios $UF^*$ using Equation 10.

Each accepted observation is required to pass several checks before it is used in the BEA-Adjusted underfunding calculation. First, the row must have clear source traceability. I record the report URL, page or table location, source type, and evidence text. The source must correspond to the same fiscal year and the same pension plan or plan components. If the report only provides an aggregate system-level table, I do not use it for a plan-level statistic unless the component mapping is explicit. I'll highlight some plan-specific cases below.

Second, the extracted values must satisfy basic pension accounting identities. When the report provides $TPL$, $FNP$, and $NPL$, I check whether $TPL \approx NPL + FNP$. Additionally, the derived $TPL_{r\pm1}$ from $TPL_{r\pm1} = NPL_{r\pm1} + FNP$ and the reported $TPL_{r\pm1}$ when available, should be close to exactly equal. Observations with large unexplained differences are not used for the analysis, because they may indicate that the scraper selected the incorrect row, component, year, or unit.

Third, the discount-rate sensitivity values must be economically monotonic. A lower discount rate should increase the liability, and a higher discount rate should reduce the liability. Therefore, after converting $NPL_{r\pm1}$ to $TPL_{r\pm1}$ when needed, the values must satisfy $TPL_{r-1} > TPL_r > TPL_{r+1}$. Rows that fail this check are not included, because they are likely to reflect a parsing error, table-column or aggregate/plan mismatch, or an incorrect year extraction.

Fourth, I compare the report values against PPD values as a cross-check. PPD is not used to choose the scraped values, but it is useful for identifying suspicious cases. If report $TPL$ differs substantially from PPD pension liability, or report $FNP$ differs substantially from PPD NetPosition (assets), the observation is flagged and not used, unless the difference can be explained by a documented reporting-basis or component-definition issue. Specifically, to be extremely conservative, the percentage difference between the PPD and CAFR total pension liabilities should be less than 1%, and the percentage difference between the PPD and CAFR assets should be less than 1%. This is key to ensure that the $NPL$ and $TPL$ sensitivities collected from the pension reports correspond to the correct pension plan-year.

Fifth, the implied duration must be calculable and reasonable. A very large and conservative filter of a duration range between 5-25 years is used. This is reasonable given that the average duration in the U.S. for 2018 is 12.5 (Lenze (2020)). Any duration outside this range is very low or very high, and might often be a component mismatch, wrong sensitivity value, or unusual plan-specific reporting. After being convinced that the values between the PPD and the CAFRs are consistent, I use the PPD supplied discount rate for the duration and adjusted TPL calculations, and supplement with the discount rates collected from the CAFRs, when not available in the PPD (85 plan-year observations). Similarly, I use the PPD NetPosition for adjusted underfunding ratio calculation, and supplement with the FNP values scraped from the CAFRs, when not available in the PPD (19 plan-year observations).

**Plan-specific notes.** Several plans required manual review because the CAFRs separate the pension components, while the PPD identifier may correspond to a broader plan scope. For these cases, I matched the reported CAFR values to the PPD plan definition before constructing the liability and duration variables. Component aggregation is accepted only when the report structure clearly identifies the full plan component set, the components share the same measurement date and discount-rate basis, and the mapping to the PPD plan is unambiguous.

- Indiana Teachers: In FY2014, the Teachers' Retirement Fund is reported through two accounts: the Pre-1996 Account and the 1996 Account. Since the PPD identifier refers to the Indiana Teachers plan as a combined plan, the two accounts are aggregated before constructing NPL, FNP, TPL, and

the discount-rate sensitivity measures.

- San Diego City ERS: The SDCERS report presents separate employer components for the City, Port, and Airport. The PPD identifier corresponds to the combined SDCERS plan scope, rather than the City component alone. The plan-year values are therefore constructed using the aggregate of the City, Port, and Airport components.

- Dallas Police and Fire: The relevant PPD identifier maps to the Combined Pension Plan. The Supplemental Plan is treated as outside the PPD plan scope. The Combined Pension Plan is used.

- Detroit Police and Fire: The Detroit Police and Fire reports disclose separate legacy and successor components. To match the PPD plan scope, the Defined Benefit / Legacy component is combined with Component I. When Component I is reported as a net pension asset, it is treated as a negative net pension liability in the component aggregation.

- Detroit General RS: The PPD identifier corresponds to the combined scope of the Component II Legacy Plan and the Component I Hybrid Plan. The accepted observation therefore aggregates both components.

- Virginia RS: The Virginia reports include a Total Virginia Retirement System row, which aggregates the State, Teacher, and Political Subdivisions components. This total VRS row is used for PPD ID 115. SPORS, VaLORS, and JRS are not added because they are separately reported systems and are outside the Virginia RS plan scope.

- Utah Public Safety: The PPD identifier corresponds to the Utah Public Safety and Firefighter Retirement Plan. The URS GASB schedules report the Public Safety Retirement System, the Firefighters Retirement System, and the Tier 2 Public Safety and Firefighter Retirement System as separate components. The plan-year values are therefore constructed using the aggregate of these three components.

**Fiscal Year End Dates.** I also collect fiscal year end dates from the CAFRs for all the plan-years in the sample to cross-check with the PPD. First, I compare if the planname collected from the CAFRs and the planname from the PPD match. Second, in three plan-year observations (corresponding to two plans), the fiscal end month-years collected from the CAFRs and the PPD do not match. For these two plans, I compare the fiscal end month-years from the CAFRs and the PPD for rest of the years, and use the

month-years from the CAFRs after confirming that it is consistent with other years. Third, I use the fiscal end month-years from the CAFRs if it is unavailable in the PPD after manually investigating rest of the fiscal end month-years of the plan. Fourth, for plan-year observations with missing fiscal end month-years in both the PPD and the CAFRs, I infer the fiscal end month-years based on rest of the years for which the data is available. Generally, there are very few observations in these cases described above. These inferences are only for completeness.

## A.3 Comparison with BEA Supplied Tables

In this section, I compare the self-constructed duration and funding ratio measures with the BEA supplied measures. The BEA has not made duration and adjusted funding ratio estimates at the plan-year level publicly available. Instead, in Lenze (2020) Table 14, the BEA provides liability weighted duration measures by state only for 2017. In Lenze (2020) Table 4, the BEA provides liability weighted funding ratios with BEA-adjusted liabilities by state only for 2018. I use these two BEA tables to compare the self-constructed measures in Table A.2 and Table A.3. Correlation for state level liability-weighted average BEA-adjusted funding ratios for 2018 between those calculated using data from CAFRs (171 plans) and those reported by the BEA (120 plans) is 94.2%, and correlation for state level liability-weighted average duration measures for 2017 between those calculated using data from CAFRs (174 plans) and those reported by the BEA (120 plans) is 88.0%.

## A.4 Robustness Results

### A.4.1 BEA-Adjusted Underfunding Positions

I calculate BEA-adjusted underfunding ratios with the durations at the plan-year level $d_{p,t}$ calculated using Equations 12 and 13 using the data collected from CAFRs described in Appendix Section A.2. This captures the maximum variation at the plan-year level. I impute the missing duration estimates step by step to preserve the maximum variation — In plan-year observations, where I am unable to calculate the duration using data from CAFRs, I use the average of the calculated duration estimates from CAFRs in that plan across years. In plan-year observations with duration estimates still missing because of no calculable duration estimate for a plan across years, I use the average of calculated duration estimates from CAFRs in that state. For states with missing calculable duration estimates, I use the duration estimates provided by Lenze (2020) Table 14.

Tables A.4 Panel A and A.7 Panel A report the results the average PE returns and average GP size respectively. The results are consistent and in fact, stronger than with using the reported underfunding ratios of pensions (Tables 3 and 4), i.e., more underfunded public pensions invest in, on average, smaller GPs and receive lower average PE returns relative to the least underfunded pensions. This is not surprising as the BEA-Adjusted underfunding ratios are larger than the reported underfunding ratios of pensions, and captures the true risk taking incentives. Tables A.4 Panel B and A.7 Panel B use the state-level duration measures from Lenze (2020) Table 14 to calculate BEA-Adjusted underfunding ratios, such that all plans within a state receive the liability-weighted state level average durations from 2017 reported by the BEA, that is, $d_{p,s,t} = d^{BEA}_{s,2017}$, $\forall p \in s$. The coefficient estimates are very close to those reported in Panel A, which is not surprising as Table A.3 shows that the liability-weighted averages of plan level durations for 2017 calculated with the CAFR collected data closely replicates the state-level BEA provided duration estimates. This provides a further reassurance on the CAFR collected data and correspondingly, the plan-level duration estimates.

### A.4.2 Robustness to Varying Fiscal Year End Dates

Pension plans have different fiscal end year dates. For instance, in the average PE return sample, 66.4% of the observations have fiscal year end date of June 30, 24.8% of observations have fiscal year end date of December 30/31, 5.8% of observations have a fiscal year end date of September 30 (one observation being September 20), and 1.5% and 1.6% of observations have fiscal year end dates of April 30 and August 31 respectively.

The BEA Methodology already accounts for varying fiscal year ends, however, to be further conservative, I do three more tests. First, instead of year fixed effects, I include month by year fixed effects for all fiscal year end month-years. Second, to ensure that the fiscal year end month-years with very few observations are not driving the results, I consider only plans with fiscal year end months of June and December, which together comprise 91% of observations. Similar to the first test, I include month by year fixed effects for fiscal end periods. Third, I subset the data on plans with only June ending periods, and include year fixed effects. Tables A.5 and A.8 show the results for average PE returns and average GP size, respectively. In all three tests, the results are consistent. The coefficient estimates are marginally smaller in magnitude but consistent with main Tables 3 and 4.

### A.4.3 Robustness to Varying Actuarial Cost Methods

Public pensions employ various actuarial cost methods for liability accounting. The most popular one is the Entry Age Normal (EAN) method. Other methods used are: Projected Unit Credit (PUC), Frozen Initial Liability (FIL), Aggregate Cost Method (AGG). Sometimes plans may also use a variation of these methods. Novy-Marx and Rauh (2011) includes a nice explanation of these methods. The PPD provides the actuarial cost methods for each plan-year. Generally, this field is well populated. In few cases where this is unavailable, I use the actuarial cost method used by the plan for the previous year. Next, for observations still missing, I use the actuarial cost method used by the plan for the next year.

The BEA Methodology already accounts for varying actuarial cost methods across plans, however, to be further conservative, I do one more test. The Entry Age Normal (EAN) method is the most frequently used. In the PE return sample, approximately 90% of the plans used the EAN method for at least five years in the sample period. Tables A.6 and A.9 report the results. The coefficient on underfunding ratio is negative, significant, and consistent with Tables 3 and 4, and in fact, a bit larger in magnitude for average PE returns. The coefficients are significant and consistent, though a bit noisy, if I subset plans using the EAN method in all years.

Table A.2: Liability-Weighted Average BEA-Adjusted Percent Funded comparison between self-constructed measure and Lenze (2020) Table 4 for 2018 at the State Level

| State | State Code | Percent funded in 2018 | |
|---|---|---|---|
| | | Own Caculation | Lenze (2020) |
| Alaska | AK | 44.6 | 41.2 |
| Alabama | AL | 49.1 | 44.0 |
| Arkansas | AR | 54.8 | 50.0 |
| Arizona | AZ | 47.5 | 44.0 |
| California | CA | 51.5 | 46.8 |
| Colorado | CO | 40.0 | 37.5 |
| Connecticut | CT | 32.8 | 36.6 |
| District of Columbia | DC | 71.4 | 58.6 |
| Delaware | DE | 62.1 | 56.6 |
| Florida | FL | 57.7 | 53.6 |
| Georgia | GA | 52.6 | 47.4 |
| Hawaii | HI | 37.8 | 35.5 |
| Iowa | IA | 58.8 | 54.7 |
| Idaho | ID | 64.4 | 59.1 |
| Illinois | IL | 26.8 | 29.8 |
| Indiana | IN | 46.9 | 45.1 |
| Kansas | KS | 49.6 | 42.6 |
| Kentucky | KY | 40.5 | 32.8 |
| Louisiana | LA | 48.8 | 48.1 |
| Massachusetts | MA | 39.9 | 38.7 |
| Maryland | MD | 48.7 | 45.5 |
| Maine | ME | 61.5 | 56.9 |
| Michigan | MI | 46.1 | 41.7 |
| Minnesota | MN | 55.1 | 50.7 |
| Missouri | MO | 51.9 | 48.6 |
| Mississippi | MS | 41.1 | 38.8 |
| Montana | MT | 48.1 | 43.8 |
| North Carolina | NC | 58.0 | 58.5 |
| North Dakota | ND | 45.3 | 40.1 |
| Nebraska | NE | 38.8 | 51.1 |
| New Hampshire | NH | 45.1 | 41.9 |
| New Jersey | NJ | 31.5 | 28.2 |
| New Mexico | NM | 44.7 | 42.8 |
| Nevada | NV | N/A | 45.7 |
| New York | NY | 64.3 | 61.8 |
| Ohio | OH | 51.0 | 50.2 |
| Oklahoma | OK | 57.2 | 52.4 |
| Oregon | OR | 57.0 | 51.9 |
| Pennsylvania | PA | 38.8 | 39.4 |
| Rhode Island | RI | 38.8 | 35.3 |
| South Carolina | SC | 37.2 | 33.7 |
| South Dakota | SD | 71.4 | 69.3 |
| Tennessee | TN | 68.4 | 58.6 |
| Texas | TX | 50.1 | 50.1 |
| Utah | UT | 59.1 | 58.0 |
| Virginia | VA | 54.8 | 52.3 |
| Vermont | VT | 43.6 | 41.5 |
| Washington | WA | 60.0 | 57.3 |
| Wisconsin | WI | 69.9 | 69.1 |
| West Virginia | WV | 57.1 | 53.7 |
| Wyoming | WY | 49.3 | 48.3 |
| United States | | 50.4 | 47.3 |

*Notes:* The table shows the BEA-adjusted average percent funded ratios of public pensions by state for 2018, where each pension plan's funded ratio is weighted by the liabilities of the plan, to arrive to the state level average. Column (3) estimates funded ratios using $TPL_{r\pm1}$ and $NPL_{r\pm1}$ collected from pension plan-year CAFR statements. Column (4) reproduces BEA-adjusted funding ratios from Lenze (2020) Table 4. Column (3) uses data on 171 state and local government defined pension plans. Column (4) uses data on 120 state and local government defined pension plans (Lenze (2020)). The last row shows the average BEA-adjusted funding ratio for the U.S. in 2018. Correlation between Column (3) and Column (4) at the state level is 94.2%.

Table A.3: Liability-Weighted Average Duration comparison between self-constructed measure and Lenze (2020) Table 14 for 2017 at the State Level

| State | State Code | Duration in 2017 | |
|---|---|---|---|
| | | Own Caculation | Lenze (2020) |
| Alaska | AK | 11.4 | 11.4 |
| Alabama | AL | 10.8 | 10.7 |
| Arkansas | AR | 12.5 | 12.7 |
| Arizona | AZ | 8.8 | 9.0 |
| California | CA | 13.8 | 13.7 |
| Colorado | CO | 14.4 | 13.8 |
| Connecticut | CT | 10.6 | 11.0 |
| District of Columbia | DC | 18.2 | 14.9 |
| Delaware | DE | 11.7 | 11.9 |
| Florida | FL | 13.0 | 12.7 |
| Georgia | GA | 12.7 | 12.6 |
| Hawaii | HI | 13.6 | 13.6 |
| Iowa | IA | 12.2 | 12.2 |
| Idaho | ID | 12.3 | 12.3 |
| Illinois | IL | 13.6 | 13.7 |
| Indiana | IN | 10.7 | 12.9 |
| Kansas | KS | 11.8 | 11.8 |
| Kentucky | KY | 13.9 | 13.4 |
| Louisiana | LA | 11.0 | 10.2 |
| Massachusetts | MA | 11.4 | 11.3 |
| Maryland | MD | 12.5 | 12.6 |
| Maine | ME | 12.5 | 12.5 |
| Michigan | MI | 10.5 | 11.7 |
| Minnesota | MN | 14.5 | 14.4 |
| Missouri | MO | 12.2 | 12.8 |
| Mississippi | MS | 12.0 | 12.0 |
| Montana | MT | 11.7 | 12.1 |
| North Carolina | NC | 13.4 | 11.2 |
| North Dakota | ND | 13.8 | 12.8 |
| Nebraska | NE | 13.4 | 12.6 |
| New Hampshire | NH | 11.9 | 11.8 |
| New Jersey | NJ | 13.3 | 13.3 |
| New Mexico | NM | 13.3 | 13.2 |
| Nevada | NV | N/A | 13.1 |
| New York | NY | 11.5 | 11.5 |
| Ohio | OH | 11.3 | 11.3 |
| Oklahoma | OK | 11.3 | 11.4 |
| Oregon | OR | 11.9 | 12.0 |
| Pennsylvania | PA | 10.7 | 10.8 |
| Rhode Island | RI | 11.7 | 11.8 |
| South Carolina | SC | 13.5 | 13.5 |
| South Dakota | SD | 14.3 | 14.0 |
| Tennessee | TN | 12.2 | 12.4 |
| Texas | TX | 12.5 | 12.2 |
| Utah | UT | 13.2 | 13.0 |
| Virginia | VA | 12.8 | 12.7 |
| Vermont | VT | 11.7 | 11.7 |
| Washington | WA | 15.6 | 14.0 |
| Wisconsin | WI | 10.5 | 10.7 |
| West Virginia | WV | 10.6 | 10.9 |
| Wyoming | WY | 12.1 | 11.8 |
| United States | | 12.4 | 12.5 |

*Notes:* The table shows the liability-weighted average durations of public pensions by state for 2017, where each pension plan's duration is weighted by the liabilities of the plan, to arrive to the state level average. Column (3) estimates duration measures using $TPL_{r\pm1}$ and $NPL_{r\pm1}$ collected from pension plan-year CAFR statements. Column (4) reproduces duration estimates from Lenze (2020) Table 14. Column (3) uses data on 174 state and local government defined pension plans. Column (4) uses data on 120 state and local government defined pension plans (Lenze (2020)). The last row shows the liability-weighted average duration for the U.S. in 2017. Correlation between Column (3) and Column (4) at the state level is 88.0%.

Table A.4: Correlations between BEA-Adjusted Underfunding Ratios and Average PE Returns for 2001-2021, BEA-Adjusted Underfunding Ratios constructed using data collected from CAFRs (Panel A) and Durations from Lenze (2020) Table 14 (Panel B), (Robustness for Table 3)

**Panel A:** BEA-Adjusted Underfunding Ratios constructed using data collected from the CAFRs

| | PE Return (%) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| BEA-Adjusted Underfunding Ratio | -0.118*** | -0.118*** | -0.116*** | -0.150*** |
| | (0.040) | (0.040) | (0.040) | (0.047) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1966 | 1966 | 1966 | 1771 |
| Adjusted R squared | 0.648 | 0.648 | 0.648 | 0.652 |
| Dependent Variable Mean | 0.116 | 0.116 | 0.116 | 0.119 |
| Dependent Variable Std | 0.152 | 0.152 | 0.152 | 0.148 |

**Panel B:** BEA-Adjusted Underfunding Ratios constructed using Durations from Lenze (2020) Table 14

| | PE Return (%) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| BEA-Adjusted Underfunding Ratio | -0.122*** | -0.122*** | -0.120*** | -0.157*** |
| | (0.041) | (0.041) | (0.041) | (0.047) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1966 | 1966 | 1966 | 1771 |
| Adjusted R squared | 0.648 | 0.648 | 0.648 | 0.653 |
| Dependent Variable Mean | 0.116 | 0.116 | 0.116 | 0.119 |
| Dependent Variable Std | 0.152 | 0.152 | 0.152 | 0.148 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021. The PE performance is measured using the net internal rate of return (IRR). Panel A constructs BEA-adjusted underfunding ratios by constructing duration at the plan-year level using data collected from CAFRs. Panel B constructs BEA-adjusted underfunding ratios using state-level 2017 duration estimates from Lenze (2020) Table 14. In both panels, Column (1) controls for LP size, Column (2) controls for LP size and includes current PE allocation, Column (3) controls for LP size and includes current alternative asset allocation, and Column (4) controls for LP size, includes the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table A.5: Correlations between Underfunding Ratios and Average PE Returns for 2001-2021 robust to varying fiscal end dates (Robustness for Table 3)

| | All plans | | | | Plans with June/December FY end | | | | Plans with June FY end | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Underfunding Ratio | -0.078** | -0.077** | -0.074** | -0.090** | -0.078** | -0.077** | -0.074** | -0.096** | -0.090*** | -0.090*** | -0.087*** | -0.103** |
| | (0.031) | (0.031) | (0.031) | (0.036) | (0.031) | (0.031) | (0.031) | (0.038) | (0.033) | (0.033) | (0.033) | (0.041) |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | | | Yes | | | | Yes | | |
| Current Alternatives Allocation | | | Yes | | | | Yes | | | | Yes | |
| Past Asset Allocation Controls | | | | Yes | | | | Yes | | | | Yes |
| Investment Consultant Dummies | | | | Yes | | | | Yes | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | | | | | | | | | Yes | Yes | Yes | Yes |
| FY End Month-Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | | | | |
| Observations | 1984 | 1984 | 1984 | 1782 | 1863 | 1863 | 1863 | 1684 | 1463 | 1463 | 1463 | 1323 |
| Adjusted R squared | 0.705 | 0.705 | 0.705 | 0.706 | 0.709 | 0.709 | 0.710 | 0.707 | 0.732 | 0.731 | 0.732 | 0.737 |
| Dependent Variable Mean | 0.115 | 0.115 | 0.115 | 0.118 | 0.115 | 0.115 | 0.115 | 0.119 | 0.119 | 0.119 | 0.119 | 0.122 |
| Dependent Variable Std | 0.153 | 0.153 | 0.153 | 0.148 | 0.152 | 0.152 | 0.152 | 0.148 | 0.157 | 0.157 | 0.157 | 0.154 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021 for varying fiscal year end dates. The PE performance is measured using the net internal rate of return (IRR) (%). Columns (1)-(4) include all pension plans and fiscal year end month by year fixed effects instead of year fixed effects. Columns (5)-(8) include plans ending in June or December, capturing 91% of observations, and fiscal year end month by year fixed effects instead of year fixed effects. Columns (9)-(12) consider plans with only June end and include year fixed effects. Columns (1), (5), (9) control for LP size, Columns (2), (6), (10) control for LP size and include current PE allocation, Columns (3), (7), (11) control for LP size and include current alternative asset allocation, and Columns (4), (8), (12) control for LP size, include the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table A.6: Correlations between Underfunding Ratios and Average PE Returns for 2001-2021 for plans with Entry Age Normal (EAN) actuarial cost method (Robustness for Table 3)

| | Plans with EAN used in >= 5 years | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.125** | -0.123** | -0.121** | -0.185*** |
| | (0.059) | (0.059) | (0.059) | (0.067) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1741 | 1741 | 1741 | 1563 |
| Adjusted R squared | 0.625 | 0.625 | 0.625 | 0.633 |
| Dependent Variable Mean | 0.115 | 0.115 | 0.115 | 0.119 |
| Dependent Variable Std | 0.153 | 0.153 | 0.153 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021 for plans with the Entry Age Normal (EAN) actuarial cost method. The PE performance is measured using the net internal rate of return (IRR) (%). Columns (1)-(4) include plans using the Entry Age Normal (EAN) method in at least 5 years. Column (1) controls for LP size, Column (2) controls for LP size and includes current PE allocation, Column (3) controls for LP size and includes current alternative asset allocation, and Column (4) controls for LP size, includes the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table A.7: Correlations between BEA-Adjusted Underfunding Ratios and Average GP Size for 2001-2021, BEA-Adjusted Underfunding Ratios constructed using data collected from CAFRs (Panel A) and Durations from Lenze (2020) Table 14 (Panel B), (Robustness for Table 4)

**Panel A:** BEA-Adjusted Underfunding Ratios constructed using data collected from the CAFRs

| | GP Size (Log) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| BEA-Adjusted Underfunding Ratio | -0.463* | -0.744*** | -0.529** | -0.619** |
| | (0.241) | (0.257) | (0.265) | (0.244) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1490 | 1261 | 1198 | 1027 |
| Adjusted R squared | 0.297 | 0.301 | 0.290 | 0.485 |
| Dependent Variable Mean | 8.882 | 8.911 | 8.966 | 8.998 |
| Dependent Variable Std | 1.213 | 1.203 | 1.135 | 1.118 |

**Panel B:** BEA-Adjusted Underfunding Ratios constructed using Durations from Lenze (2020) Table 14

| | GP Size (Log) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| BEA-Adjusted Underfunding Ratio | -0.531** | -0.782*** | -0.582** | -0.661*** |
| | (0.235) | (0.249) | (0.255) | (0.231) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1490 | 1261 | 1198 | 1027 |
| Adjusted R squared | 0.297 | 0.302 | 0.291 | 0.486 |
| Dependent Variable Mean | 8.882 | 8.911 | 8.966 | 8.998 |
| Dependent Variable Std | 1.213 | 1.203 | 1.135 | 1.118 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Panel A constructs BEA-adjusted underfunding ratios by constructing duration at the plan-year level using data collected from CAFRs. Panel B constructs BEA-adjusted underfunding ratios using state-level 2017 duration estimates from Lenze (2020) Table 14. In both panels, Column (1) controls for LP AUM, Column (2) controls for LP AUM and conditions on having a positive allocation to PE, Column (3) also adds LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Column (2), and Column (4) adds GP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Column (3).

Table A.8: Correlations between Underfunding Ratios and Average GP Size (Logarithmic) for 2001-2021 robust to varying fiscal end dates (Robustness for Table 4)

| | All plans | | | | Plans with June/December FY end | | | | Plans with June FY end | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Underfunding Ratio | -0.498*** | -0.614*** | -0.516*** | -0.590*** | -0.593*** | -0.730*** | -0.630*** | -0.592*** | -0.546*** | -0.680*** | -0.584*** | -0.614*** |
| | (0.176) | (0.181) | (0.187) | (0.177) | (0.178) | (0.184) | (0.189) | (0.176) | (0.192) | (0.198) | (0.205) | (0.192) |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes | | | Yes | Yes | | | Yes | Yes |
| Fund Strategy | | | | Yes | | | | Yes | | | | Yes |
| Fund Industry Focus | | | | Yes | | | | Yes | | | | Yes |
| Fund Region Focus | | | | Yes | | | | Yes | | | | Yes |
| Fund Benchmark Returns | | | | Yes | | | | Yes | | | | Yes |
| Fund Domicile | | | | Yes | | | | Yes | | | | Yes |
| Vintage Year FE | | | | | | | | | Yes | Yes | Yes | Yes |
| Pension FY End Month-Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | | | | |
| Positive PE Allocation | | Yes | Yes | Yes | | Yes | Yes | Yes | | Yes | Yes | Yes |
| Observations | 1493 | 1248 | 1176 | 1002 | 1426 | 1216 | 1144 | 994 | 1194 | 1021 | 969 | 837 |
| Adjusted R squared | 0.310 | 0.315 | 0.298 | 0.480 | 0.310 | 0.316 | 0.299 | 0.470 | 0.302 | 0.300 | 0.295 | 0.457 |
| Dependent Variable Mean | 8.875 | 8.893 | 8.956 | 8.993 | 8.865 | 8.892 | 8.956 | 8.997 | 8.928 | 8.947 | 8.995 | 9.033 |
| Dependent Variable Std | 1.218 | 1.213 | 1.143 | 1.123 | 1.216 | 1.205 | 1.131 | 1.112 | 1.190 | 1.177 | 1.126 | 1.114 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021 for varying fiscal year end dates. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Columns (1)-(4) include all pension plans and fiscal year end month by year fixed effects instead of year fixed effects. Columns (5)-(8) include plans ending in June or December, capturing 91% of observations, and fiscal year end month by year fixed effects instead of year fixed effects. Columns (9)-(12) consider plans with only June end and include year fixed effects. Columns (1), (5), (9) control for LP AUM, Columns (2), (6), (10) control for LP AUM and condition on having a positive allocation to PE, Columns (3), (7), (11) also add LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Columns (2), (6), (10), and Columns (4), (8), (12) add GP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Columns (3), (7), (11).

Table A.9: Correlations between Underfunding Ratios and Average GP Size (Logarithmic) for 2001-2021 for plans with Entry Age Normal (EAN) actuarial cost method (Robustness for Table 4)

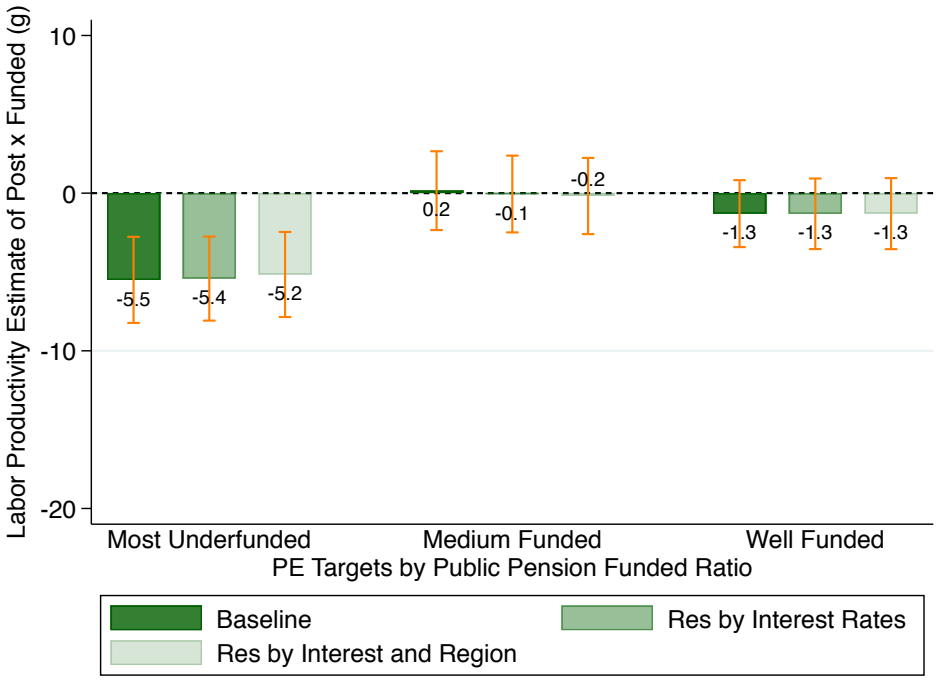
| | Plans with EAN used in >= 5 years | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.385* | -0.656*** | -0.523** | -0.566** |
| | (0.231) | (0.248) | (0.266) | (0.232) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1345 | 1116 | 1050 | 904 |
| Adjusted R squared | 0.284 | 0.285 | 0.269 | 0.480 |
| Dependent Variable Mean | 8.872 | 8.902 | 8.967 | 9.002 |
| Dependent Variable Std | 1.219 | 1.210 | 1.141 | 1.125 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021 for plans with the Entry Age Normal (EAN) actuarial cost method. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Columns (1)-(4) includes plans using the Entry Age Normal (EAN) method in at least 5 years. Column (1) controls for LP AUM, Column (2) controls for LP AUM and conditions on having a positive allocation to PE, Column (3) also adds LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Column (2), and Column (4) adds GP and LP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Column (3).

# B Supporting Figures and Tables for Real Outcomes (Section 4)

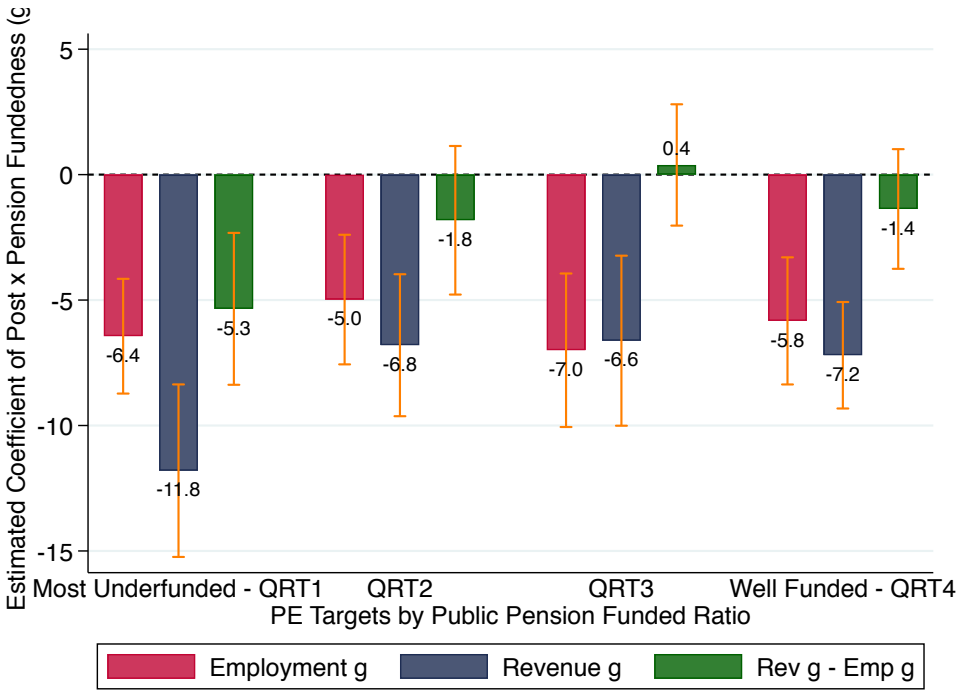
## B.1 Supporting Results by Public Pension Underfunding Ratios

Figure B.1: Labor Productivity Estimates of Post Buyout × Underfunding Ratio, Residualized for Macroeconomic Conditions, Private Equity Deals 1997-2018




*Notes:* The figure plots estimated coefficients for labor productivity yearly growth rates of Post × Pension Underfunding Split in Equation 4 for each tercile of firms. Different shades of bars correspond to underfunding ratios of pensions post *residualizing* for different macroeconomic conditions. Firms are split based on the weighted average of underfunding positions of public pensions financing the firms. Weights are capital commitments by each pension in the firm via PE funds. All specifications control for industry, size, age, and type of the firm, in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level to account for potential heterogeneity. Orange lines represent 90% confidence intervals.

Figure B.2: Estimates of Post Buyout × Quartile Split Based on Underfunding Ratios of Public Pensions, Private Equity Deals 1997-2018




*Notes:* The above figure plots estimated (total) coefficients for employment (red), revenue (blue), and labor productivity (green) yearly DHS growth rates of Post × Quartile Split based on underfunding ratios of public pensions in Equation 4 for each quartile of firms. Firms are split based on the weighted average of underfunding positions of public pensions financing the firms. Weights are capital commitments by each pension in the firm via PE funds. All specifications control for industry, size, age, and type of the firm, in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level to account for potential heterogeneity. Orange lines represent 90% confidence intervals.

Table B.1: Estimated Coefficients for Post Buyout × Quartile Split Based on Underfunding Ratios of Public Pensions, Private Equity Deals 1997-2018 (Differential Estimates Relative to Base: Least Underfunded)

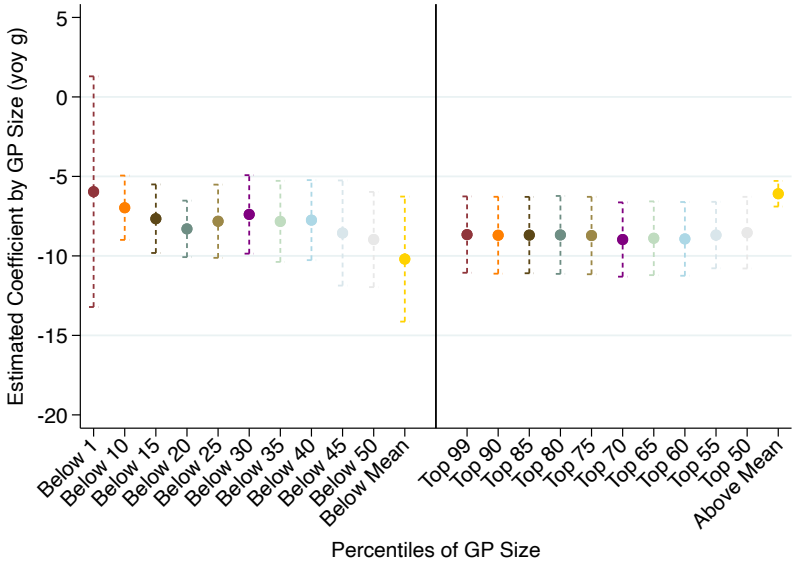
| Dependent Variable: | Emp $g$ (1) | Rev $g$ (2) | Rev $g$-Emp $g$ (3) |
|---|---|---|---|
| Post Buyout (Base: Least Underfunded Pensions) | $-0.0583^{***}$ (0.0154) | $-0.0720^{***}$ (0.0129) | $-0.0137$ (0.0145) |
| Post Buyout × Most Underfunded Pensions (Quartile 1 in Figure B.2) | $-0.00611$ (0.0163) | $-0.0459^{**}$ (0.0226) | $-0.0398^{*}$ (0.0220) |
| Post Buyout × Medium Underfunded Pensions (Quartile 2 in Figure B.2) | 0.00849 (0.0153) | 0.00402 (0.0172) | $-0.00447$ (0.0190) |
| Post Buyout × Medium Underfunded Pensions (Quartile 3 in Figure B.2) | $-0.0117$ (0.0192) | 0.00584 (0.0206) | 0.0176 (0.0183) |
| Observations | 44,500 | 44,500 | 44,500 |
| Adjusted $R^2$ | 0.2020 | 0.1460 | 0.0196 |
| Dependent Variable Mean | 0.0237 | 0.0223 | $-0.0014$ |
| Year FE | Y | Y | Y |
| Firm Size FE | Y | Y | Y |
| Firm Age FE | Y | Y | Y |
| Industry FE | Y | Y | Y |
| Type of Unit FE | Y | Y | Y |
| Lagged Firm $g$ | Y | Y | Y |
| Weighted Emp $t_0$ | Y | Y | Y |

*Notes:* The table displays coefficients $\alpha_0$ and $\beta^r$ from Equation 4. The regression consists of four categories: most underfunded, medium underfunded (QRT 2), medium underfunded (QRT 3), and least underfunded public pension financed firms. Firms are split based on the weighted average of underfunding positions of public pensions financing the firms. Weights are capital commitments by each pension in the firm via PE funds. All specifications control for industry, size, age, and type of the firm, in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are in parentheses, clustered at the firm level to account for potential heterogeneity. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

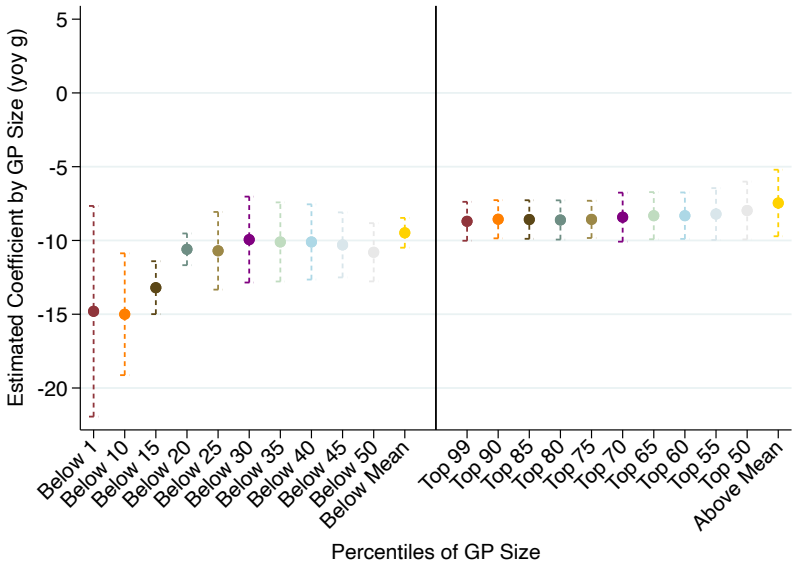
## B.2 Additional Results by GP

### B.2.1 Supplemental Figures

Figure B.3: Employment $g$ and Revenue $g$ Estimates of Post Buyout $\times$ GP Size Percentile, PE Deals 1997-2018



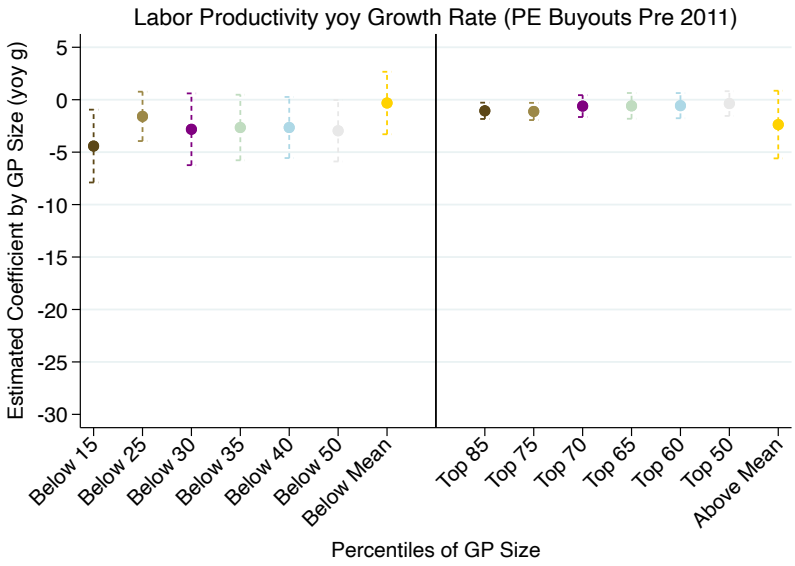

**(A)** Employment $g$



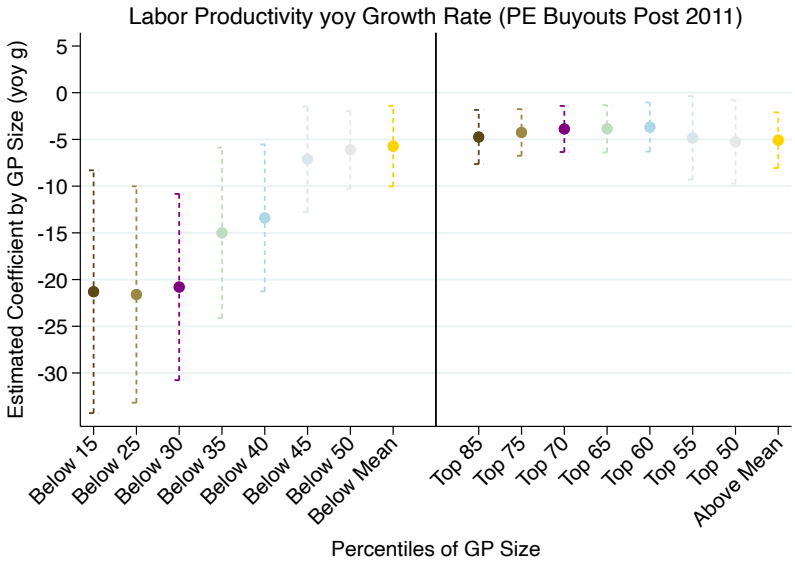

**(B)** Revenue $g$

*Notes:* The figure shows yoy growth rate estimates from Equation 4, where $\left(\text{Post}_{it} \times \mathbb{I}_i^{UIFr}\right)$ is substituted with $\left(\text{Post}_{it} \times \mathbb{I}_i^{GPx}\right)$. $\mathbb{I}_i^{GPx}$ is an indicator variable which takes the value 1 for target firms financed by GPs of size less than or equal to $x$th percentile. I estimate the regression for different percentiles: $x = 1, 10, 15, 20, 25, 30, 35, 40, 45, 50,$ mean, where each color shows estimates from a different percentile cut. Panel A shows employment growth rates, and Panel B shows revenue growth rates. These are total coefficient estimates for each category. All specifications control for industry, size, age, and type of the firm, in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level. Bars show 90% confidence intervals.

Figure B.4: Labor Productivity $g$ Estimates of Post Buyout $\times$ GP Size Percentile over time, PE Deals 1997-2018



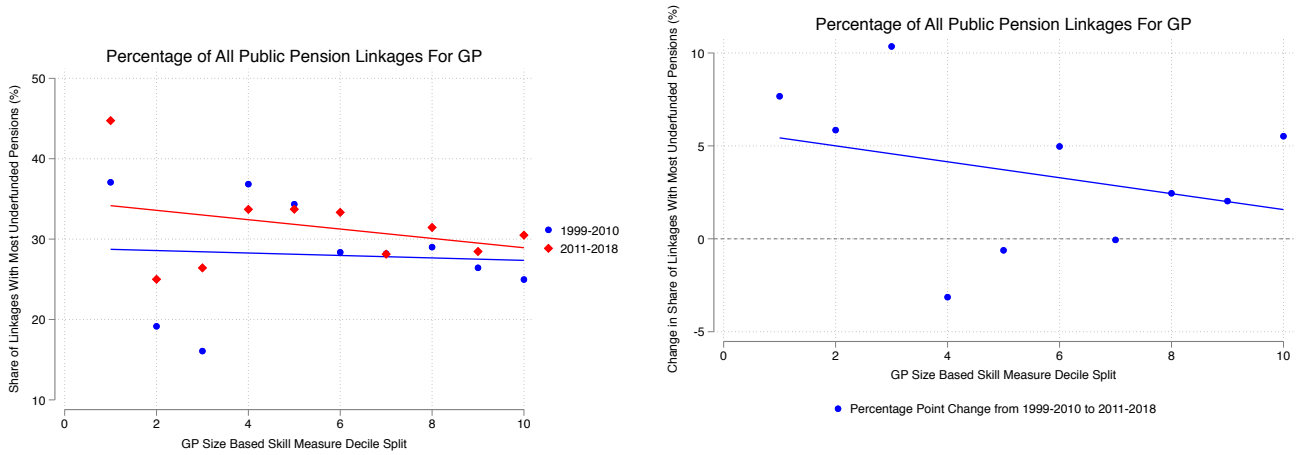

**(A)** Pre-2011




**(B)** Post-2011

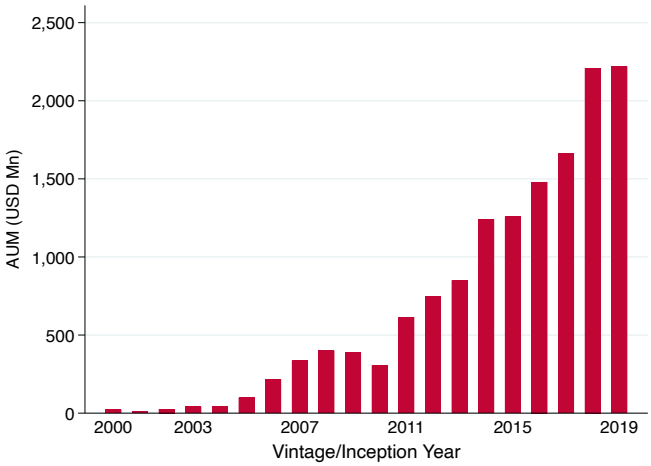
*Notes:* The figure shows labor productivity yoy growth rate estimates from Equation 4, where $(\text{Post}_{it} \times \mathbb{I}_i^{UF^r})$ is substituted with $(\text{Post}_{it} \times \mathbb{I}_i^{GP^x})$. $\mathbb{I}_i^{GP^x}$ is an indicator variable which takes the value 1 for target firms financed by GPs of size less than or equal to $x$th percentile. I estimate the regression for different percentiles: $x = 15, 25, 30, 35, 40, 45, 50$, mean, where each color shows estimates from a different percentile cut. Note that the size cut offs are less than the previous figure, due to U.S. Census disclosure requirements. I use the market-based size measure. The ranking of GPs in terms of size remains consistent over time. Panel A considers deals pre-2011, and Panel B considers deals post-2011. These are total coefficient estimates for each category. All specifications control for industry, size, age, and type of the firm, in addition to year fixed effects. Regressions control for pre-buyout firm growth. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level. Bars show 90% confidence intervals.

Figure B.5: Percent of Investment Linkages Between the Most Underfunded Public Pensions and GP Size Across Time




*Notes:* This figure shows the percentage of investment linkages, measured as the number of capital commitments by a public pension LP to a PE fund, between the most underfunded public pension tercile (as a share of all public pension commitments) and the GPs, split into deciles based on AUM, over time. Panel A shows this for two time periods: (1) 1999-2010 and (2) 2011-2018. Panel B depicts the change over time. The year of commitment is the vintage year of the PE fund. The underfunding ratio of pension is considered at the time of capital commitment to the PE fund. Data are sourced from Preqin.

Figure B.6: Capital Inflow: Average Yearly Total AUM Raised by GPs Across PE Funds




*Notes:* The vintage year or the inception year is the year the fund is set up in. The figure shows the average of total AUM raised by GPs across all their PE funds by vintage year. Data are sourced from Preqin.

### B.2.2 Book Value Measure of GP size in PE

I construct the book value measure of GP size in private equity, interpreted broadly for fund quality, using the total capital raised by the fund. Assets of a GP $\mathcal{J}$ in private equity is the sum of assets of its component funds $j$ existing in that year.

$$\text{Assets}_{\mathcal{J},t} = \sum_{j \in \mathcal{J}} \text{Assets}_{j,t} \tag{14}$$

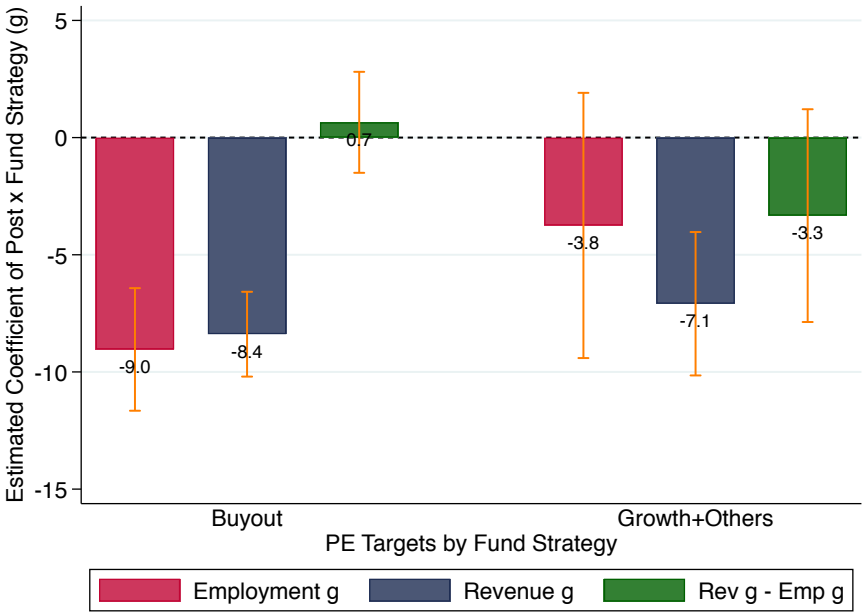
I use the lifespan of the fund to determine its start and end period. The lifespan is the fund duration including first 1-2 years of capital commitments, next 5-6 years of investment, followed by 1-2 years of liquidation. The median lifespan of the funds in the sample is 10 years, similar as suggested in Kaplan and Strömberg (2009). I consider the median when the fund lifespan is not available. If the time period of the fund is given in half years, I round up to the next year. To measure the accurate significance of a GP in the private equity industry, I consider all private equity funds existing in that year. The ranking of private equity GPs based on the market value size measure and the book value size measure is consistent. The ranking of the top GPs is also consistent over time. For the regression analyses, GP size is aggregated at the pension-year level by taking the average across the GPs invested in by the pension for that year.

### B.2.3 PE Fund Heterogeneity based on Fund Strategy

In the sample, most of the funds are regular balanced buyout funds. This is not surprising as the focus of the study is PE buyout. However, different funds within a GP can vary by granular strategies. While most of the granular fund strategies are also balanced buyout, a small percentage are growth, fund of funds, turnaround, multi-strategy etc. As a robustness check, I study differences on target firms based on granular fund strategies. The target firms are classified as “buyout" if all funds financing the firm are balanced buyout. To be extremely conservative, I define a target as financed by “growth+others" if at least one of the funds financing the firm is a growth or other fund.

Figure B.7 shows employment, revenue, and labor productivity year over year growth rates post buyout for target firms financed by the PE buyout category and by the PE “growth+others" category. Target firms financed by “growth+others" funds experience an insignificant decrease in labor productivity post buyout. The standard errors are large for “growth+others" as majority of the funds are balanced buyout strategy funds.

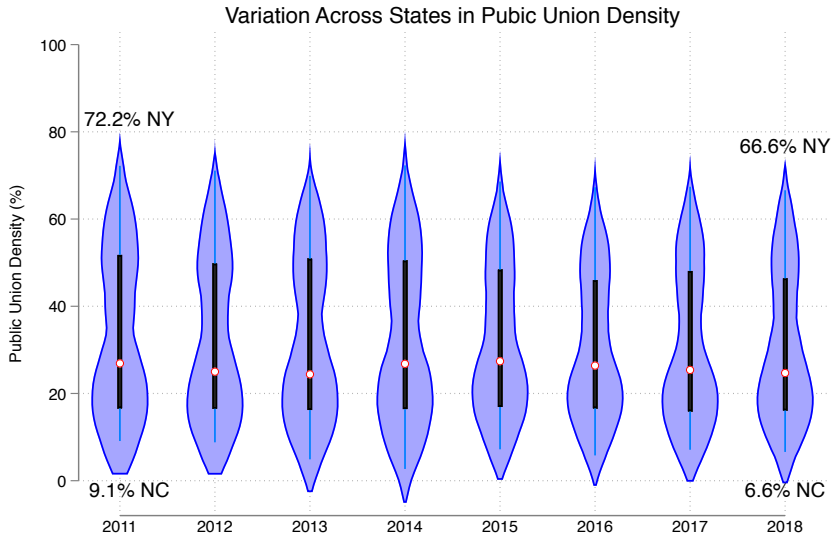
Figure B.7: Estimated Coefficients of Post Buyout × Fund Strategy Relative to Pre-Buyout, PE Deals from 1997-2018




*Notes:* The figure plots estimated coefficients for employment (red), revenue (blue), and labor productivity (green) growth rates of the (Post × Fund Strategy) version of Equation 4. "Buyout" includes target firms financed by only balanced buyout strategy funds. "Growth+Others" includes target firms financed by at least one of growth, multi-strategy, and other funds. Standard errors are clustered at the firm level. Orange lines show 90% confidence intervals.

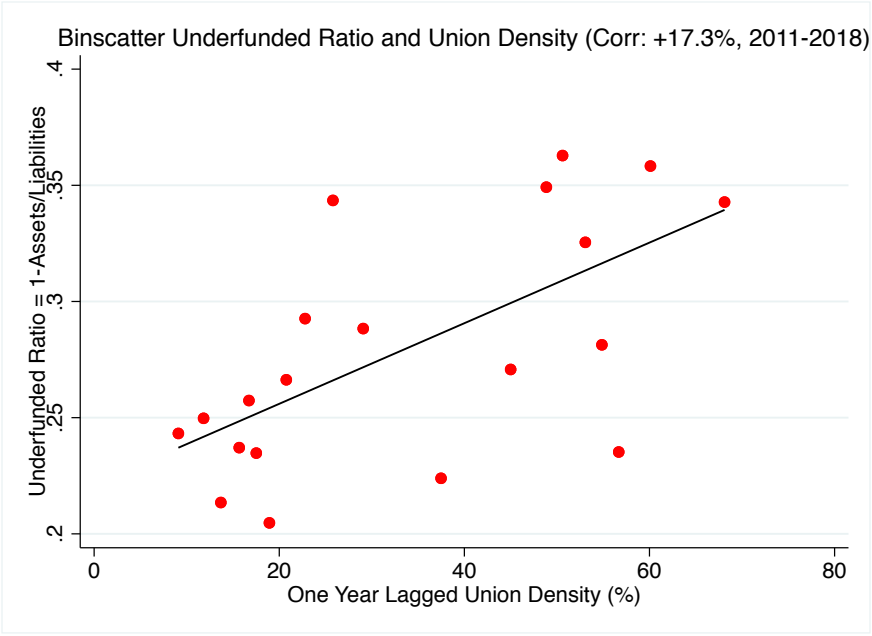
# C Supporting Figures and Tables for Mechanism (Section 5)

Figure C.1: Variation in Public Union Density Across States, 2011-2018




*Notes:* Figure shows variation in public union density across states over time. Public union density is defined as the percentage of public workers which are part of a union. Dispersion is similar for years not reported. Data are sourced from CPS and Union Stats.

Figure C.2: Correlation Between Underfunding Ratio and One Year Lag Public Union Density




*Notes:* Figure plots a binscatter of underfunding ratio against one year lag of union density among public workers. Public union density is defined as the percentage of public workers which are part of a union. Underfunding ratio is one minus assets divided by liabilities for public pension plans. Figure uses the time period 2011-2018, correlation is positive +17.3% and significant. For the time period 1997-2018, correlation is positive +6.3% and significant. Balance sheet fundamentals of public pensions are sourced from Public Plans Data and FOIA requests. Union density is sourced from CPS and Union Stats.

Table C.1: Correlations between Underfunding Ratios and Average PE Returns for 2001-2021 excluding plans with ex officio participant board members, generally union representatives (Robustness for Table 3 and extension of Table 9)

| | Plans without ex officio participant board members (generally union representatives) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.101*** | -0.100*** | -0.099*** | -0.134*** |
| | (0.032) | (0.032) | (0.032) | (0.036) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1946 | 1946 | 1946 | 1750 |
| Adjusted R squared | 0.651 | 0.651 | 0.651 | 0.654 |
| Dependent Variable Mean | 0.115 | 0.115 | 0.115 | 0.118 |
| Dependent Variable Std | 0.154 | 0.154 | 0.154 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021 *excluding* plans with ex officio participant board members, which are generally union representatives (Andonov et al. (2018)). The plans excluded are: Illinois SERS, Louisiana Schools, New York City ERS, New York City Fire, New York City Police, and Oklahoma Fire. The PE performance is measured using the net internal rate of return (IRR) (%). Column (1) controls for LP size, Column (2) controls for LP size and includes current PE allocation, Column (3) controls for LP size and includes current alternative asset allocation, and Column (4) controls for LP size, includes the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table C.2: Correlations between Underfunding Ratios and Average GP Size (Logarithmic) for 2001-2021 excluding plans with ex officio participant board members, generally union representatives (Robustness for Table 4 and extension of Table 10)

| | Plans without ex officio participant board members (generally union representatives) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.530*** | -0.731*** | -0.610*** | -0.578*** |
| | (0.185) | (0.192) | (0.199) | (0.176) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1434 | 1218 | 1147 | 991 |
| Adjusted R squared | 0.293 | 0.301 | 0.287 | 0.481 |
| Dependent Variable Mean | 8.869 | 8.893 | 8.957 | 8.985 |
| Dependent Variable Std | 1.228 | 1.217 | 1.146 | 1.125 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021 *excluding* plans with ex officio participant board members, which are generally union representatives (Andonov et al. (2018)). The plans excluded are: Illinois SERS, Louisiana Schools, New York City ERS, New York City Fire, New York City Police, and Oklahoma Fire. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Column (1) controls for LP AUM, Column (2) controls for LP AUM and conditions on having a positive allocation to PE, Column (3) also adds LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Column (2), and Column (4) adds GP and LP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Column (3).

Table C.3: Correlations between Underfunding Ratios and Average PE Returns for 2001-2021 robust to excluding major states
(Robustness for Table 3 and extension of Table 11)

| | Excluding California | | | | Excluding Texas | | | | Excluding Massachusetts | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Underfunding Ratio | -0.127*** | -0.127*** | -0.125*** | -0.143*** | -0.097*** | -0.097*** | -0.096*** | -0.124*** | -0.104*** | -0.104*** | -0.102*** | -0.137*** |
| | (0.032) | (0.032) | (0.032) | (0.040) | (0.032) | (0.032) | (0.032) | (0.036) | (0.032) | (0.032) | (0.032) | (0.036) |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | | | Yes | | | | Yes | | |
| Current Alternatives Allocation | | | Yes | | | | Yes | | | | Yes | |
| Past Asset Allocation Controls | | | | Yes | | | | Yes | | | | Yes |
| Investment Consultant Dummies | | | | Yes | | | | Yes | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 1727 | 1727 | 1727 | 1531 | 1900 | 1900 | 1900 | 1715 | 1942 | 1942 | 1942 | 1753 |
| Adjusted R squared | 0.628 | 0.628 | 0.628 | 0.626 | 0.658 | 0.658 | 0.658 | 0.666 | 0.642 | 0.642 | 0.642 | 0.650 |
| Dependent Variable Mean | 0.116 | 0.116 | 0.116 | 0.119 | 0.116 | 0.116 | 0.116 | 0.119 | 0.114 | 0.114 | 0.114 | 0.117 |
| Dependent Variable Std | 0.152 | 0.152 | 0.152 | 0.147 | 0.152 | 0.152 | 0.152 | 0.147 | 0.152 | 0.152 | 0.152 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021 when excluding major states. The PE performance is measured using the net internal rate of return (IRR) (%). Columns (1)-(4) exclude California. Columns (5)-(8) exclude Texas. Columns (9)-(12) exclude Massachusetts. Columns (1), (5), (9) control for LP size, Columns (2), (6), (10) control for LP size and include current PE allocation, Columns (3), (7), (11) control for LP size and include current alternative asset allocation, and Columns (4), (8), (12) control for LP size, include the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table C.4: Correlations between Underfunding Ratios and Average PE Returns for 2001-2021 with State Political Board Members $< 50\%$ (Robustness for Table 3 and extension of Table 11)

| | Total % of Political State Board Members $< 50\%$ | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.130** | -0.127** | -0.130** | -0.156** |
| | (0.063) | (0.063) | (0.063) | (0.069) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1464 | 1464 | 1464 | 1366 |
| Adjusted R squared | 0.550 | 0.550 | 0.550 | 0.538 |
| Dependent Variable Mean | 0.100 | 0.100 | 0.100 | 0.102 |
| Dependent Variable Std | 0.136 | 0.136 | 0.136 | 0.133 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021 with state political board members $< 50\%$, using data from Andonov et al. (2018). State political board members are comprised of state appointed and state ex officio board members. The PE performance is measured using the net internal rate of return (IRR) (%). Column (1) controls for LP size, Column (2) controls for LP size and includes current PE allocation, Column (3) controls for LP size and includes current alternative asset allocation, and Column (4) controls for LP size, includes the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table C.5: Correlations between Underfunding Ratios and Average PE Returns for 2001-2021 excluding states with $>$ 20% in-state investments: California, Illinois, Massachusetts, New York (Robustness for Table 3 and extension of Table 11)

| | Excluding States with In-State Investments $>$ 20%: CA, IL, MA, NY | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.131*** | -0.132*** | -0.130*** | -0.148*** |
| | (0.034) | (0.034) | (0.034) | (0.042) |
| LP AUM | Yes | Yes | Yes | Yes |
| Current PE Allocation | | Yes | | |
| Current Alternatives Allocation | | | Yes | |
| Past Asset Allocation Controls | | | | Yes |
| Investment Consultant Dummies | | | | Yes |
| Pension Fund FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 1516 | 1516 | 1516 | 1347 |
| Adjusted R squared | 0.607 | 0.607 | 0.607 | 0.615 |
| Dependent Variable Mean | 0.114 | 0.114 | 0.114 | 0.117 |
| Dependent Variable Std | 0.152 | 0.152 | 0.152 | 0.149 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average PE returns for 2001-2021 excluding states with $>$ 20% in-state investments across alternative assets: California, Illinois, Massachusetts, New York, as defined by Andonov et al. (2018), Online Appendix Table IAII corresponding to the 1990-2011 period. As investments are sticky and long dated, it is reasonable to assume that the cross-sectional trend across states persists beyond 2011 as well. The PE performance is measured using the net internal rate of return (IRR) (%). Column (1) controls for LP size, Column (2) controls for LP size and includes current PE allocation, Column (3) controls for LP size and includes current alternative asset allocation, and Column (4) controls for LP size, includes the average past three-year PE, equity, fixed income allocations, and investment consultant dummies.

Table C.6: Correlations between Underfunding Ratios and Average GP Size (Logarithmic) for 2001-2021 robust to excluding major states (Robustness for Table 4 and extension of Table 12)

| | Excluding California | | | | Excluding Texas | | | | Excluding Massachusetts | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
| Underfunding Ratio | -0.387** | -0.561*** | -0.454** | -0.533*** | -0.449** | -0.610*** | -0.533*** | -0.542*** | -0.419** | -0.596*** | -0.477** | -0.517*** |
| | (0.193) | (0.203) | (0.209) | (0.183) | (0.178) | (0.184) | (0.191) | (0.171) | (0.177) | (0.184) | (0.188) | (0.167) |
| LP AUM | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes | | | Yes | Yes | | | Yes | Yes |
| Fund Strategy | | | | Yes | | | | Yes | | | | Yes |
| Fund Industry Focus | | | | Yes | | | | Yes | | | | Yes |
| Fund Region Focus | | | | Yes | | | | Yes | | | | Yes |
| Fund Benchmark Returns | | | | Yes | | | | Yes | | | | Yes |
| Fund Domicile | | | | Yes | | | | Yes | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes | | Yes | Yes | Yes | | Yes | Yes | Yes |
| Observations | 1283 | 1070 | 1000 | 844 | 1384 | 1184 | 1115 | 949 | 1493 | 1264 | 1199 | 1028 |
| Adjusted R squared | 0.292 | 0.294 | 0.276 | 0.488 | 0.316 | 0.323 | 0.309 | 0.480 | 0.298 | 0.302 | 0.292 | 0.486 |
| Dependent Variable Mean | 8.850 | 8.866 | 8.937 | 8.978 | 8.880 | 8.900 | 8.965 | 8.996 | 8.882 | 8.910 | 8.965 | 8.999 |
| Dependent Variable Std | 1.252 | 1.255 | 1.178 | 1.159 | 1.201 | 1.194 | 1.126 | 1.110 | 1.215 | 1.205 | 1.137 | 1.118 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021 when excluding major states. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Columns (1)-(4) exclude California. Columns (5)-(8) exclude Texas. Columns (9)-(12) exclude Massachusetts. Columns (1), (5), (9) control for LP AUM, Columns (2), (6), (10) control for LP AUM and condition on having a positive allocation to PE, Columns (3), (7), (11) also add LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Columns (2), (6), (10), and Columns (4), (8), (12) add additional GP and LP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Columns (3), (7), (11).

Table C.7: Correlations between Underfunding Ratios and Average GP Size (Logarithmic) for 2001-2021 with State Political Board Members $<$ 50% (Robustness for Table 4 and extension of Table 12)

| | Total % of Political State Board Members $<$ 50% | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.364* | -0.594*** | -0.441* | -0.445** |
| | (0.217) | (0.229) | (0.231) | (0.204) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1172 | 983 | 947 | 843 |
| Adjusted R squared | 0.257 | 0.263 | 0.260 | 0.490 |
| Dependent Variable Mean | 8.864 | 8.900 | 8.942 | 8.988 |
| Dependent Variable Std | 1.192 | 1.177 | 1.129 | 1.108 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021 with state political board members $<$ 50%, using data from Andonov et al. (2018). State political board members are comprised of state appointed and state ex officio board members. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Column (1) controls for LP AUM, Column (2) controls for LP AUM and conditions on having a positive allocation to PE, Column (3) also adds LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Column (2), and Column (4) adds GP and LP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Column (3).

Table C.8: Correlations between Underfunding Ratios and Average GP Size (Logarithmic) for 2001-2021 excluding states with $>$ 20% in-state investments: California, Illinois, Massachusetts, New York (Robustness for Table 4 and extension of Table 12)

| | Excluding States with In-State Investments $>$ 20%: CA, IL, MA, NY | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Underfunding Ratio | -0.417* | -0.602** | -0.451* | -0.576** |
| | (0.238) | (0.254) | (0.268) | (0.225) |
| LP AUM | Yes | Yes | Yes | Yes |
| Past Asset Allocations | | | Yes | Yes |
| Fund Strategy | | | | Yes |
| Fund Industry Focus | | | | Yes |
| Fund Region Focus | | | | Yes |
| Fund Benchmark Returns | | | | Yes |
| Fund Domicile | | | | Yes |
| Vintage Year FE | Yes | Yes | Yes | Yes |
| Positive PE Allocation | | Yes | Yes | Yes |
| Observations | 1056 | 877 | 827 | 717 |
| Adjusted R squared | 0.270 | 0.262 | 0.245 | 0.484 |
| Dependent Variable Mean | 8.897 | 8.918 | 8.974 | 9.000 |
| Dependent Variable Std | 1.265 | 1.258 | 1.188 | 1.165 |

*Notes:* Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Robust standard errors in parentheses. The table shows the relation between underfunding ratios of public pensions and average GP size for 2001-2021 excluding states with $>$ 20% in-state investments across alternative assets: California, Illinois, Massachusetts, New York, as defined by Andonov et al. (2018), Online Appendix Table IAII corresponding to the 1990-2011 period. As investments are sticky and long dated, it is reasonable to assume that the cross-sectional trend across states persists beyond 2011 as well. GP size cumulates sizes of constituent PE funds existing in each year for each GP. In the regression analyses, GP Size is in logarithmic terms. Column (1) controls for LP AUM, Column (2) controls for LP AUM and conditions on having a positive allocation to PE, Column (3) also adds LP's past allocations as controls (average past three-year PE, equity, fixed income allocations) to Column (2), and Column (4) adds GP and LP characteristics (fund strategy, industry and region focus, benchmark returns, and fund domicile) to Column (3).

# D Additional Aggregate Results

## D.1 Difference in Differences Estimation

I compare all target firms in the sample (treated firms) with a comparable group of firms in the U.S., which have never been under PE influence (control firms), based on a tight granular matching following Davis et al. (2014) using U.S. Census micro data. It would be ideal to have an exogenous shock to capital flow to private equity funds across all industries. However, in the absence of that, the literature has relied on constructing control firms using a matched sample.

The difference in differences empirical specification takes the following form,

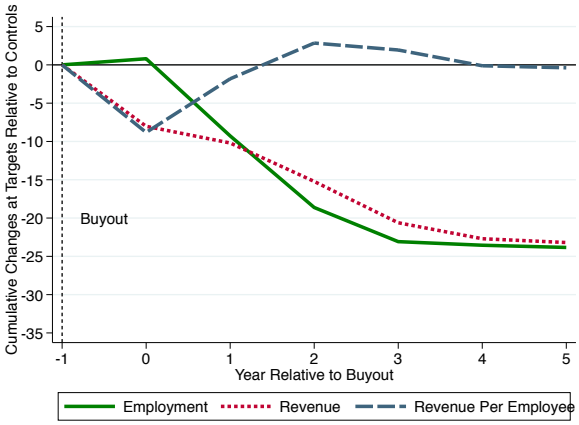
$$y_{it} = \alpha_t + \sum_{j=-5, j\neq -1}^{j=+5} \gamma_j \left(\text{PE}_i \times \text{Buyout Year}_{it_0+j}\right) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it} \tag{15}$$

where, $\gamma_j$ represents the coefficient of interest. The control firms consist of active firms in the buyout transaction year, which are in the same industry, firm size, firm age, and single-/multi-unit status group as the target firms, but are not bought by private equity during their entire history. The difference in differences design does not suffer from bias as in settings of staggered treatments (Goodman-Bacon (2021), Sun and Abraham (2021), Athey and Imbens (2022)), as the control group consists of firms never bought by PE. The control firms are constructed based on the cross product of the following categories: firm size categories are 1-4, 5-9, 10-19, 20-49, 50-99, 100-249, 250-499, 500-999, 1,000-2,499, 2,500-4,999, 5,000-9,999, and greater than 10,000 employees; firm age categories are 0-5, 6-10, 11-15, 16-20, and 21 or more years; 22 industries are defined based on two-digit NAICS codes; a dummy for single-/multi-unit status, and the year of buyout transaction. I use NAICS code because of good coverage in the Revenue Enhanced LBD. The regression is saturated with 5,600 dummies $D_{cit}$ capturing industry $\times$ size $\times$ age $\times$ type $\times$ buyout year groups. The regression controls for lagged firm growth from $t_0 - 3$ to $t_0 - 1$, denoted as $\text{LFIRM}_i$. $\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the control firms. $\text{Buyout Year}_{i,t_0+j}$ is a dummy for each $j$ taking a value of 1 in the year $t_0 \pm j$ relative to buyout year, with $j = -5, \cdots, 5$. The year before buyout $t_0 - 1$ is the omitted category, and years beyond 5 years pre- and post-buyout are binned with year +/-5 relative to buyout. To capture the relative business significance of entities, the empirical specification is weighted by employment at the time of buyout. Appendix D.2 compares raw data patterns in treated and control firms with prior research as a sanity check. As the control group exceeds the treated group, comprising of most firms in the U.S. economy,

I run into computing constraints. Following standard approach while using U.S. Census micro-data, I select a 10% random sample from the universe of control firms for each group. Appendix D.3 shows robustness to different random samples, and Appendix D.4 shows robustness with additional controls.

I track employment, revenue, and revenue minus employment, all in terms of growth rates, for treated and control firms five years pre- and post-buyout. Figure 5 Panels A-C, in the main text, show year over year growth rates five years pre- and post-buyout for employment, revenue, and labor productivity. Figure D.1 uses those estimates from Figure 5 to show the cumulative changes.

Figure D.1: Cumulative Changes of Outcome Variables for Target Firms Relative to Control Firms Over Time based on Difference in Differences Estimated Coefficients, PE Deals 1997-2018



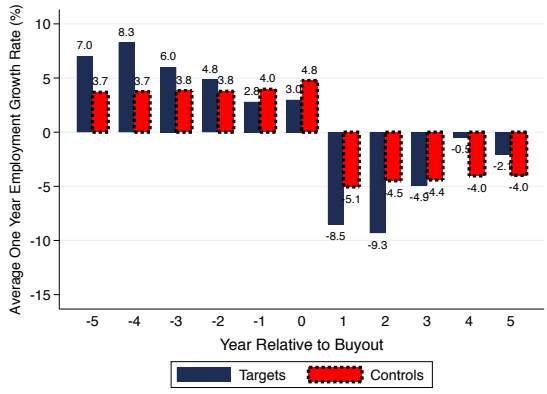

*Notes:* This figure cumulates difference in differences estimated coefficients $\gamma_j$ from equation: $y_{it} = \alpha_t + \sum_{j=-5, j\neq -1}^{j=+5} \gamma_j (\text{PE}_i \times \text{Buyout Year}_{it_0+j}) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it}$ for employment, revenue, and revenue minus employment. The cumulative changes are from estimates in Figure 5 Panels A-C, and normalized to 0 in year -1 relative to buyout. The transaction takes place between year -1 and 0. $\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the control firms. $\text{Buyout Year}_{i,t_0+j}$ is a dummy for each $j$ taking a value of 1 in the year $t_0 \pm j$ relative to buyout year, with $j = -5, \cdots, 5$. $D_{cit}$ are dummies for a fully saturated interacted fixed effects of firm characteristics: industry, age, size, type of unit, and buyout year. $\text{LFIRM}_i$ is the lagged firm growth from $t_0 - 3$ to $t_0 - 1$. The year before buyout $t_0 - 1$ is the omitted category, and years beyond 5 years pre- and post-buyout are binned with year +/-5 relative to buyout. Standard errors are clustered at the firm level, for potential heterogeneity.

## D.2 Non-Parametric Comparison

Here, I present a non-parametric comparison of growth rates in target firms and control firms five years before and after the private equity transaction. Figure D.2 Panel A shows year over year growth rates for employment in this sample for deals from 1997 to 2018. It is seen that firms in the control group also shrink post buyout, but less than the target firms. This is not surprising as the control group is constructed on an extremely tight granular matched sample approach. The industries and types of firms targeted by PE are those which require substantial restructuring. Regression estimates are capturing the difference between treated and control firms. Prior literature (see Davis et al. (2014) in the American

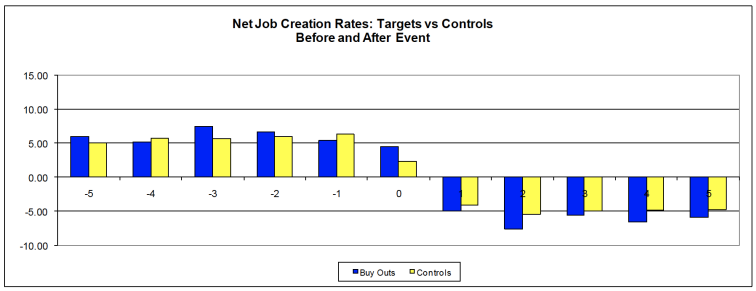
Economic Review online appendix, reproduced below in Panel B) finds a similar pattern of employment growth rates for target and control firms for PE deals from 1980 to 2000. This presents a sanity check for the data patterns being captured in the treated and control firms in this sample, as I am borrowing the methodology. Revenue was not introduced in the past versions of the LBD.

Figure D.2: Employment Growth Rates (yoy) at U.S. Target and Control Firms Compared with Earlier Literature




**(A)** My sample, deals 1997-2018

**Figure C.1: Employment Growth Rates before and after the Buyout Year, Targets and Controls Compared, Buyouts from 1980 to 2000**




**(B)** Reproduced from Davis et al. (2014) Online Appendix published in the AER

*Notes:* Panel A plots year over year employment growth rates in target firms and control firms five years pre- and post-buyout for private equity deals from 1997-2018 in the sample. Blue bars represent target firms and red (dotted) bars represent control firms. Panel B reproduces the Figure C.1 from Online Appendix of Davis et al. (2014) published in the American Economic Review, which shows employment growth rates at buyouts and controls for 1980-2000. Seeing similar patterns is a good sanity check, as I am borrowing the methodology.

## D.3 Robustness: Different Random Samples using U.S. Census Bureau Micro-Data

Table D.1: Difference in Differences Estimated Coefficients, PE Deals 1997-2018

| Dependent Variable: | Emp $g$ (1) | Pay $g$ (2) | Rev $g$ (3) | Pay $g$-Emp $g$ (4) | Rev $g$-Emp $g$ (5) |
|---|---|---|---|---|---|
| | | | *Random Sample 2* | | |
| Treatment × Post Buyout | $-0.0247^{***}$ | $-0.0243^{***}$ | $-0.0269^{***}$ | 0.0004 | $-0.0022$ |
| | (0.0099) | (0.0105) | (0.0081) | (0.0029) | (0.0079) |
| Observations | 25,440,000 | 25,440,000 | 25,440,000 | 25,440,000 | 25,440,000 |
| Adjusted $R^2$ | 0.0395 | 0.0561 | 0.0454 | 0.0101 | 0.0076 |
| | | | *Random Sample 3* | | |
| Treatment × Post Buyout | $-0.0266^{***}$ | $-0.0241^{**}$ | $-0.0281^{***}$ | 0.0025 | $-0.0015$ |
| | (0.0103) | (0.0106) | (0.0078) | (0.0026) | (0.0079) |
| Observations | 25,430,000 | 25,430,000 | 25,430,000 | 25,430,000 | 25,430,000 |
| Adjusted $R^2$ | 0.0362 | 0.0524 | 0.0422 | 0.0094 | 0.0081 |
| | | | *Random Sample 4* | | |
| Treatment × Post Buyout | $-0.0223^{***}$ | $-0.0211^{**}$ | $-0.0254^{***}$ | 0.0013 | $-0.0031$ |
| | (0.0084) | (0.0089) | (0.0077) | (0.0030) | (0.0075) |
| Observations | 25,440,000 | 25,440,000 | 25,440,000 | 25,440,000 | 25,440,000 |
| Adjusted $R^2$ | 0.0368 | 0.0541 | 0.0401 | 0.0096 | 0.0076 |
| | | | *Random Sample 5* | | |
| Treatment × Post Buyout | $-0.0290^{***}$ | $-0.0253^{***}$ | $-0.0293^{***}$ | 0.0037 | $-0.0003$ |
| | (0.0086) | (0.0091) | (0.0069) | (0.0029) | (0.0076) |
| Observations | 25,450,000 | 25,450,000 | 25,450,000 | 25,450,000 | 25,450,000 |
| Adjusted $R^2$ | 0.0378 | 0.0529 | 0.0415 | 0.0078 | 0.0077 |
| Industry × Age × Size × Type × Transaction Year FE | Y | Y | Y | Y | Y |
| Year FE | Y | Y | Y | Y | Y |
| Lagged Firm $g$ | Y | Y | Y | Y | Y |
| Weighted Emp $t_0$ | Y | Y | Y | Y | Y |

*Notes:* The table displays coefficients $\gamma$ of the difference in differences Equation:

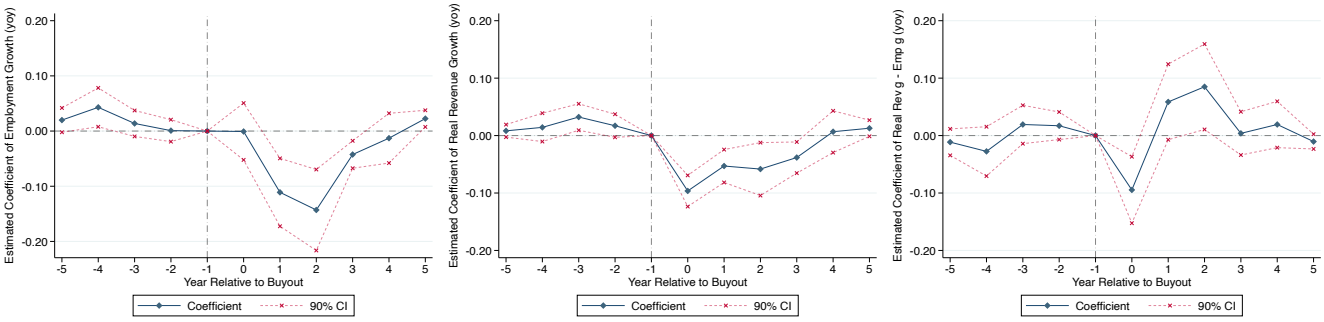
$$y_{it} = \alpha_t + \gamma(\text{PE}_i \times \text{Post}_{it}) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it}$$

$\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the control firms. $\text{Post}_{it}$ is a dummy taking a value of 1 in the years following the buyout year. $D_{cit}$ are dummies for a fully saturated interacted fixed effects of firm characteristics: industry, age, size, type of unit, and buyout year. $\text{LFIRM}_i$ is the lagged firm growth from $t_0 - 3$ to $t_0 - 1$. The year before buyout $t_0 - 1$ is the omitted category. Regression estimates are weighted by employment in buyout year $t_0$. For robustness, regressions are also weighted by employment in year $t_0 - 3$ relative to buyout, and give similar results (not reported). Standard errors are clustered at the firm level to account for potential heterogeneity. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## D.4 Using Additional Controls

In addition to the lagged firm growth from period $t_0 - 3$ to $t_0 - 1$ relative to buyout used in Figure 5, I also control for lagged one year revenue controls for revenue growth effects. Figure D.3 shows similar results for both versions of the difference in differences estimation.

Figure D.3: Difference in Differences Estimated Coefficients $\gamma_j$ Over Time Relative to Buyout Year, PE Deals 1997-2018




**(A)** Employment $g$ **(B)** Revenue $g$ **(C)** Rev $g$ - Emp $g$

| Dependent Variable: | Emp $g$ (1) | Rev $g$ (2) | Rev $g$-Emp $g$ (3) |
|---|---|---|---|
| Treatment × Post Buyout | −0.0288* | −0.0295*** | −0.0009 |
| | (0.0152) | (0.0099) | (0.0123) |
| Industry × Age × Size × Type × Transaction Year FE | Y | Y | Y |
| Year FE | Y | Y | Y |
| Controls | Y | Y | Y |
| Weighted Emp $t_0$ | Y | Y | Y |
| Observations | 19,030,000 | 19,030,000 | 19,030,000 |
| Adjusted $R^2$ | 0.0407 | 0.0556 | 0.0080 |
| Dependent Variable Mean | 0.0202 | 0.0251 | 0.0049 |

**(D)** Long Run

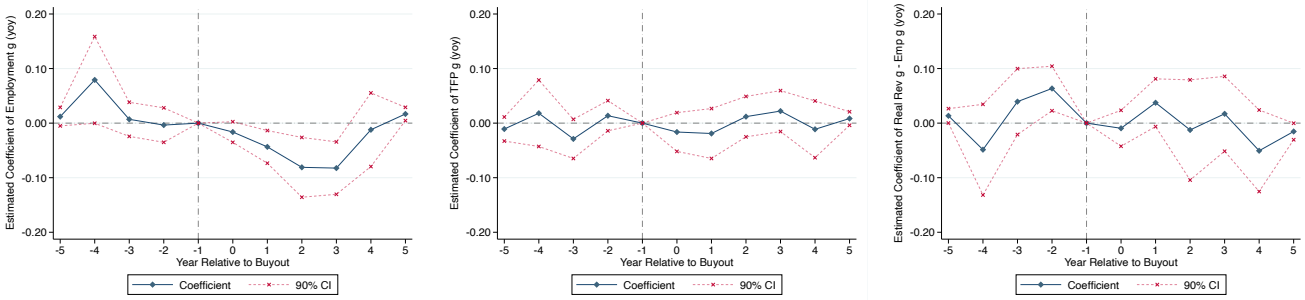
*Notes:* Panels A, B, and C show coefficients $\gamma_j$ of the difference in difference Equation:

$$y_{it} = \alpha_t + \sum_{j=-5, j\neq -1}^{j=+5} \gamma_j (\text{PE}_i \times \text{Buyout Year}_{it_0+j}) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it}$$

$\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the control firms. $\text{Buyout Year}_{i,t_0+j}$ is a dummy for each $j$ taking a value of 1 in the year $t_0 \pm j$ relative to buyout year, with $j = -5, \cdots, 5$. $D_{cit}$ are dummies for a fully saturated interacted fixed effects of firm characteristics: industry, age, size, type of unit, and buyout year. $\text{LFIRM}_i$ is the lagged firm growth from $t_0 - 3$ to $t_0 - 1$. The year before buyout $t_0 - 1$ is the omitted category, and years beyond 5 years pre- and post-buyout are binned with year +/-5 relative to buyout. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level, for potential heterogeneity. Dotted red lines show 90% confidence intervals. Panel D shows coefficient $\gamma$ for the long run, where $\text{Post Buyout}_{it}$ captures all years post buyout for firm $i$, and 0 otherwise. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## D.5 Manufacturing Target Firms

Figure D.4: Difference in Differences Estimated Coefficients $\gamma_j$ for Manufacturing Firms Over Time Relative to Buyout Year, PE Deals 1997-2018




**(A)** Employment Growth **(B)** TFP Growth **(C)** Labor Productivity

| Dependent Variable: | Emp $g$ (1) | Pay $g$ (2) | Rev $g$ (3) | Rev $g$-Emp $g$ (4) | $\Delta \log(TFP)$ (5) |
|---|---|---|---|---|---|
| Treatment × Post Buyout | −0.0076 | −0.0082 | −0.0183 | −0.0107 | 0.0038 |
| | (0.0080) | (0.0082) | (0.0119) | (0.0120) | (0.0055) |
| Industry × Age × Size × Type × Buyout Year FE | Y | Y | Y | Y | Y |
| Year FE | Y | Y | Y | Y | Y |
| Controls | Y | Y | Y | Y | Y |
| Weighted Emp $t_0$ | Y | Y | Y | Y | Y |
| Observations | 4,106,000 | 4,106,000 | 4,106,000 | 4,106,000 | 4,106,000 |
| Adjusted $R^2$ | 0.0373 | 0.0523 | 0.0369 | 0.0158 | 0.0110 |
| Dependent Variable Mean | −0.0019 | 0.0059 | 0.0081 | 0.0100 | 0.0018 |

**(D)** Long Run

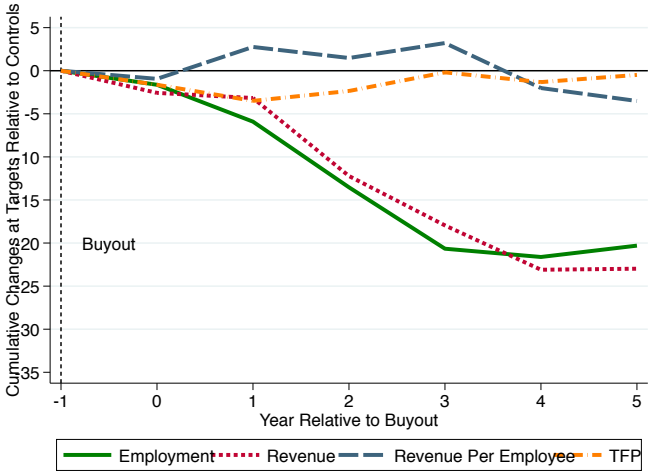
*Notes:* Panels A, B, and C display coefficients $\gamma_j$ of the difference in differences Equation:

$$y_{it} = \alpha_t + \sum_{j=-5, j\neq -1}^{j=+5} \gamma_j \left(\text{PE}_i \times \text{Buyout Year}_{it_0+j}\right) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it}$$

for manufacturing firms. $\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the control firms. $\text{Buyout Year}_{i,t_0+j}$ is a dummy for each $j$ taking a value of 1 in the year $t_0 \pm j$ relative to buyout year, with $j = -5, \cdots, 5$. $D_{cit}$ are dummies for a fully saturated interacted fixed effects of firm characteristics: industry, age, size, type of unit, and buyout year. $\text{LFIRM}_i$ is the lagged firm growth from $t_0 - 3$ to $t_0 - 1$. The year before buyout $t_0 - 1$ is the omitted category, and years beyond 5 years pre- and post-buyout are binned with year +/-5 relative to buyout. Regression estimates are weighted by employment in buyout year $t$. Standard errors are clustered at the firm level, for potential heterogeneity. Dotted red lines show 90% confidence intervals. Panel D shows coefficient $\gamma$ for the long run, where $\text{Post Buyout}_{it}$ captures all years post buyout for firm $i$, and 0 otherwise. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

**Description of Measures.** Using detailed cost metrics from the ASM and the CMF, I use total factor productivity measures constructed following the neoclassical production function. Establishment $e$'s real gross ouput at time $t$, $Y_{eit}$ can be written as a function of labor $L_{eit}$, capital $K_{eit}$, and materials $M_{eit}$: $Y_{eit} = A_{eit} \cdot F\left(K_{eit}, L_{eit}, M_{eit}\right)$. $A_{eit}$ represents the plant level productivity (TFP). Following Baily et al. (1992), $\ln \text{TFP}_{eit}$, the log of total factor productivity at the plant level is $\ln \text{TFP}_{eit} =$

Figure D.5: Cumulative Changes of Outcome Variables for Manufacturing Target Firms Relative to Control Firms Over Time, Private Equity Deals 1997-2018




*Notes:* The figure cumulates difference in differences estimated coefficients $\gamma_j$ from Figure D.4, estimated from Equation: $y_{it} = \alpha_t + \sum_{j=-5, j\neq -1}^{j=+5} \gamma_j (\text{PE}_i \times \text{Buyout Year}_{it_0+j}) + \sum_c \theta_c D_{cit} + \lambda_0 \text{LFIRM}_i + \varepsilon_{it}$, for manufacturing firms over time relative to buyout year. Estimates are normalized to 0 in year -1 relative to buyout. The transaction takes place between year -1 and 0. $\text{PE}_i$ takes the value of 1 for firms bought by private equity, and 0 for the control firms. $\text{Buyout Year}_{i,t_0+j}$ is a dummy for each $j$ taking a value of 1 in the year $t_0 \pm j$ relative to buyout year, with $j = -5, \cdots, 5$. $D_{cit}$ are dummies for a fully saturated interacted fixed effects of firm characteristics: industry, age, size, type of unit, and buyout year. $\text{LFIRM}_i$ is the lagged firm growth from $t_0 - 3$ to $t_0 - 1$. The year before buyout $t_0 - 1$ is the omitted category, and years beyond 5 years pre- and post-buyout are binned with year +/-5 relative to buyout. Standard errors are clustered at the firm level, for potential heterogeneity.

$\ln Y_{eit} - \alpha_K \ln K_{eit} - \alpha_L \ln L_{eit} - \alpha_M \ln M_{eit}$. Operationally, plant level output is shipment plus change in finished and work-in-progress inventories, deflated by the four-digit industry-level shipment deflator. Capital is calculated separately for equipment and structures using the perpetual inventory method. Labor includes production and non-production worker hours. Materials include both, energy and other materials, deflated by their respective industry-level price indices. Factor elasticities are industry-level cost shares. I use the Census computed TFP measure, and confirm it with my own construction of TFP following Abraham and White (2006), Giroud (2013).[52] I aggregate ln TFP at the firm level using employment at establishments as weights.

[52] A detailed description on capital stock and elasticities is given in the Data Appendix of Abraham and White (2006) and Foster et al. (2014) respectively.

# E Data Description

This section describes in detail the datasets used in the paper.

**Public Plans Data (PPD):** The PPD contains detailed annual data on the largest state and local pension plans in the U.S. The data ranges from 2001 to 2021 and covers 204 plans. The statistics include balance sheet variables, such as, assets, liabilities, funded positions, asset allocations, investment returns and horizon. More information can be found here: https://publicplansdata.org/public-plans-database/.

**Comprehensive Annual Financial Reports (CAFRs) of Public Pensions:** I collect data on public pensions' financials from the CAFRs. From 2014 onwards, I obtain the total and net pension liabilities for pension plan rate plus/minus one percentage point, which enables me to construct duration estimates and market value of underfunding ratios (BEA-Adjusted underfunding ratios) of public pensions, described in Appendix A and used in Section 3.4. I also collect the total and net pension liability measures at the pension plan rate, the plan rate, and the total assets of the pension from the CAFRs, to match and compare them with those provided by the PPD. Sometimes, pension systems aggregate assets and liabilities across plans. This comparison is crucial to ensure that the total and net pension liabilities are collected for the appropriate plan. Appendix Section A.2 describes this in detail.

Further, I collect fiscal year end periods for each plan-year, and cross-check the values with the PPD. I also cross-check PE returns from the CAFRs with those reported by the PPD for a subset of plan-years.

**Board Composition of U.S. Public Pensions:** I obtain data on board composition of U.S. public pensions from Andonov et al. (2018) and the National Association of State Retirement Administrators (NASRA). Data from Andonov et al. (2018) is used for robustness analyses in Sections 5.1.2 and 5.2. Data from NASRA is used for sanity check.

**FOIA Requests:** The Public Plans Data has good coverage of public pension fundamentals from 2001. I supplement data on public pension assets and liabilities going back to 1983 from FOIA requests to individual pensions. This data is shared by Korteweg et al. (2023).

**Preqin:** Preqin is a dataset on alternative assets providing detailed information on investments in pri-

vate markets across all asset classes: private equity (PE), venture capital (VC), hedge funds (HF), real estate (RE), infrastructure (INF). Preqin sources its data mainly from FOIA requests and relationships with GPs/funds. More information can be found on https://pro.preqin.com/. I use the Preqin portal instead of the Wharton Research Data Service (WRDS) to download the data, as the portal has more detailed information than the WRDS access.

Preqin has multiple tables, which can be mainly classified into “investors", “GPs", “funds", “performance", and “companies and deals". To clarify, “funds" refer to a PE fund, for example Blackstone Capital Partners VI and “GPs" refer to the parent PE firm, for example Blackstone Group. I download all tables for the PE asset class category. In addition to the above mentioned main segments, Preqin also provides sub tables within each segment. This is tedious to get as one cannot download all these tables at once, and have to do it investor by investor. For example, for each investor, I download the “historical allocations", “fund portfolio", “fund family relationships", and “buyout deals exposure".

Next, I merge different tables of Preqin using investor, PE fund, GP, and target firm identifiers. This gives me linkages across the multiple players in private markets and helps me observe the entire chain of capital flow to the most granular level. Specifically, I observe CalPERs (LP) investing in Blackstone Capital Partners VI (PE fund) which belongs to Blackstone Group (GP), and the Blackstone Capital Partners VI fund buys Cordis, a medical device manufacture company (target firm) based out of Florida in 2021.

On the deal side – between the sub PE fund and the firm, I observe detailed geographic and website identifiers of the portfolio companies. I manually did web searches and visited websites of firms, to fix data discrepancies in firm location (zip codes, states) and websites. The exact terms of the deal are sparsely populated and not needed for the analysis. Additionally, I obtain fund performance measures like IRR, geographic focus, strategy, fund size, industry focus, and management fee (sparse coverage).

The uniqueness of the data comes from its granularity. First, I observe not only the relationships between the LP and GP which is mostly studied in previous literature, but also the linkages between LPs and PE funds within a GP. This allows me to study PE funds within a GP, and construct private equity specific GP size measures. Second, for a subset of LP-GP linkages, I observe the committed capital amounts, which is the amount committed by LPs to PE funds generally at the time of fund inception. This is extremely sensitive information. First, I observe this at the LP-sub PE fund level, and second, I see the exact amount committed by the LP. For the purpose of this paper, I consider deals where the country of the PE target is the U.S. and the investor is a U.S. public pension.

**Standard Statistical Establishment Listing (SSEL):** The SSEL is sourced from The Business Register (BR), which is the backbone of all U.S. Census Bureau administrative micro-data and economic surveys. The BR is a central repository maintained by the U.S. Census Bureau which tracks statistical and administrative records of all active employer business administrations having payroll during the past three years, or having an indication to hire in the future. It is the most current and compresensive database being maintained in the U.S. since 1972.

The SSEL has detailed information on establishment names and addresses including zip code and finer geographic identifiers such as the census tract and block-level. The smallest unit of observation is an establishment or a place of business. The SSEL also provides linkages across firms and employments over time. The data is continuously updated every year, and an annual snap-shot of establishments is made available to the researcher. More information about the BR and SSEL can be found in the following Center for Economic Studies (CES) working papers: https://www2.census.gov/ces/wp/2016/CES-WP-16-17.pdf and https://www2.census.gov/ces/wp/2002/CES-WP-02-17.pdf.

**Revenue Enhanced Longitudinal Business Database (LBDREV):** The LBD covers all business establishments in the U.S. private non-farm sector with at least one paid employee (Jarmin and Miranda (2002)). An establishment is the lowest level of aggregation in the LBD.

The database links establishments and firms over time, tracking entry and exit of establishments, employment, pay, and detailed industry and state codes. This enables accurate measurement of changes in business activity. This is especially crucial since firms often change their Employer Identification Number (EIN) while filing taxes, or entites change because of merger or re-organization. The main contribution of the revised LBD is to create time consistent longitudinal establishment and firm identifiers, especially for small, single-establishment firms which had broken links in prior versions. The Census Bureau re-programmed and re-examined the original LBD for such inconsistencies, and republished a revenue enhanced LBD (LBDREV) in September 2020.

In this paper, I use the revised LBD. I will refer to LBDREV as LBD. A good reference for the LBD and the changes made is https://www2.census.gov/ces/wp/2021/CES-WP-21-08.pdf.

**Census of Manufactures (CMF):** The Economic Censuses provide more detailed statistics on employment, costs, capital expenditures, value of shipments, and revenues. The CMF covers all manufacturing

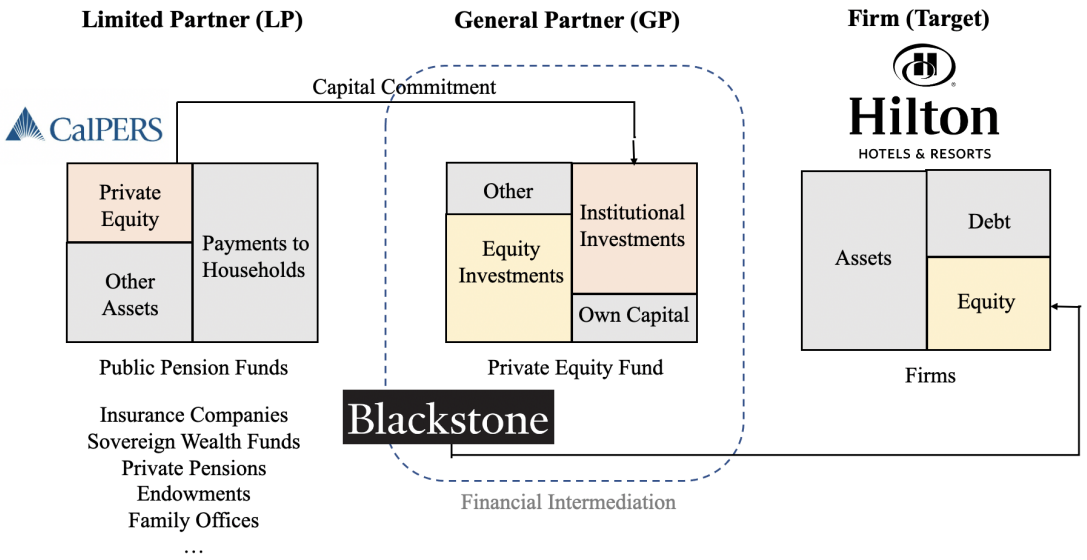
establishments and firms (NAICS Sector 31-33) with at least one paid employee. The Census is conducted every five years — years ending in '2 and '7. More information on the CMF can be found here: https://www.census.gov/data/tables/2017/econ/economic-census/naics-sector-31-33.html.

**Annual Survey of Manufactures (ASM):** The ASM provides detailed estimates of statistics for manufacturing establishments and firms with at least one paid employee. The manufacturing firms in the survey are sampled from the CMF, which covers the universe of manufacturing firms in the U.S. The ASM is conducted annually except for years ending in '2 and '7, when the CMF is carried out.

The ASM provides statistics on employment, payroll, detailed cost measures on labor, materials consumed, and energy, capital expenditures, and value of shipments. More details about the data is here: https://www.census.gov/programs-surveys/asm/about.html.

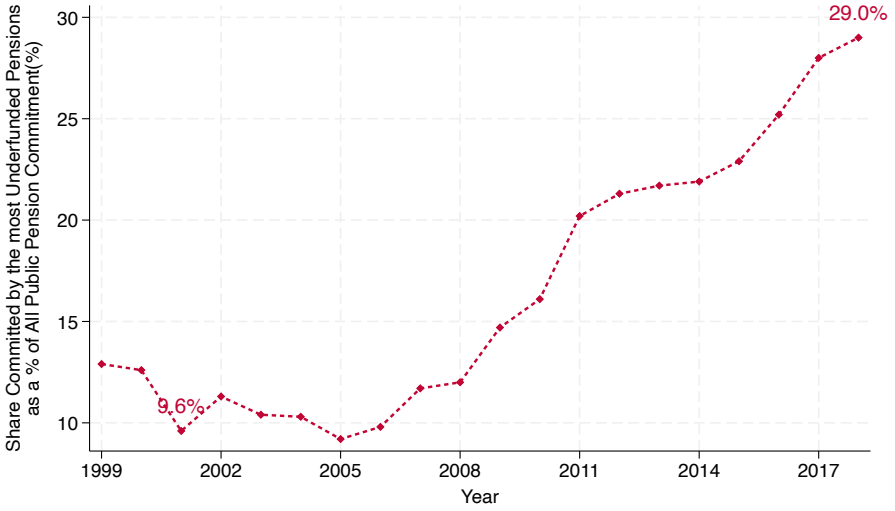
**Current Population Survey (CPS) from Union Stats and BLS:** Union Stats is the Union Membership and Coverage Database providing public and private sector labor union membership and density statistics. Union statistics are available by state, metropolitan area, and industry from 1983 to 2021. I also verify the union data from the Current Population Survey (CPS) releases on the BLS website. More information on union stats can be found here: http://www.unionstats.com; Hirsch et al. (2001) gives background on the construction of estimates.

Figure E.1: Connection between Public Pensions, PE Funds, and Target Firms




*Notes:* The figure depicts transfer of capital in private capital markets from the supplier (investor, LP on the left hand side) to the receiver (firm, target on the right side) via the intermediary (PE fund, GP in the middle).

Figure E.2: Capital Commitments by the Most Underfunded Public Pensions Over Time




*Notes:* The figure plots three year moving averages of shares committed by the most underfunded public pension tercile as a percentage of all public pension commitment to PE. The underfunding position of pension is of the year prior to commitment. The figure uses all PE buyouts. Data are sourced from Preqin and Public Plans Database.

# F Sample Construction

## F.1 Cleaning Preqin and Merging Across Preqin Datasets

### F.1.1 Investor Files

The main investor files contains investor characteristics such as name, type, indicating whether it is a public pension, private pension, sovereign wealth fund, family office, insurance company, or a bank, assets under management, allocation to private asset classes, and geographic location.

### F.1.2 Fund Portfolio Files

This data consists of investor-fund pairs. I observe the connections between investors and funds, including detailed information on investor and fund characteristics. I obtain industry focus, fund domiciles, fund vintage, and parent PE fund connections. Further, I observe the dollar amounts of capital committed between the investor and the fund. The main advantage of the study is that I observe connections between the investor and the PE fund within the GP.

### F.1.3 Deals and Portfolio Companies

The “deals" tables depict investments made by PE funds (within a GP) to firms. The firms are also known as portfolio companies. The tables have detailed geographic identifiers for the firms. The main focus of the analysis is the connection between the funds and the firms via the deal date.

### F.1.4 Cleaning and Merging

I apply the following cleaning approach:

1. In Step 1, I clean the Preqin data on portfolio companies. In many instances, the states are coded incorrectly. Preqin also has two fields of states and addresses, which don't match at times. For instance, a company might have a headquarter office and a regional office, which can be a reason for discrepancy. For companies with inconsistent states and addresses across fields, I manually search the websites of individual companies and clean the states.

2. I apply two main filters. First, I keep only those targets and deals which have at least one of the asset class designations as “PE".[53] I control for granular fund strategies in the analysis. Second, I keep targets in the U.S.

3. I standardize names and addresses of all companies in Preqin.

4. I drop observations where the deal date is not available.

5. In few cases, an investor-fund pair might be involved in multiple deals with the same target in multiple years. This is can generally happen when one PE fund sells the target to another PE fund in a secondary market. To cleanly identify the effects of buyouts, I consider the first buyout. Correspondingly, I only consider real outcome effects with respect to the first deal before the second deal. In the same spirit, in case there are multiple buyout deals for the same company in the same year, I consider the first deal by date. This can happen if different establishments within a firm undergo an LBO by different PE funds. These are very few cases and do not alter the result.

For the next part of the paper, I only consider deals which have a public pension LP or GP connection associated with them, which are majority of the matched firms.

[53] A deal can have more than one asset class designation. This can happen when a fund focuses on more than one asset class.

I merge tables from Section F.1.1, F.1.2, and F.1.3 to get the LP - GP - PE fund - target firm (or "portfolio company") chain. In order to study the effects on firms post buyout, and heterogeneity in outcomes due to LPs and GPs, I merge this chain of data with the U.S. Census datasets described below.

### F.2 Merging Private Equity Buyouts with SSEL

SSEL has names and exact addresses of all establishments in the U.S. Each establishment in the U.S. Census micro data is linked to a firm, so I have access to the full establishment-firm hierarchical structure in the U.S. The SSEL is the main Census dataset which is used to connect external datasets with the Census Bureau micro-data. I merge firms in Preqin with SSEL based on state, name, city, and address. The objective is to match the buyout targets with firms in the Census, which can be either multi- or single-unit. In a few cases, it might happen that more than one establishment in the same Census firm identifier is part of different buyout deals. I drop them, as it is not possible to ascertain which target the unmatched establishments from the Census firm belong to. I follow a step-by-step methodological approach to merge private equity targets with the Census. I perform this match within the primary state of the firm identified from Preqin, and then combine the state-by-state merged results.

1. From the output of Section F.1, I extract a list of unique PE targets in the U.S. along with their full name, address, other geographic identifiers, and deal dates. I consider the first deal date as the point of reference for targets involved in multiple deals. Additionally, one target might have two identifiers in the Preqin data. This might happen if the company changed its structure and is given a new identifier (few cases). I consider only one of the identifiers to get a unique set of target names and identifiers, which is necessary for merge with the Census micro data.

2. The SSEL establishment-firm data is sourced from the Business Register (BR). I use the SSEL yearly files from 1976 to 2019 for merging the targets with the U.S. Census micro data. Specifically, I match the targets to the SSEL file one year before the buyout deal.[54] I consider a year before the buyout, following existing literature, as the targets might undergo a name or entity change, or might dissolve some years post buyout. The number of establishments in the SSEL range from 5.2 mn. in 1976 to 9 mn. in 2019, and the number of firms from 4.5 mn. in 1976 to 7 mn. in 2019. I take the following cleaning approach:

   (a) I consider the state code from CBP. This state code is available for most establishments. This

[54] I redo the match using two years before the deal, it does not change the result.

code also matches with the state fips code based on the physical and mailing addresses for majority of the establishments. When the state code from CBP is not available, I consider the physical state code, followed by the mailing state code. I do not consider establishments which do not have a state associated to them for merge accuracy.

(b) I standardize names and addresses of all establishments in the Census. I consider both the main name ("name1") and the pseudo name ("name2"), and the street and physical addresses. I standardize both versions of the names and addresses. For merge accuracy, I do not consider establishments which have no name.[55]

3. I match on exact state and names, exact state and addresses. I do multiple checks to make sure the match is accurate. First, for the address matches, I check for zip code and city matches. I do not impose stringent restrictions for city matches. To get accurate matches, I make sure the city in the Preqin data approximately matches the city of at least one establishment in the Census data. Second, I omit all "PO Box" matches.

4. It might be the case that one portfolio company is matched to multiple Census firm identifiers. This can happen for two reasons. First, when multiple firms have the same address, for instance in a large complex. Second, when a firm has different Census firm identifiers but the same headquarter address for its various subsidiaries. This gives false matches. In such a situation, ideally I would want to find the closest Census-firm subsidiary to the target. However, it is not feasible to distinguish between the two cases. To get the cleanest possible sample, I drop cases where one target is matched to multiple firms within a state.

5. The reverse might also be possible, in which multiple targets might be matched to the same Census firm. This might happen when different establishments of a firm are parts of different buyout deals. These situations are rare. In such situations, I am unable to identify the parent firm from the buyout data for the unmatched establishments in the SSEL. To get a clean sample, I omit such buyout targets with multiple matches.

6. Next, I combine all the links between targets and matched establishments in the SSEL year files.

[55] More information on the variables can be found here: https://www2.census.gov/ces/wp/2021/CES-WP-21-08.pdf.

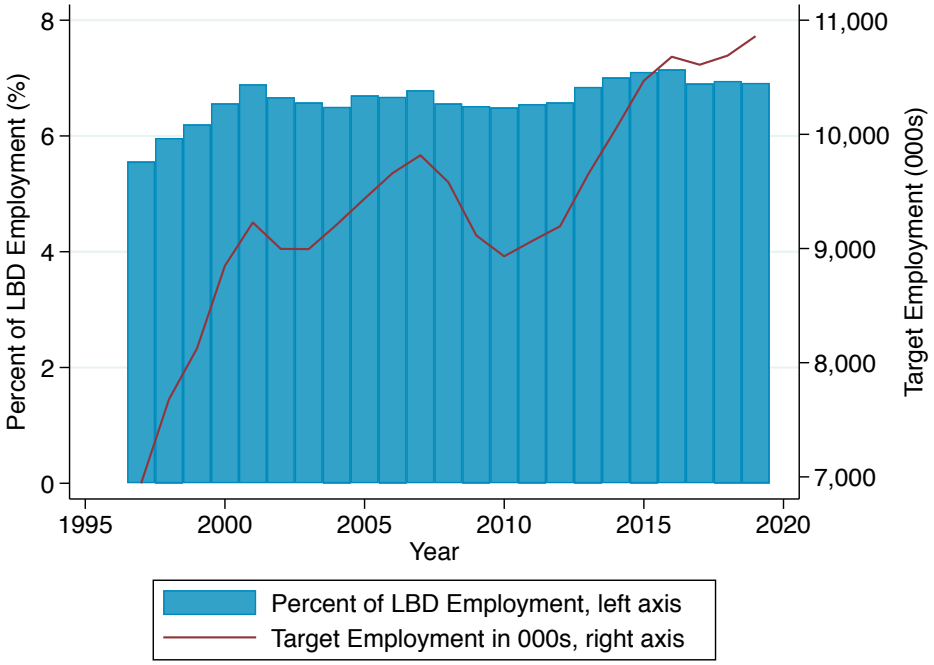
### F.3 Merging Private Equity Buyouts with Revenue Enhanced LBD

I combine all the LBD revised establishment year files. Next, I merge the output of Section F.2 with the appended LBD files by year and establishment identifier.

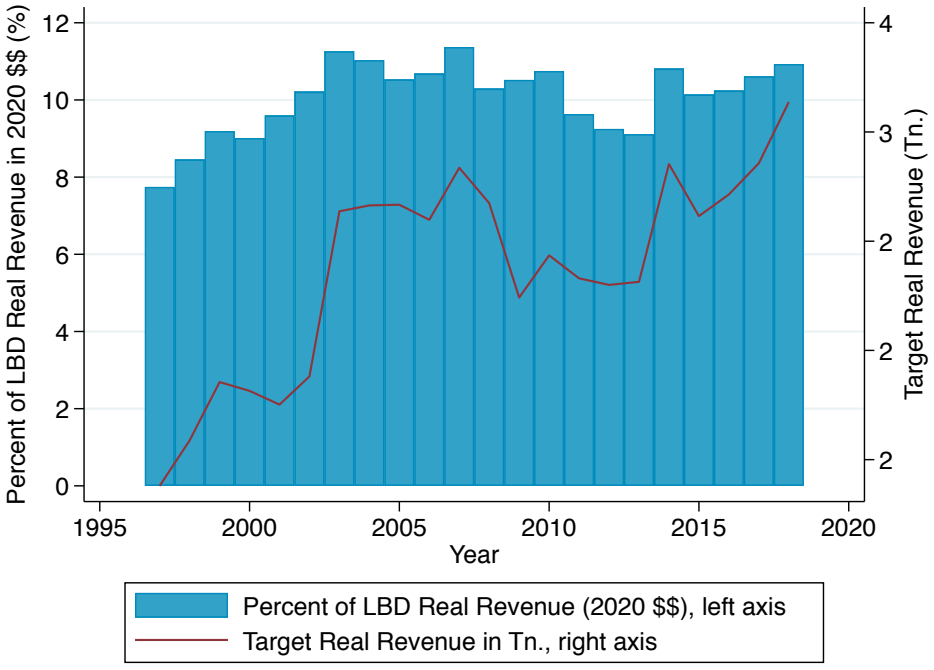
In few cases, Census firm identifiers in the SSEL and LBD do not match. I drop these to maintain consistency across datasets. In the end, the matched sample is such that the firm identifiers have a one to one mapping across datasets. Next, I pull all the unmatched establishments of the matched firms between Preqin and SSEL.

Figure F.1 shows PE target employment and revenue as a percentage of total U.S. non-farm payroll employment and revenue over time for the matched target firms to the Census micro-data. Figure F.2 is similar to Figure F.1, but limits attention to the year of buyout. Figure F.3 shows the matched and unmatched target firms by year. Figure F.4 shows matched and unmatched target firms by industry and state. The stringent match methodology explains the conservative matches.

Figure F.1: PE Target Employment and Revenue for matched Target Firms as a Percentage of Total U.S. Non-Farm Payroll Employment and Revenue Over Time



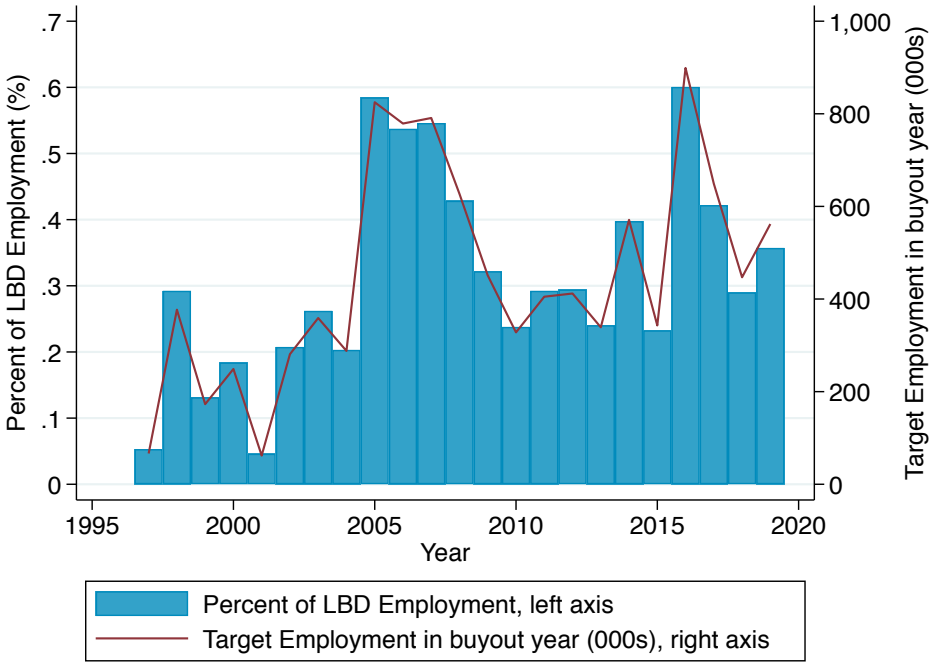

**(A)** Employment



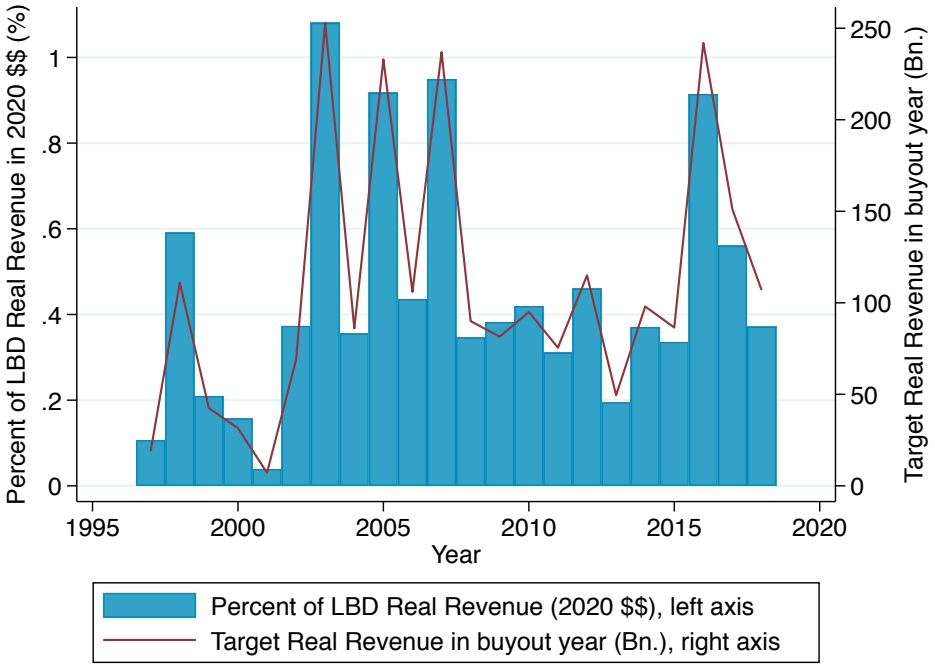

**(B)** Revenue

*Notes:* The above figures plot employment and revenue of PE targets matched with the U.S. Census micro-data. Panel A shows employment. Panel B shows revenue. The blue bars represent employment (revenue) in PE targets as a percent of total LBD employment (revenue) over time on the left axis. The red line shows total employment (revenue) in raw numbers for matched PE targets on the right axis. Revenue is in real 2020 dollars.

Figure F.2: PE Target Employment (Revenue) for matched Target Firms as a Percentage of Total U.S. Non-Farm Payroll Employment (Revenue) in Buyout Year



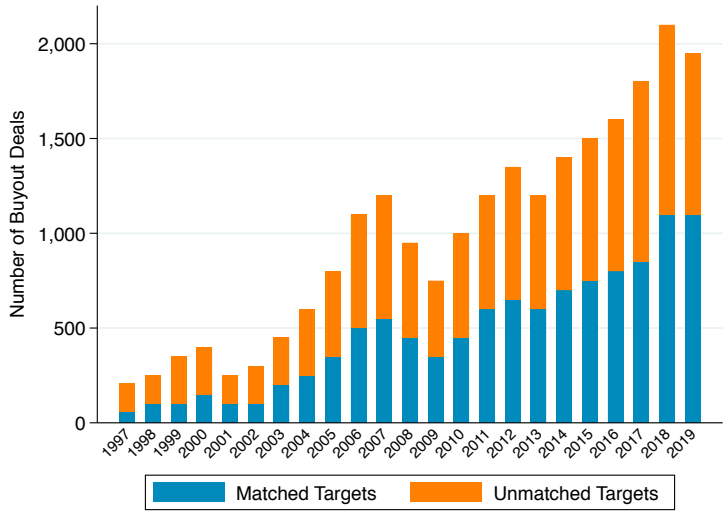

**(A)** Employment




**(B)** Revenue

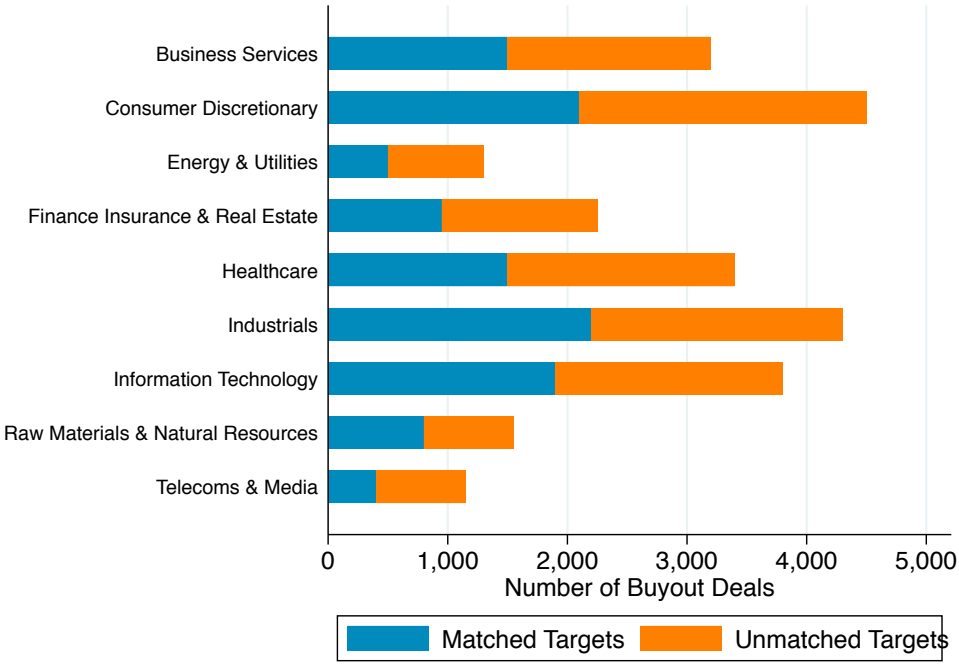
*Notes:* The figures plot employment and revenue of PE targets matched with the U.S. Census micro-data in the year of buyout. Panel A shows employment and panel B shows real revenue in 2020 dollars. Blue bars plot target employment (revenue) as a percent of total LBD employment (revenue) over time on the left axis. The red line shows total matched employment (revenue) in raw numbers on the right axis. The figure represents numbers as of the buyout year.

Figure F.3: Matched and Unmatched U.S. Target Companies Over Time

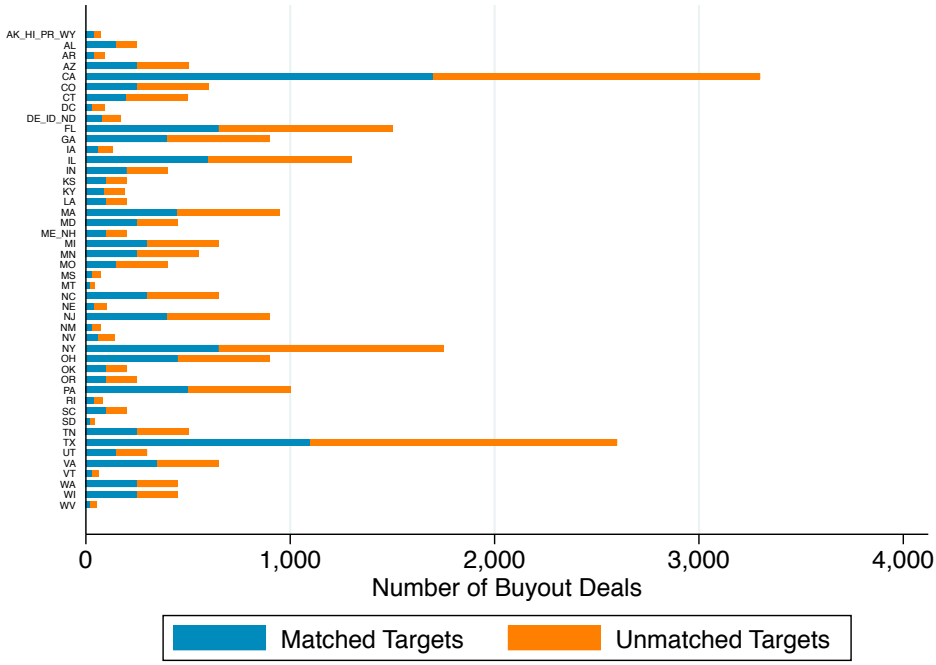



*Notes:* The figure plots the number of buyout deals involving U.S. target companies over years from 1997 to 2019. Matches prior to 1997 are not disclosed from the Census yet. The buyout deals are sourced from Preqin. Blue bars represent the number of targets matched with the U.S. Census micro-data, and orange bars represent the unmatched targets. The unmatched firms are due to a strict merge criteria considering firm characteristics one year pre-buyout to get a clean match. This reduces noise from possible incorrect addresses in external datasets.

Figure F.4: Matched and Unmatched Targets by Industry and State




**(A)** By Target Industry




**(B)** By Target State

*Notes:* The figures plot total number of buyout deals involving U.S. PE target companies across industries (panel A) and states (panel B). PE buyout deals are from 1979 to 2019. Blue bars represent the number of targets matched with the U.S. Census micro-data, and orange bars represent the unmatched targets. Some states are grouped together to meet Census disclosure requirements. Buyout deals are sourced from Preqin.

### F.4 Merging PE Target Firms with ASM and CMF

This section describes the merge of private equity buyouts with the Census of Manufactures (CMF), which exists for years ending in ′2 and ′7, and the Annual Survey of Manufactures (ASM), which is carried out every year other than years ending in ′2 and ′7.

With the revised LBD, there exists an LBDREV linkage file which connects LBDREV identifiers to the Economic Censuses and Surveys. I use this link file as a bridge to connect the LBD with the ASM and the CMF. This is especially useful as there are multiple versions of the establishment identifier in the LBD.

I use the main files from the CMF and the ASM, which have detailed information on establishment-level costs and sales. Additionally, the Census has ASM-CMF total factor productivity (TFP) files which compute TFP at the establishment level. These measures were originally used in Foster et al. (2014). The bridge file is used to merge both these datasets to the LBD.

I also merge the NBER-CES Manufacturing Database to the ASM and the CMF via four-digit SIC codes and years. For this purpose, it is important to get a comprehensive link of the establishments with the industry codes. I use the industry codes in the LBD as the base, and supplement it with industry codes in the ASM and the CMF, when missing. The coverage of the LBD industry codes is better than that of the ASM and the CMF.

### F.5 Merging PE Target Firms with Public Plans Database (PPD) and FOIA Requested Data

First, I supplement financials from the Public Plans Data (PPD) with FOIA requests from 75 individual public pensions. I complement the data going back until 1983 for these pensions.

I manually match U.S. public pension investors in the private equity dataset to the public pensions in the PPD and FOIA combined dataset, by name. I manually search the websites of each state pension. Often times, a state pension will have different subsidiaries for teachers, employees, firemen, maintaining separate balance sheets. I match financials and individual PE investments on the subsidiary – i.e., I match California Teachers' financials with California Teachers' individual PE investments. In cases where I do not have the exact subsidiary, I match financials of the parent plan, e.g. Colorado Public Employee Retirement Association for its Local, State, and School division.